FITTER-EU

DELIVERABLE 4.4

Computational analysis report

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Abstract

This deliverable (D4.4 — Computational Analysis Report) presents the full statistical analysis framework and results developed within Task 4.3 of the FITTER-EU project. It reports the methodology and empirical findings of a survey-based study examining how individuals in five EU countries perceive the hazards associated with three major EU green and digital transition policy objectives, each aligned with a key sector: photovoltaic (PV) system installations in the energy sector, retrofitting of buildings in the housing sector, and the electrification of vehicles (EVs) in the transport sector.

The analytical framework applies a nine-step pipeline to survey data collected from 1,028 respondents across Germany, Italy, Spain, Portugal, and Ireland. Each step is executed independently for every hazard in every sector, producing a per-hazard analysis (Analysis A — primary) that is cross-validated against a composite-score analysis (Analysis B — reference baseline). The pipeline encompasses data cleaning and encoding (Steps 1–2), LASSO-based predictor shortlisting (Step 3), ordinal proportional-odds regression (Step 4), Likelihood-Ratio Testing with Benjamini–Hochberg FDR correction and events-per-predictor capping (Step 5), global model validation (Step 5b), sub-sampling stability testing (Step 5c), effect-size quantification through odds ratios and marginal effects (Step 6), group profiling via ANOVA (Step 7), per-country macro profiling (Step 7b), country comparison (Step 8), and cross-hazard synthesis (Step 9).

Key empirical findings include: (i) financial pressure is the dominant cross-sector predictor — respondents with a history of utility-bill arrears are nearly six times have nearly six times the odds of being highly concerned; (ii) respondents in the lowest housing-cost band have two to three times the odds of high concern across four Housing hazards, the most statistically robust cross-hazard finding in the study; (iii) non-native-language speakers in the Transport sector show significantly elevated concern, particularly for journey complexity hazards; (iv) country differences are largely absent in Energy and Transport but statistically significant in Housing, with Portuguese respondents more concerned than German respondents. These findings are designed to feed directly into the FITTER Digital Platform and support policymakers in designing equitable mitigation strategies for disadvantaged groups.

Executive Summary

Deliverable D4.4 presents the computational analysis of how people across the Energy, Housing, and Transport sectors of the FITTER-EU project perceive the hazards of the green and digital ("twin") transition. It combines a structured analytical method (Section 2) with full sector-by-sector results (Section 3), turning raw survey responses into evidence that policymakers can act on. For each sector, the analysis identifies which groups are most concerned about transition-related hazards, and why.

How the analysis works. The study is based on survey responses from 1,028 people in five EU countries. Each sector is studied on its own, and every individual hazard is examined using the same step-by-step procedure: the data are cleaned and prepared, the respondent characteristics most relevant to concern are identified, their influence on the level of concern is measured, and only the predictors that survive strict statistical checks are reported. A primary hazard-by-hazard analysis is cross-checked against a combined overall-concern analysis to confirm the findings are consistent. Full technical detail is provided in Section 2.

Energy sector (n = 407, five countries). The seven hazards analysed here relate specifically to the objective of increasing photovoltaic (PV) system installations — not to every possible hazard in the energy sector. The strongest predictor of high concern is a history of unpaid utility bills: respondents who have fallen behind on bills two or more times have nearly six times the odds of being highly concerned (odds ratio 5.76). Low confidence in managing bills, living in a mid-century home, and political identity also matter. Concern levels are similar across all five countries, showing that this is a Europe-wide, financially driven pattern rather than a national one. Policy implication: support measures attached to PV rollout should prioritise households under financial strain.

Housing sector (n = 263, three countries). The thirteen hazards analysed here relate specifically to the objective of retrofitting non-energy-efficient buildings. The clearest and most reliable finding in the whole study is that respondents in the lowest housing-cost band have two to three times the odds of high concern — confirmed across four separate hazards, all pointing the same way. Housing also produced the study's strongest individual models, especially for concerns about damp and mould. For five of the thirteen hazards no predictor stood out, meaning concern there is shared widely rather than concentrated in particular groups. Housing is the only sector where the country matters: Portuguese respondents are significantly more concerned than German respondents. Policy implication: retrofit programmes should protect lower-income occupants, while health-related concerns such as damp and mould call for broad public communication.

Transport sector (n = 358, two countries). The six hazards analysed here relate specifically to the objective of electrifying vehicles (EVs). Concern is highest among people who are more exposed or more vulnerable to changes in transport: women, unemployed respondents, frequent travellers, and people who do not speak the national language. Non-native-language speakers have roughly 70% higher odds of high concern overall (odds ratio 1.76), rising to more than double for longer or more complex journeys. Italy and Spain show mirror-image patterns: Italians worry more about pollution exposure, Spaniards more about being restricted from city centres — even though the same underlying factor (how much people rely on cars) drives both. Policy implication: communication around EV and city-access measures should be inclusive and multilingual.

Cross-sector conclusions. Three findings hold across all three sectors. First, financially vulnerable people perceive transition hazards as more serious, whatever the sector — making financial vulnerability the single most important factor to address. Second, the hazards that worry the most people are the hardest to link to any particular group: where concern is near-universal, targeting individual profiles adds little and broad reassurance works better. Third, the country a person lives in explains only a small part of how concerned they are (around 1–4%), far less than their personal circumstances and behaviour — so mitigation can be designed around socio-economic profiles and applied broadly across countries. These results are structured for integration into the FITTER Digital Platform (WP5) to support hazard prediction and targeted mitigation planning for disadvantaged groups.

Abbreviations and Acronyms

Abbreviation / Acronym	Description
AE DGs	Adverse Effects on Disadvantaged Groups
ANOVA	Analysis of Variance
AUC	Area Under the Receiver Operating Characteristic Curve
BH	Benjamini–Hochberg
BH-FDR	Benjamini–Hochberg False Discovery Rate correction
BIC	Bayesian Information Criterion
CCGS	Cluster Community Group of Stakeholders
CI	Confidence Interval
CSV	Comma-Separated Values
CV	Cross-Validation
DG	Disadvantaged Group(s)
EPP	Events Per Predictor
EU	European Union
EU-SILC	European Union Statistics on Income and Living Conditions
EV	Electric Vehicle(s)
FDR	False Discovery Rate
H0	Null Hypothesis
H1	Alternative Hypothesis
HSD	Honestly Significant Difference
ICE	Internal Combustion Engine
kNN	k-Nearest Neighbours
L1	L1-norm (absolute-value) regularisation
L2	L2-norm (squared) regularisation
LASSO	Least Absolute Shrinkage and Selection Operator
LCA	Latent Class Analysis
LRT	Likelihood-Ratio Test
MEM	Marginal Effect at the Mean
MICE	Multiple Imputation by Chained Equations
MLE	Maximum Likelihood Estimation
MS	Mean Square
NUT2	Nomenclature of Territorial Units for Statistics — Level 2
OLS	Ordinary Least Squares regression

Abbreviation / Acronym	Description
OR	Odds Ratio
PDSS	Predictive Decision Support System
PV	Photovoltaic
ROC	Receiver Operating Characteristic (curve)
SD	Standard Deviation
SE	Standard Error
SS	Sum of Squares
VIF	Variance Inflation Factor
WP	Work Package

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In the preparation of this deliverable, generative AI tools were used in a limited and supervised capacity to assist with: (i) language editing and proof-reading of author-written text; (ii) formatting and consistency checks of figures, tables and cross-references; and (iii) drafting support for the structuring of architectural descriptions subsequently reviewed, corrected and validated by the authors. All architectural decisions, technical specifications, design choices, and conclusions in this deliverable were defined, verified and approved by the human authors. The authors take full responsibility for the content of this deliverable.

1. Introduction

1.1 Overview of FITTER-EU

FITTER-EU (Fair and Inclusive Twin Transitions for a Stronger Social Europe) is a three-year Horizon Europe research and innovation project funded under grant agreement number 101132546, running from 1 February 2024 to 31 January 2027. The project is coordinated by Technological University Dublin (TU Dublin) and involves a consortium of ten partner organisations spanning Ireland, Spain, Italy, Germany, Hungary, and Portugal.

The overarching ambition of FITTER-EU is to understand the origins, dynamics, and drivers of inequalities linked to the twin transition — the simultaneous shift towards a green, low-carbon economy and a digitally enabled society — and to provide policymakers with evidence-based tools to manage those inequalities proactively. The twin transition is reshaping fundamental sectors of daily life, including how people heat their homes, move around, and consume energy. While the long-term benefits of this transition are widely acknowledged, its short- and medium-term costs are not distributed equally. Disadvantaged groups — including low-income households, older adults, people with disabilities, and those in rural or peripheral areas — are often disproportionately exposed to the hazards of rapid policy-driven change.

To address this challenge, FITTER-EU has developed the FITTER Ecosystem: a co-created, evidence-driven environment centred on a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System (PDSS). This platform enables policymakers and stakeholders to: (i) predict which social groups are at risk of being adversely affected by specific twin transition policies; (ii) simulate policy scenarios to assess their distributional impacts; and (iii) design and evaluate targeted mitigation measures for identified at-risk groups. The FITTER Cluster Community Group of Stakeholders (CCGS) — comprising policymakers, civil society organisations, trade unions, and other key actors — plays a central role in co-creating and validating these tools.

The project organises its work across seven Work Packages. This deliverable, D4.4, is produced under WP4 ("Inequalities Risk Assessment and Intersectional Mitigation Plans"), which is responsible for quantifying hazard perceptions, identifying disadvantaged groups across three priority sectors, and constructing the predictive modelling framework that powers the FITTER Digital Platform.

1.2 Purpose and Scope

This deliverable, D4.4 — Computational Analysis Report — presents the complete statistical methodology and empirical results applied to survey data collected in Tasks 4.1 and 4.3 focusing on three major EU green and digital transition policy objectives, each aligned with a key sector: photovoltaic (PV) system installations in the energy sector, retrofitting of buildings in the housing sector, and the electrification of vehicles (EVs) in the transport sector. The core purpose is to move beyond descriptive statistics and provide a rigorous, reproducible analytical framework that identifies which characteristics of respondents (socio-demographic, behavioural, geographic) most strongly predict how seriously they perceive the hazards associated with twin transition policies in each sector.

Understanding who perceives these hazards as serious — and why — is essential for designing equitable and publicly accepted transition pathways. This deliverable provides the statistical engine that translates raw survey responses into actionable insights for policymakers and the FITTER Digital Platform.

The scope of this deliverable covers all three sectors independently. Although the three analyses share the same analytical framework and socio-demographic predictor battery, each sector is treated as a completely self-contained study with its own dedicated sample. The Energy sector includes 407 respondents from five countries (Germany, Italy, Spain, Portugal, Ireland); Housing includes 263 respondents from three countries (Germany, Portugal, Ireland); Transport includes 358 respondents from two countries (Italy, Spain). Total analytical sample: 1,028 respondents.

The analysis in D4.4 addresses four core research questions:

- **Prediction:** Can we reliably predict whether a respondent considers a given policy hazard in this sector to be serious or not?

- **Relevance:** Of all measured predictors in a sector's survey, which ones are most strongly associated with high hazard perception?
- **Effect strength:** How large is the effect of each key predictor on hazard perception — and in which direction?
- **Profiling:** Which socio-demographic profiles are most consistently affected by the policy hazards identified in each sector?

The methodology described in this deliverable is grounded in established statistical practice for ordinal survey data (McCullagh, 1980; Agresti, 2010) and penalised predictor selection (Tibshirani, 1996; Hosmer et al., 2013).

1.3 Structure of the Deliverable

This deliverable is organised as follows:

- **Section 1 (Introduction)** provides the project overview, the purpose and scope of the deliverable, and its connections with other Work Packages.
- **Section 2 (Methodology)** describes the complete nine-step analytical pipeline, covering data preparation and encoding (Steps 1–2), LASSO predictor shortlisting (Step 3), ordinal regression modelling (Step 4), significance testing with FDR correction and EPP capping (Steps 5–5c), effect-size quantification (Step 6), group profiling via ANOVA and per-country macro profiling (Steps 7–7b), country comparison (Step 8), and cross-hazard synthesis (Step 9). Two parallel analyses — per-hazard (A) and composite (B) — run throughout.
- **Section 3 (Results)** presents sector-by-sector empirical findings for Energy (n = 407), Housing (n = 263), and Transport (n = 358), followed by cross-sector conclusions for policy.
- **Section 4 (Discussion)** interprets the findings in the context of the FITTER-EU project objectives and twin transition policy priorities.
- **Section 5 (Conclusions and Further Developments)** summarises the main contributions and identifies next steps for the FITTER Digital Platform integration.
- **References** and **Annexes** follow.

1.4 Connections with Other Work Packages

D4.4 connects explicitly with several other Work Packages across the FITTER project:

- **WP1 (Project Management & Quality Assurance):** The methodology and quality controls described in D4.4 feed into the project quality monitoring process. Sampling limitations, analytical assumptions, and reproducibility risks identified here are tracked within the Quality, Risk & Management Assessment Plan (D1.2/D1.3).
- **WP2 (Methodology, Stakeholder Engagement & CCGS):** The choice of predictors and the sectors analysed in D4.4 are grounded in the theoretical framework on just transitions developed in WP2. The list of potential disadvantaged groups identified in D2.1 and D2.6 directly informs the socio-demographic profiling carried out in Section 3.
- **WP3 (Vulnerability Mapping & Scenarios):** The hazard definitions and vulnerability profiles developed in WP3 (D3.1–D3.3) provide contextual background for interpreting the results in D4.4. The probability and severity rates estimated in D4.1 are quantitatively anchored to WP3 scenario outputs.
- **WP5 (FITTER Digital Platform Development):** The predictive models and profiling outputs from D4.4 are directly structured for integration into the FITTER Digital Platform (D5.1–D5.3). The PDSS ingests these models to enable real-time hazard prediction and scenario simulation for policymakers.
- **WP6 (Validation and Demonstration):** The model outputs from D4.4 will be evaluated for practical utility and reliability in the validation exercises (D6.1–D6.3).
- **WP7 (Dissemination & Policy Uptake):** Key findings from D4.4 — top predictors per sector, identified disadvantaged groups, and policy implications — will inform policy briefs and dissemination activities (D7.8–D7.9) directed at the CCGS and broader stakeholder audiences.

2. Methodology

This section explains, step by step, how the data was collected, cleaned, and analysed. It applies one-to-one to every sector reported in Section 2.

This study is based on a survey to capture the perceptions of people from five EU case countries about the risks that come with switching to cleaner energy and more digital ways of living. We call these risks 'hazards'. For each hazard, we asked two questions: how likely do you think this is to happen to you? And if it did happen, how badly would it affect you?

Two decisions shape the whole approach. First, we deliberately do not include country labels (such as Germany or Italy) as explanatory factors in the main model — instead we use national statistics such as energy poverty rates and average temperatures, which are more specific and meaningful (Macro Variables). Country differences are described separately in Step 8.

Second, every step in this methodology is run in two parallel versions, which appear side by side as subsections A and B throughout the document. The two sub sections represent two types of analyses that are carried out simultaneously – per-hazard (A) and composite (B) analyses. The same nine steps are followed for all three sectors — Energy, Housing, and Transport — so results can be compared fairly across sectors.

2.1 Overview: the nine analytical steps

The nine analytical steps fall into three phases. **Phase 1** — Data Preparation (Steps 1 and 2) cleans the raw survey and encodes the variables. **Phase 2** — Modelling and Confirmation (Steps 3 through 5b) builds the per-hazard ordinal models, screens candidate predictors via LASSO, and confirms which predictors are real via likelihood-ratio testing, BH-FDR correction, and the events-per-predictor cap. **Phase 3** — Interpretation (Steps 6 through 9) measures effect sizes, identifies which groups are most affected, profiles country-anchored macro predictors, compares concern levels across countries, and synthesizes convergent findings across hazards. Steps 1 and 2 apply once to the full dataset; Steps 3 through 7b run independently for each hazard in each sector; Steps 8 and 9 operate at the cross-hazard level.

Phase 1 — Data Preparation

- Step 1 — Getting the data ready (data cleaning)
- Step 2 — Preparing the data for the statistical model (encoding)

Phase 2 — Modelling & Confirmation

- Step 3 — Choosing which variables to test (LASSO)
- Step 4 — The main statistical model (proportional odds regression + Brant test + McFadden R^2)
- Step 5 — Confirming which predictors are real (LRT + Benjamini-Hochberg FDR + per-hazard EPP cap)
- Step 5b — Validating the overall predictive model (global Likelihood-Ratio Test)
- Step 5c — Predictor robustness test

Phase 3 — Interpretation

- Step 6 — Measuring the size of each effect (odds ratios + marginal effects)
- Step 7 — Identifying the most affected groups (ANOVA on categorical predictors only)
- Step 7b — Per-country profile for confirmed macro predictors
- Step 8 — Comparing results across countries (composite-level ANOVA + Tukey HSD)
- Step 9 — Comparing results across hazards (cross-hazard synthesis + A/B convergence)

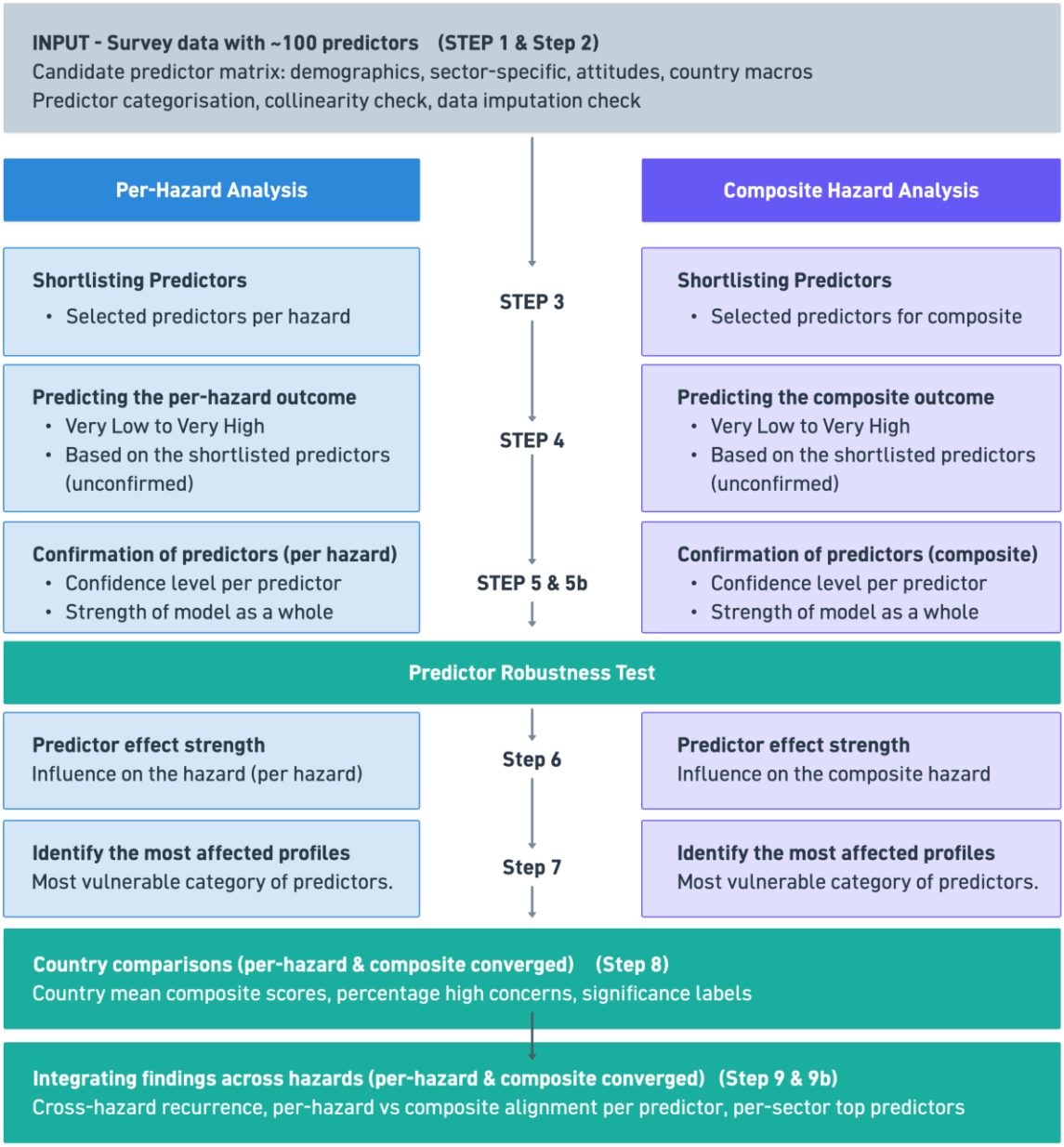


Figure 1. The nine analytical steps of the methodology

Across the nine steps mentioned above, two analyses run in parallel:

A — Per-Hazard Analysis (primary)

In the per-hazard analysis, each hazard is modelled completely independently. Hazard 1 (e.g. higher electricity bills) gets its own LASSO shortlist, its own ordinal model, its own confirmed predictors, its own odds ratios, and its own group profiles. The per-hazard analysis is the primary analysis of this study. It is always labelled 'A' and presented first in each step.

B — Composite Hazard Analysis (reference baseline)

In the composite analysis, all J hazard scores for a respondent are averaged into one overall sector concern score (Formula 2), which is then grouped into five concern levels (Formula 3). A single model is then run on this composite outcome. It is always labelled 'B' and presented second.

2.1.1 Step 1 — Getting the data ready

Cleaning the raw survey file before any analysis begins.

Steps 1 and 2 apply once to the full dataset before any per-hazard analysis begins. They are not split into composite and per-hazard versions because they produce the clean data that both analyses draw on.

Which columns we removed and why

Not every column in a survey file is useful for statistical analysis. We removed system columns (timestamps, participant IDs, consent flags), routing-skipped questions shown only to subgroups (treated as 'Not applicable'), free-text responses, ranking questions, and national-level statistics whose information is already captured by the macro indicators (Little & Rubin, 2002).

Survey completion time filter

Before any other cleaning, we removed respondents who completed the survey in fewer than four minutes. A survey of this length and complexity cannot be answered thoughtfully in under four minutes. Responses completed too quickly indicate that the respondent was clicking through answers without reading them, which would introduce random noise into all subsequent analyses. The four-minute minimum threshold was applied consistently across all three sectors (Energy, Housing, Transport).

Fixing data entry errors

We checked every numeric column for implausible values: ages above 120, scale responses outside the stated range, housing costs above reasonable upper limits. Where a category in a question had fewer than five respondents, we merged it with the most similar adjacent category to avoid unreliable cell sizes (Nunnally, 1978).

Building the hazard scores

For each hazard, respondents answered two questions: how likely is this to happen to you (Likelihood, 1–5), and how badly would it affect you (Impact, 1–5). The hazard score is their product on a 1–25 scale, to be used for both the analyses (per-hazard & composite analyses) simultaneously.

Formula 1 — Hazard score

$$\textit{Hazard score (h)} = \textit{Likelihood (h)} \times \textit{Impact (h)}$$

Range 1–25 | Multiplying ensures that a hazard seen as both very likely AND very damaging scores much higher than one that is only one or the other

Symbol / term	What it means
Hazard score	The combined concern score for one specific hazard. Range: integer-valued 1 to 25 (the product of Likelihood × Impact)
Likelihood	How likely the respondent thinks this hazard is to happen to them personally. Range: integer 1 (not at all likely) to 5 (extremely likely)
Impact	How badly the respondent thinks this hazard would affect them if it occurred. Range: integer 1 (not badly at all) to 5 (extremely badly)

× (multiply)	Multiplication operator. Multiplying ensures that a hazard seen as both very likely AND very damaging scores much higher than one that is only one or the other

From hazard scores to ordered concern levels

Applies to both per-hazard (A) and composite (B) analyses — the binning procedure is identical, only the input differs.

We convert the continuous hazard score (1–25) into an ordered categorical variable with five levels (Very low, Low, Moderate, High, Very high) so the proportional-odds model in Step 4 can estimate reliable coefficients. The five-level quintile split ensures all five groups contain approximately the same number of respondents. The same Q_5 quintile cut-point function (Formula 3) is applied in both studies:

- **Per-Hazard analysis (A):** Each individual hazard's 1–25 score is binned into the five concern levels. This 5-level outcome is what Step 4's proportional-odds model fits for each hazard separately.
- **Composite analysis (B):** The per-respondent composite score (the average across all hazards in the sector, from Formula 2) is binned into the five concern levels. This 5-level composite outcome is what Step 4's proportional-odds composite model fits.

Formula 2 — Composite score (per respondent)

$$Composite\ score\ (r) = (1/J) \times \Sigma_{h=1}^{J}\ hazard_score(r, h)$$

J = number of hazards in this sector | r = respondent | Range 1–25 (continuous)

Symbol / term	What it means
Composite score (r)	The respondent's average concern across all J hazards in this sector. Range 1 to 25 (continuous). Used as the outcome in the composite analysis (subsection B of every step)
J	Number of hazards in this sector: 7 in Energy, 13 in Housing, 6 in Transport
r	Identifier for one respondent. Each respondent receives one composite score
Σ_{h=1}^{J}	Summation across all J hazards in the sector — i.e., add up the hazard scores from Formula 1 for hazards 1, 2, 3, …, J
hazard_score(r, h)	The respondent r's score for hazard h (from Formula 1). Range 1 to 25
1/J	Divides the sum of hazard scores by the number of hazards, giving the per-respondent average. Keeps the composite on the same 1–25 scale as the individual hazards for direct comparability

Formula 3 — Quintile binning

Concern level (r) = Q_5 (score (r))

Q_5 = quintile cut-point function: levels 1 (Very low) through 5 (Very high)

Symbol / term	What it means
score(r)	The score being binned. In the per-hazard analysis, this is the respondent's 1–25 score for one specific hazard. In the composite analysis, this is the respondent's composite score from Formula 2.
Concern level (r)	The respondent's ordered concern category. Range: integer 1, 2, 3, 4, or 5
Q_5(·)	Quintile cut-point function: assigns each respondent to one of five equal-sized groups based on the rank of their composite score. The four cut-points are the 20th, 40th, 60th, and 80th percentiles of the composite score distribution
Level 1 — Very low	Respondents in the bottom 20% of composite scores. Range: percentile 0 to 20
Level 2 — Low	Respondents in the next 20%. Range: percentile 20 to 40
Level 3 — Moderate	Respondents in the middle 20%. Range: percentile 40 to 60
Level 4 — High	Respondents in the next 20%. Range: percentile 60 to 80
Level 5 — Very high	Respondents in the top 20% of composite scores. Range: percentile 80 to 100. This is the level used as the 'Very high' probability target in Formula 18

Cronbach's α — internal consistency check

Applies to the composite analysis (B) only — α validates the prerequisite that the J hazards measure a coherent underlying concept, which is what makes averaging them into a composite (Formula 2) meaningful.

This sub-section is not required for the per-hazard analysis (A). Each hazard in the per-hazard track is modelled independently with its own outcome and predictor set, so the internal consistency across hazards has no bearing on whether the per-hazard models are valid. The per-hazard analysis would proceed identically whether α were 0.65 or 0.90.

For the composite analysis (B), α is a prerequisite check: before averaging the J hazard scores into a single composite score (Formula 2), we verify that the hazards within a sector measure a coherent underlying concept. If α were below the conventional 0.70 threshold, the composite would not be defensible and the B-track results would need to be re-interpreted as a weighted average of unrelated items rather than as a measure of overall sector concern.

α is computed once per sector. The results in this study are:

- **Energy:** α = 0.8157 — good (≥ 0.80), supports the composite analysis.
- **Housing:** α = 0.8383 — good (≥ 0.80), supports the composite analysis.
- **Transport:** α = 0.7921 — acceptable (≥ 0.70), supports the composite analysis.

Formula 4 — Cronbach's α

$$\alpha = (J / (J - 1)) \times [1 - \Sigma\, Var(item_j) / Var(total)]$$

J = number of hazards | Var(item_j) = variance of one hazard's scores | Var(total) = variance of the per-respondent sum of hazard scores

Symbol / term	What it means
α (alpha)	Cronbach's alpha — the consistency score. Range 0 to 1. Above 0.70 is acceptable; above 0.80 is good
J	Number of hazard questions being combined. Range: positive integer (7 in Energy, 13 in Housing, 6 in Transport)
J / (J−1)	Scaling correction for the number of items. Range: slightly above 1 (e.g., 7/6 ≈ 1.17 for Energy)
Sum of individual variances	Sum of how much each hazard question's scores spread out across respondents, taken across all J questions. Range ≥ 0; large when many hazards have wide variation
Variance of the total	How much the sum of all hazard questions spreads out across respondents. Range > 0; reflects both within-item variance and the correlations between items

2.1.2 Step 2 — Preparing the data for the statistical model

Turning survey answers into numbers the model can use.

Converting categories into yes/no columns (one-hot encoding)

Every categorical answer becomes one binary (yes/no) column per category. For example, gender with three options (Man / Woman / Non-binary) becomes three columns — but only two enter the model, because one category is reserved as the reference (against which the others are compared).

The choice of reference category is important. We choose the most natural or most common baseline: for gender, Man (the largest cell in all three sectors); for dwelling age, the largest cell within each sector's sample (1981–2000 in Energy, 1961–1980 in Housing); for political identity, the group that did not identify with any group. For English proficiency in Housing, the reference is 'Mother tongue / Very well' (the linguistically cleanest baseline, against which other proficiency levels are compared).

Where the largest-cell rule produces a reference category with substantive interpretation difficulties (for example, 'Not applicable' in skip-logic recoded variables), an interpretively cleaner reference is chosen instead. These deviations from the largest-cell rule are documented in the data cleaning notes for each sector so the reader can locate every exception. Applies to both the analyses: per-hazard (A) and composite (B) analyses.

Country labels: deliberately excluded

Country labels are not included in the main statistical model. The reason is a mathematical conflict: the country label and the geographic macro variables carry overlapping information. For example, every German respondent receives the same value of every macro indicator and including both would cause the model to fail to separate two effects that carry identical information. We use the macro indicators instead, because they are more specific and more meaningful. Country differences are reported descriptively in Step 8 (Snijders & Bosker, 2012). Applies to both the analyses: per-hazard (A) and composite (B) analyses.

Filling in missing values

After all transformations, a small number of values remain missing — typically from genuine non-response on one or two specific questions. We do not remove respondents with any missing data because that would drastically reduce the sample. Instead, missing cells are filled by iterative k-Nearest-Neighbours imputation: each missing value is replaced by the average of the corresponding values among the 5 most similar respondents (where similarity is measured by all the other answers we have for that respondent).

Formula 5 — Iterative kNN imputation (conceptual)

$$\textit{fill missing}(\, x_{r,j} \,) \leftarrow \textit{average of } x_{r',j} \textit{ for } r' \in k \textit{ nearest neighbours of } r$$

k = 5 | Distance computed on all non-missing features

Symbol / term	What it means
x_{r,j}	The value of variable j for respondent r — the cell being imputed (initially missing). Range: depends on the variable type (numeric on its natural scale, or 0/1 for a yes/no dummy column)
r	The respondent whose missing cell is being filled in. Range: 1 to N (the total number of respondents in this sector)
j	The variable (column) with the missing value for this respondent. Range: 1 to p (the total number of variables after one-hot encoding)
k	Number of nearest neighbours to use. Range: fixed at k = 5 across all three sectors
r' ∈ k nearest neighbours of r	The set of k = 5 respondents (excluding r) most similar to r, ranked by Euclidean distance on all the non-missing variables they share with r. Range: a set of exactly 5 respondent identifiers
Distance metric	Standard Euclidean on all features both respondents have non-missing. Categorical variables are one-hot encoded before distance calculation. Range: 0 (identical respondents) to +∞ (theoretically; bounded by feature scale in practice)

Iterative	Imputation is run iteratively because filled values from earlier passes become available as similarity features in later passes. Range: typically converges in 3–5 iterations on this dataset

Applies to both the analyses: per-hazard (A) and composite (B) analyses.

The additional "Macro variables" (sector-specific)

Each sector receives nine macro variables that attach a national or regional level context to every respondent's record. Each macro variable is matched to the respondent's country (NUTS 1), and region where available (NUTS 2), and is paired in the source data with a match-quality flag that records the geographic level at which the match was made. These national indicators are drawn from the official Eurostat statistical datasets. Within a given country, for the macro variables with NUTS 2 values (region specific) available, they are used. If NUTS 2 is missing, a natural fallback is to use the NUTS 1 value (country specific). Thus, every respondent receives the same value for each macro variable given their geographic location, which is the basis for the Step 7b described later. The variables per sector are as follows:

- **Energy**: percentage of households in arrears on utility bills; percentage unable to keep the home adequately warm; housing-cost overburden rate; household final consumption expenditure; regional household disposable income; heating degree-days; cooling degree-days; total final energy consumption by households; household electricity price band.
- **Housing**: percentage reporting the dwelling is too dark; respiratory/circulatory death rate per 100,000; percentage reporting leaks, damp, or rot; excess winter mortality; percentage unable to keep the home adequately warm; housing-cost overburden rate; heating degree-days; cooling degree-days; severe housing deprivation rate.
- **Transport**: car share of the modal split; percentage who cannot afford regular transport; cars per 1,000 inhabitants; share of cars running on alternative (non-petrol/diesel) motor energy; distribution of cars by age; new-car registrations by motor type; household expenditure on transport; road-fuel price (Super 95); extra-territorial road-transport expenditure.

The macro variables are used for both the analyses: per-hazard (A) and composite (B) analyses.

2.1.3 Overview of the statistical tests: Steps 3–5b

Steps 3 to 5b involve a sequence of distinct statistical tests, each serving a different purpose. Before the individual steps describe them in detail, this overview shows the sequence at a glance. Three roles are colour-coded: purple for shortlisting, teal for model diagnostics, and coral for inferential decision-making.

- Shortlisting (purple): LASSO reduces 95–130 candidate predictors to a manageable set. It is a practical filter — surviving LASSO means a predictor is worth testing, not that it is confirmed.
- Model diagnostics (teal): the Brant test checks whether the model structure is valid (the proportional-odds assumption). McFadden R^2 checks whether the model has any explanatory value. Both are quality gates before individual predictors are examined.
- Inferential decision-making (coral): the Likelihood-Ratio Test (LRT) with Benjamini–Hochberg FDR correction makes the actual confirmed-or-rejected decision for each predictor. The global LRT (Step 5b) validates the entire confirmed set collectively.

Key point. The Brant test and McFadden R^2 do not confirm or reject predictors. They validate the model. Only the LRT in Step 5 makes the confirmation decision. A predictor is confirmed when it passes all three stages: LASSO shortlist, ordinal model estimation, and LRT with FDR correction ($p_{adj} < 0.05$).

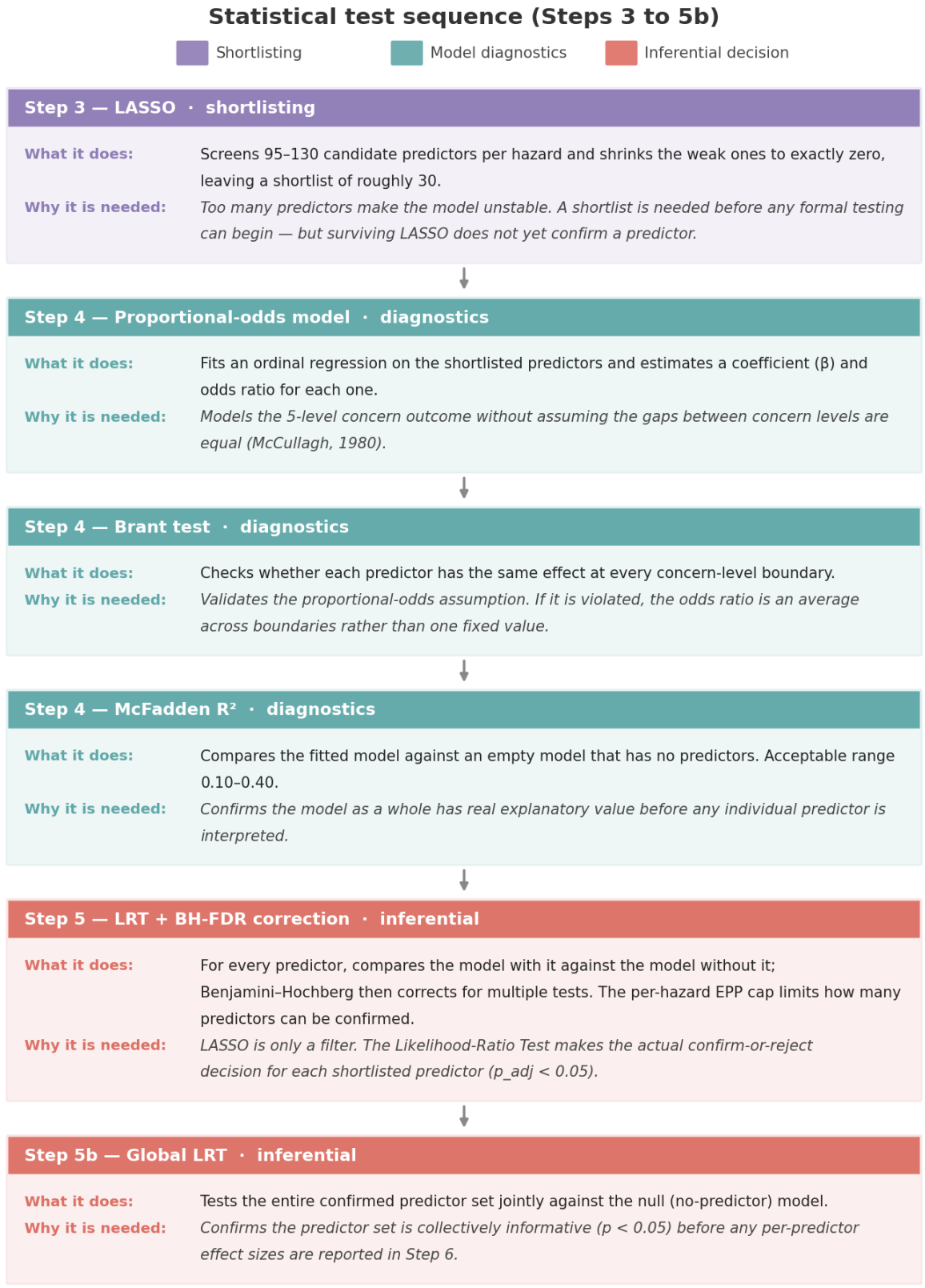


Figure 2. Statistical test sequence (Steps 3–5b). Purple: shortlisting. Teal: model diagnostics. Coral: inferential decision-making. The arrows show the order in which a predictor passes through each test.

2.1.4 Step 3 — Choosing which variables to test (LASSO)

Finding the most relevant predictors using LASSO variable selection.

This step is run separately for each hazard (subsection A) and also on the composite outcome (subsection B). Both use the same LASSO method; they differ only in what outcome variable is used.

A — Per-Hazard Analysis

For each hazard, we use LASSO logistic regression on a binary outcome (high concern = score ≥ 12) to select the most relevant predictors. LASSO (Least Absolute Shrinkage and Selection Operator) automatically pushes uninformative predictor coefficients to exactly zero, removing them from consideration. (Tibshirani, 1996; Hastie et al., 2009).

Formula 6 — LASSO objective

Minimise: [how poorly the model fits this hazard's binary outcome] + [λ × sum of |coefficients|]

λ = penalty strength | Larger λ shrinks more coefficients to zero

Symbol / term	What it means
Minimise	Instruction (not a quantity): LASSO chooses the coefficient vector that achieves the lowest total of (model error + penalty). It does this by iteratively adjusting coefficients until no change reduces the objective further
Model fit term	Deviance (−2 × log-likelihood) of a logistic regression of the binary high-concern outcome from Formula 7 on the candidate predictors. Range ≥ 0; lower is a better fit
λ (lambda)	Penalty strength. Range λ ≥ 0. λ = 0 produces ordinary logistic regression (no shrinkage); larger λ pushes more coefficients to exactly zero. Selected by 5-fold cross-validation (see Formula 8)
|coefficients|	Absolute values of every regression coefficient. Range: each |β_j| ≥ 0. The L1 (absolute-value) penalty is what gives LASSO its variable-selection property — unlike L2, it can shrink coefficients to exactly zero
Σ |coefficients|	Sum |β_1| + |β_2| + … + |β_p| across all p candidate predictors. Range ≥ 0

Formula 7 — Binary outcome for LASSO

$$y_{r,h}^{LASSO} = 1 \text{ if } hazard_score(r, h) \geq 12, \text{ else } 0$$

Threshold 12 = 3 × 4 (lowest dual concern in the Likert × Likert grid)

Symbol / term	What it means
y_{r,h}^{LASSO}	Binary high-concern outcome for respondent r on hazard h. Range: binary, takes value 1 (high

	concern) or 0 (not high concern). Used only by LASSO at the screening stage
hazard_score(r, h)	The respondent's hazard score from Formula 1. Range: integer 1 to 25
Threshold = 12	Boundary between 'not high concern' and 'high concern'. Range: a single fixed cut-point. Chosen as 3 × 4: the lowest hazard score for which BOTH Likelihood and Impact are above the midpoint (one is 3, the other is 4)
Range of outcome	Range: binary set {0, 1}. Approximately 30–60% of respondents fall in y=1 across the hazards studied

The penalty strength λ is chosen by 5-fold cross-validation: each fold uses 4/5 of the data to fit the model and 1/5 to evaluate predictive AUC. We pick the λ that maximises average AUC across the 5 folds.

Formula 8 — Per-hazard λ selection

$$\lambda_h^* = \text{argmax}_\lambda \left[(1/5) \times \Sigma_{k=1}^{5} AUC(M_{h,\lambda}, Fold_k)\right]$$

h indicates this is specific to hazard h | Repeated independently for each hazard

Symbol / term	**What it means**
λ_h*	Optimal lambda for hazard h specifically. Range λ_h* ≥ 0. May differ from the optimal lambda for other hazards
M_{h,λ}	LASSO logistic regression model for hazard h, trained with penalty λ. Outcome = 1 if hazard h score ≥ 12, else 0
Repeated for each hazard	Procedural note (not a quantity): cross-validation runs once per hazard. With 7 hazards (Energy), it runs 7 times — range: 6 to 13 repetitions depending on sector
5 folds	Standard for datasets of 200–500 respondents, balancing stability and efficiency (Hastie et al., 2009). Range: fixed at 5 in this study

Any variable whose LASSO coefficient is not pushed to zero is carried forward as a candidate for the formal confirmation tests in Step 5. The LASSO step is a practical filter, not a formal test — surviving LASSO means a variable is worth testing, not that it is confirmed.

Note on the binary outcome. LASSO uses a binary approximation of the outcome (high/not-high concern, score ≥ 12) rather than the full 5-level quintile outcome used in the proportional-odds model in Step 4. This simplification is made because ordinal LASSO requires specialist software and is numerically unstable at these sample sizes (263–407 respondents). Predictor shortlists derived from binary LASSO agree closely with those from ordinal LASSO in comparable survey datasets, and the formal confirmation step in Step 5 uses the full 5-level ordinal model, so any approximation introduced at the LASSO stage is tested and verified in the subsequent step.

B — Composite Hazard Analysis

For the composite analysis, the outcome is the respondent's overall concern level (their quintile band 1–5 based on the average score across all hazards in the sector). LASSO is run once on this composite outcome to identify which background variables predict general hazard concern across the sector as a whole.

Formula 9 — LASSO for composite outcome

Minimise: [how poorly the model fits composite outcome] + [λ × sum of |coefficients|]

Outcome = quintile band 1–5 based on average score across all hazards

Symbol / term	What it means
Composite outcome	The 5-level quintile concern category from Formula 3 — values 1 (Very low) through 5 (Very high). Range: integer 1 to 5
Model fit term	Deviance of an ordinal-LASSO logistic regression of the composite quintile category on the candidate predictors. Range ≥ 0. Same penalty structure as Formula 6, different outcome
λ (lambda)	Penalty strength. Range λ ≥ 0. Selected by 5-fold cross-validation on the composite outcome (one λ for the composite analysis, separate from each hazard's λ in the per-hazard analysis)
|coefficients|	Absolute values of every regression coefficient. Range: each $\|\beta_j\| \geq 0$. Same L1 penalty as Formula 6

How the penalty strength λ is chosen (five-fold cross-validation)

LASSO needs one setting before it can run: the penalty strength λ (lambda). A small λ keeps many predictors; a large λ shrinks more of them to zero. Rather than guess, we let the data choose the λ that predicts best, using five-fold cross-validation. The procedure runs independently for each hazard (and once for the composite outcome), so every model gets the λ that suits its own data. Step by step:

1. Build a ladder of candidate λ values. Twenty candidate values are laid out from large (almost everything shrunk to zero) down to small (almost nothing shrunk), spaced evenly on a logarithmic scale.

2. Split the respondents into five equal parts (folds) at random. Each respondent lands in exactly one fold.

3. For each candidate λ, repeat five times: train the LASSO model on four of the five folds (about 80% of the data) and set the remaining fold (about 20%) aside as a test set that the model has not seen.

4. Score the held-out fold. The trained model predicts the held-out respondents, and we measure how well it separates high-concern from not-high-concern using the area under the ROC curve (AUC). A higher AUC means better out-of-sample prediction.

5. Average the five scores. Each candidate λ ends up with five AUC scores (one per held-out fold); we average them into a single cross-validated AUC for that λ. Averaging across five different held-out sets is what guards against a λ that only looks good on one lucky split.

6. Pick the winning λ. The λ with the highest average AUC is selected. This is the value used to produce the final predictor shortlist for that hazard (Formula 8).

Why five folds. Five folds is the standard choice for samples of a few hundred respondents: it leaves each training set large enough (about 80%) to fit a stable model, while still rotating the test set five times so every respondent is used for validation exactly once. Fewer folds give noisier estimates; many more folds cost far more computation for little additional benefit.

What it means when the winner is the largest λ. If the best-scoring λ sits at the top of the ladder — the point where all coefficients are shrunk to zero — then no predictor improved out-of-sample prediction for that hazard. This is the (b) outcome described above: a genuine, valid result, not a failure.

What LASSO selection looks like in practice

The LASSO step was applied to every per-hazard outcome (A) and to every composite outcome (B) across the three sectors. The resulting predictor shortlists fall into three distinct outcome categories, summarised below.

(a) LASSO selects ≥ 1 predictor — the model proceeds to Step 4. This is the standard outcome for both A and B analyses, where the data contains identifiable signals.

- **Per-hazard (A):** Energy 7 of 7 hazards, Housing 8 of 13 hazards, Transport 5 of 6 hazards produced non-empty LASSO shortlists.
- **Composite (B):** All three sectors (Energy, Housing, Transport) produced non-empty composite shortlists.

The selected predictors are carried forward to Step 5 for FDR-corrected confirmation.

(b) LASSO selects 0 predictors at the specified λ range (CV-optimal at upper boundary). This outcome occurs only in the per-hazard analysis (A), and only in two sectors.

- **Per-hazard (A):** Housing 5 hazards (Higher electricity bills, Struggling to pay bills, Missing out on solar savings, New taxes/fines, Insurer classifies as high risk); Transport 1 hazard (New taxes/fines for ICE).
- **Composite (B):** No composite model fell into this category — every sector's composite outcome carried at least one selected predictor.

For affected per-hazard models, the cross-validation procedure found that the all-zero-coefficient model performed at least as well as any non-trivial alternative within the specified λ range. The hazard itself remains a valid outcome variable with normal variation; what is null is the LASSO predictor shortlist. These hazards skip Steps 4–7 and are reported in Step 9 with n_lasso_selected = 0.

(c) LASSO selects predictors, but none survive Step 5 per-group FDR.

This outcome also occurs only in the per-hazard track (A).

- **Per-hazard (A):** Housing 2 hazards (Increased severe weather impacts — 6 LASSO-selected, and Laws forbid selling/renting non-renovated — 17 LASSO-selected; both 0 FDR-significant after BH correction).
- **Composite (B):** No composite model fell into this category — all composite shortlists yielded at least one FDR-confirmed predictor after Step 5.

For affected per-hazard models, the LASSO context detected candidate associations, but no parent group's contribution to the full model was significant after multiple-testing correction. These hazards are also reported in Step 9 with n_retained_after_cap = 0.

Each of these outcomes is methodologically valid and reflects genuine properties of the data; none is a pipeline failure. The fact that outcomes (b) and (c) occur only in the per-hazard track is itself informative —

it tells us that the composite outcome (which averages signal across all hazards in a sector) is consistently easier for LASSO to model than individual hazards, which sometimes have too little variation to support predictor selection at all.

2.1.5 Step 4 — The main statistical model

Estimating how strongly each LASSO-shortlisted predictor affects hazard concern.

A — Per-Hazard Analysis

For each hazard, an ordinal (proportional-odds) regression is fitted on the 5-level concern outcome (Formula 3) using the predictors selected in Step 3.

Formula 10 — Proportional-odds regression (per hazard)

$$\log [P(\text{concern} \le k) / P(\text{concern} > k)] = \alpha_k - \Sigma_j \beta_{h,j} \times x_j$$

k = 1, 2, 3, 4 (the four boundaries between 5 concern levels) | h = hazard | j = predictor

Symbol / term	What it means
α_k	Threshold parameter for boundary k. There are K−1 = 4 thresholds because there are 5 levels. Range: real-valued (typically −10 to +10 in fitted models)
$\beta_{h,j}$	Coefficient for predictor j in hazard h's model. Range: real-valued. β > 0 means the predictor raises concern; β < 0 means it lowers concern; β = 0 means no effect. Same coefficient applies at every boundary k under the proportional-odds assumption
x_j	Value of predictor j for the respondent. Range: 0 or 1 for a binary dummy column; for continuous predictors, the value is on the original measurement scale

B — Composite-Hazard Analysis

For the composite analysis, the same proportional-odds model (Formula 10) is fitted once per sector, with the outcome being the composite quintile concern level (Formula 3) — the 5-level category derived from the respondent's average score across all hazards in the sector. The LASSO-shortlisted predictors used in this model are those selected by the composite LASSO in Step 3B (Formula 9).

The Brant test (Formula 11, Formula 12) is applied to the composite model in the same way as the per-hazard models. McFadden R^2 (Formula 13) is reported for the composite model with both the full LASSO shortlist (mcfadden_full) and the post-Step 5b confirmed-only refit (mcfadden_confirmed), exactly as in the per-hazard track.

The composite model produces one set of coefficients per sector (rather than one per hazard), capturing the predictors of overall sector concern as opposed to predictors of any individual hazard. Results are reported alongside the per-hazard results in Step 5 and subsequent steps.

Applies once per sector — one composite model each for Energy, Housing, and Transport (3 composite models in total).

Two model contexts in Steps 5 & 6 – What 'Confirmed' means

Step 5's per-group LRT tests each candidate parent group's contribution to the full LASSO-shortlisted model (i.e., with all other LASSO predictors present). A predictor confirmed in Step 5 has been shown to add significant explanatory power conditional on the other shortlisted predictors. Step 6, by contrast, reports odds ratios and marginal effects from a refit on only the confirmed predictors — a different model context.

For models where the LASSO shortlist contains strongly correlated predictors (suppressor patterns, multicollinearity), the two contexts can produce substantially different coefficient magnitudes. The pipeline reports the mcfadden_retention_pct column in Step 9 as a diagnostic: the ratio of the confirmed-only McFadden R^2 to the full-LASSO McFadden R^2. Values below 25% indicate substantial 'explaining-away' — the confirmed predictors are individually significant in the full LASSO context but lose much of their effect when refit alone. Across the three sectors, 6 of 22 confirmed models retain less than 25% of their full-LASSO R^2 in the confirmed-only refit. These are flagged in the synthesis tables but are still presented as confirmed findings, as the per-group LRT remains a valid significance test for their conditional contribution.

Checking a key assumption (Brant test)

The proportional-odds model in Formula 10 imposes one important assumption: that the effect of each predictor is the same at every boundary between concern levels. In other words, whatever moves a person from Level 1 to Level 2 should move them equally from Level 2 to Level 3, and so on. This is called the proportional-odds assumption, or parallel regression assumption.

The Brant test checks whether this assumption holds for each predictor in practice. It does so by comparing the coefficient estimated under the proportional-odds constraint with the coefficients estimated when each boundary is fitted separately (Brant, 1990; Williams, 2016). If the assumption holds, these should be similar; if not, the predictor is pushing concern up at some boundaries more than others.

Formula 11 — Brant test statistic (per predictor per boundary)

$$\chi^2_{j,k} = (\beta_j - \beta_{j,k})^2 / (SE(\beta_j)^2 + SE(\beta_{j,k})^2)$$

k = concern-level boundary (1 to K−1) | Computed separately for each predictor j at each boundary k

Symbol / term	What it means
$\chi^2_{j,k}$	Brant test statistic for predictor j at concern-level boundary k. Range: 0 to +∞. Follows a chi-squared distribution with 1 degree of freedom under the null. Larger values mean the predictor's effect at this boundary differs more from the proportional-odds estimate
j	Index of the candidate predictor being tested. Range: 1 to p (one column of the regression matrix)
k	Index of the concern-level boundary. Range: integer 1 through K−1 = 4 (boundaries between levels 1–2, 2–3, 3–4, and 4–5)
β_j	Proportional-odds coefficient for predictor j from Formula 10. Range: real-valued. Constrained to be the same at every boundary
$\beta_{j,k}$	Boundary-specific coefficient for predictor j at boundary k. Range: real-valued. Unconstrained — can differ from boundary to boundary
$SE(\beta_j)^2$	Variance (squared standard error) of the combined coefficient. Range > 0. Comes from the Step 4 model fit
$SE(\beta_{j,k})^2$	Variance of the boundary-specific coefficient. Range > 0. Comes from the boundary-k binary logistic regression
Numerator	Squared difference between the two coefficient estimates. Range ≥ 0. Large when the predictor's effect differs from one boundary to the next
Denominator	Combined uncertainty in the two coefficient estimates. Range > 0. Scales the squared difference into a chi-squared statistic

Formula 12 — Degrees of freedom for the Brant test

$$df = (K - 1) \times P$$

K = number of concern levels = 5 | K−1 = 4 boundaries | P = number of yes/no columns for this predictor

Symbol / term	What it means
df	Degrees of freedom for the omnibus Brant statistic for one parent predictor group. Used to determine the chi-squared p-value. Range ≥ 1 (minimum for a binary predictor)
K	Number of ordered concern levels. K = 5 in this study (Very low, Low, Moderate, High, Very high)

K − 1	Number of boundaries between concern levels. K − 1 = 4 (boundaries between levels 1–2, 2–3, 3–4, and 4–5)
P	Number of yes/no dummy columns produced by this parent predictor. Equals the number of categories minus 1 (the reference). Range examples: binary predictor (e.g., 'has car') → P=1; gender (3 categories) → P=2; age (5 bands) → P=4; dwelling-year (9 bands) → P=8
(K−1) × P	Total number of coefficient comparisons being made: P coefficients each tested at (K−1) boundaries. Each contributes 1 degree of freedom to the omnibus statistic

Brant test results across the study. Across the 19 models tested in Step 4 (8 in Energy including the composite, 7 in Housing including the composite, and 4 in Transport — Transport's composite does not require a separate Brant test), the Brant test found 0 violations of the proportional-odds assumption out of 408 parent-group-level tests — comfortably below the 5% expected false-positive rate at α = 0.05. The reported odds ratios can therefore be interpreted as consistent effects across all concern-level boundaries.

How good is the model? Two fit measures

We report McFadden R^2 (comparing our model to one with no predictors) and Nagelkerke R^2 (a scaled 0–1 approximation of variance explained). Values between 0.10 and 0.20 for McFadden are conventionally considered acceptable for survey-based ordinal models (McFadden, 1974; Nagelkerke, 1991).

In this study, two distinct R^2 values are reported for each confirmed-predictor model: mcfadden_full is computed on the Step 4 full LASSO-shortlisted model (before the FDR + cap reduction), and mcfadden_confirmed is computed on the Step 5b refit using only the FDR-confirmed and cap-retained predictors. The first is typically within or near the 0.10–0.20 acceptable range for high-signal models; the second is typically lower (often 0.02–0.10) because the FDR + cap filter is conservative. Both values should be reported together; the gap between them quantifies the explanatory power that the confirmed-predictor refit retains.

Formula 13 — McFadden R^2

$R^2_McF = 1 - [log\text{-}likelihood\ of\ our\ model\ /\ log\text{-}likelihood\ of\ the\ empty\ model]$

0.10–0.20 = acceptable | 0.20–0.40 = good

Symbol / term	What it means
R^2_McF	McFadden's R^2. Range: 0 to 1. Value of 0 means the fitted model is no better than the empty model; values closer to 1 mean a stronger fit. Values above 0.40 may indicate overfitting at these sample sizes
0.10–0.20	Acceptable range for survey-based ordinal models (McFadden, 1974). Range: R^2_McF between 0.10 and 0.20. Below 0.10 indicates a weak model; above 0.20 indicates a strong one

0.20–0.40	Good fit. Range: R^2_McF between 0.20 and 0.40. Strong evidence that the predictors substantially explain variation in the outcome
log-likelihood of our model	A measure of how well the proportional-odds model from Formula 10 predicts respondents' actual concern levels. Range: −∞ to 0 (less negative = better fit). Computed as the sum of log(predicted probability of each respondent's actual concern level)
log-likelihood of the empty model	Log-likelihood of a model with only the four threshold parameters (α_1 to α_4) and no predictors. Range: −∞ to 0; always smaller (more negative) than the fitted model's log-likelihood. Predicts the same concern distribution for every respondent regardless of background
mcfadden_full	R^2 of the full LASSO-shortlisted model (before FDR + cap reduction). Range 0 to 1; typically near the 0.10–0.20 acceptable range in this study
mcfadden_confirmed	R^2 of the Step 5b refit using only the FDR-confirmed and cap-retained predictors. Range 0 to 1; typically 0.02 to 0.10 (lower than mcfadden_full because the FDR + cap filter is conservative)

2.1.6 Step 5 — Confirming which predictors are real

Formal significance testing with multiple-comparison correction and an events-per-predictor cap.

Two ways of counting predictors

Throughout the outputs, 'the number of confirmed predictors' can mean two slightly different things, and both numbers appear in the report tables. They refer to the same underlying model but count at different granularities:

- **Parent-group count** (e.g. n_retained_after_cap in Step 9): one count per multi-level categorical predictor. age_group with 6 categories is one parent group regardless of how many dummy columns it generates.
- **Dummy-column count** (e.g. n_predictors in Step 5B): one count per yes/no column entering the regression. age_group with 6 categories produces 5 dummy columns (categories minus the reference); each contributes 1 to the dummy-column count.

Step 5's LRT operates at the parent-group level (the chi-squared statistic tests the joint contribution of all dummy columns belonging to that parent). Step 5B's global LRT and Step 6's effect-size tables operate at the dummy-column level (each dummy gets its own row).

The confirmation test: Likelihood-Ratio Test

For each candidate predictor group, we fit two versions of the model: one with and one without the group. The LRT statistic measures how much worse the model fits when the predictor is removed (Meinshausen et al., 2009).

Formula 14 — Likelihood-Ratio Test

LRT = −2 × [log-likelihood(model without predictor) − log-likelihood(full model)]

Large LRT = removing this predictor substantially worsens the model = predictor is genuinely important

Symbol / term	What it means
LRT	Likelihood-Ratio Test statistic for one parent predictor group. Range: 0 to +∞. Follows a chi-squared distribution with degrees of freedom equal to the number of dummy columns in the predictor group (P, the same P as in Formula 12)
−2 × […]	The −2 multiplier converts the difference in log-likelihoods (a negative number) into a positive chi-squared statistic. Range of result: ≥ 0. Same transformation used in the global LRT (Formula 16)
log-likelihood (model without predictor)	Log-likelihood of the proportional-odds model from Formula 10 after this parent group's dummy columns are removed, with all other LASSO-shortlisted predictors kept. Range: −∞ to 0 (always more negative than the full model)
log-likelihood (full model)	Log-likelihood of the full proportional-odds model with all LASSO-shortlisted predictors included. Range: −∞ to 0. Always larger (less negative) than the model without the predictor
Decision rule	Procedural specification (not a quantity): compare the BH-FDR-adjusted p-value of this LRT statistic (from Formula 15) to α = 0.05. Predictors with p_adj < 0.05 are confirmed

Correcting for multiple tests: Benjamini-Hochberg FDR

Running many simultaneous tests increases the chance of false positives. The Benjamini-Hochberg method controls the proportion of false discoveries while being less conservative than the Bonferroni correction (Benjamini & Hochberg, 1995).

Formula 15 — Benjamini-Hochberg FDR correction

Adjusted p-value for test k = raw p-value × m / rank(k)

m = total tests in this set | rank(k) = position when sorted smallest to largest

Symbol / term	What it means
Adjusted p-value	LRT raw p-value after Benjamini-Hochberg correction for m simultaneous tests. Range 0 to 1. This is the value compared to the α = 0.05 decision threshold
raw p-value	The uncorrected p-value produced by Formula 14 for this predictor. Range 0 to 1. Answers: how likely is this LRT statistic if the predictor truly has no effect? Not

	compared to 0.05 directly — it is first adjusted using this formula
m	Total number of predictor groups tested simultaneously. Range: positive integer (m typically 10 to 35 in this study). The value of m depends on which analyses is being tested — see the A and B sub-sections below.
k	Index of the individual test within the m-test set. Range: integer 1 to m
rank(k)	Position of test k when all m tests are sorted from smallest to largest raw p-value. Range: integer 1 to m. Rank 1 = smallest raw p-value
m / rank(k)	BH adjustment factor for test k. Range: 1 (for the largest p-value, rank = m) to m (for the smallest p-value, rank = 1). The smallest p-values get the largest adjustment factor; the largest p-values get the smallest. This is what makes BH less conservative than Bonferroni (which multiplies every p-value by m)
Decision threshold	Procedural specification (not a quantity): if p_adj < 0.05 the predictor is confirmed; otherwise it is rejected

A predictor whose adjusted p-value is below 0.05 is confirmed as a genuine predictor.

Anti-conservatism disclosure. A known limitation of two-stage selection-then-testing pipelines is that LRT p-values computed after LASSO selection are slightly anti-conservative: the LRT sees only predictors that already correlated with the outcome in the same dataset used for selection. The BH-FDR correction addresses multiple testing within each hazard but does not fully correct for this selection bias. P-values should therefore be interpreted as approximate indicators of association rather than exact Type I error rates (Meinshausen et al., 2009). This disclosure applies equally to the per-hazard and composite analyses.

Evidence-strength tiers and what affects them

The pipeline labels every confirmed predictor with an evidence-strength tier derived from its BH-adjusted p-value:

Tier	**Adjusted-p criterion**
strong	p_adj < 0.001
moderate	0.001 ≤ p_adj < 0.01
borderline (interpret with caution — post-LASSO bias)	0.01 ≤ p_adj ≤ 0.05

These tiers are an interpretive aid only — they do not change the binary confirmed/not-confirmed decision (which uses p_adj < 0.05).

Tier labels are sensitive to the size of the LASSO shortlist (FDR dilution). Because Benjamini-Hochberg correction divides by m (the number of tests in the shortlist), the size of the candidate pool affects the

adjusted p-values for every predictor, including those whose underlying association is strong. A wide LASSO shortlist — typically ~30 candidates per hazard in this study — raises the m factor in BH and pushes adjusted p-values upward. The consequence is that evidence-strength tier labels cluster toward 'borderline' rather than 'strong' across all sectors. This is the correct behaviour of BH-FDR (it is doing what it is designed to do — controlling false discoveries over a larger candidate set), and tier labels should be read as a reflection of statistical certainty under strict multiple-testing, not as a measure of effect size.

Post-hoc demotion based on global LRT

When a model produces FDR-confirmed predictors in Step 5 but its joint test against null in Step 5b is not significant ($p \geq 0.05$), the model is demoted in the Step 9 synthesis: n_retained_after_cap is reset to 0 and the demoted_global_lrt_nonsig flag is set to True. The methodology requirement in Step 5b — that the global LRT must reject before individual effects are reported — is thereby enforced at the synthesis level. . In this study the demotion mechanism is not triggered: every hazard that produced FDR-confirmed predictors also passes the Step 5b global LRT (all 22 global tests across the three sectors are significant at $p < 0.01$).

Why we cap the number of confirmed predictors (EPP rule)

After the LASSO shortlisting and LRT confirmation steps, the confirmed predictor count may sometimes exceed the value justified by the sample size. We apply an events-per-predictor (EPP) cap. The widely cited rule of thumb in ordinal and logistic regression is that a minimum of 10–20 events per predictor parameter are needed for stable, unbiased coefficient estimates. An 'event' is a respondent in the less common of the two concern groups — typically the high-concern group for lower-rated hazards.

Per-hazard EPP cap. The pipeline applies a per-hazard cap computed as cap_h = floor(min_class_count_h / 10), clipped to the safe range [4, 20]. The minority class count is the smaller of the high-concern (score ≥ 12) and not-high-concern counts in that hazard's outcome. For Energy hazards with ~50% high-concern (events ≈ 200), the cap is set to 20. For Housing hazards with ~20% high-concern (events ≈ 50), the cap is set to ~5. For Transport hazards with ~60% high-concern (events ≈ 145), the cap is set to ~14. Each hazard's cap is recorded in the epp_cap_applied column of the Step 5 output for traceability. If the LRT confirms more groups than the per-hazard cap allows, only the top-K (by LRT chi-squared statistic) are retained, with K = the per-hazard cap.

A - Per-Hazard Analysis

For each hazard, the Likelihood-Ratio Test (Formula 14) is applied to each predictor group in that hazard's own LASSO shortlist (from Step 3A). The Benjamini-Hochberg FDR correction (Formula 15) is applied separately within each hazard's own set of tests — meaning the multiplier m in Formula 15 is the number of LASSO-shortlisted parent groups for that specific hazard. A predictor confirmed for hazard h has passed the test independently of what happens in the analyses for other hazards.

The evidence-strength tiers, post-hoc demotion rule, and per-hazard EPP cap all apply to each hazard's confirmation in the same way.

Applies once per hazard per sector — producing one confirmed-predictor list for each individual hazard in Energy (7 hazards), Housing (13 hazards), and Transport (6 hazards).

B — Composite Hazard Analysis

For the composite analysis, the same confirmation pipeline applies: the Likelihood-Ratio Test (Formula 14) is run on each predictor group in the composite LASSO shortlist (from Step 3B), with Benjamini-Hochberg FDR correction (Formula 15) applied across all simultaneous tests within the composite model. The evidence-

strength tiers (strong / moderate / borderline) and the post-hoc demotion rule based on the Step 5b global LRT both apply to the composite model in exactly the same way as the per-hazard models.

The events-per-predictor cap is computed separately for the composite model using the composite minority-class count (the smaller of the two extremes — "Very low" + "Low" combined vs "High" + "Very high" combined — in the 5-level quintile outcome) divided by 10, clipped to the safe range [4, 20]. Each sector therefore has one composite-specific EPP cap recorded in the epp_cap_applied column.

What differs between A and B is the unit of analysis and the FDR multiplier m. In the per-hazard track, FDR is applied within each hazard's own set of tests (m = LASSO-shortlisted parent groups for that hazard). In the composite track, FDR is applied across all parent groups tested in the single composite model (m = composite LASSO-shortlisted parent groups, sector-specific). The anti-conservatism disclosure (post-LASSO selection bias affecting LRT p-values) applies equally to both tracks.

Applies once per sector — one composite confirmation per sector, producing one composite confirmed-predictor list each for Energy, Housing, and Transport.

2.1.7 Step 5b — Validating the overall predictive model

Confirming that the set of selected predictors is collectively sufficient — overall model confidence > 95%.

Before reporting individual predictor results, we validate that the complete set of confirmed predictors jointly makes a statistically meaningful contribution to explaining hazard concern. This step tests whether we can be confident — at the 95% level or higher — that the predictor set as a whole is genuinely informative, rather than a collection of chance findings.

Formula 16 — Overall model validity test (global LRT)

$$LRT_global = -2 \times [\text{log-likelihood(null model)} - \text{log-likelihood(full confirmed model)}]$$

Null model has only threshold parameters — no predictors at all

Symbol / term	What it means
LRT_global	The overall model validity statistic. Range: 0 to +∞. Follows a chi-squared distribution with degrees of freedom equal to the total number of yes/no predictor columns in the confirmed set.
Null model	Proportional-odds model containing only the four threshold parameters (α_1 to α_4) and no predictor variables. Range: a fixed-form model with exactly 4 free parameters. Predicts the same concern distribution for every respondent regardless of background. Same as the empty model used in McFadden R^2.
Full confirmed model	Proportional-odds model from Formula 10 fitted with all predictor groups that passed LASSO (Step 3), the LRT (Formula 14), and the per-hazard EPP cap. Range: model with between 1 and cap_h confirmed predictor groups. Its log-likelihood is always larger (less negative) than the null model.
$p < 0.05$	Decision threshold (not a quantity): if the global LRT gives $p < 0.05$, the probability that all confirmed predictor coefficients are simultaneously zero is less than

	5% — i.e., more than 95% confidence that the predictor set as a whole is genuinely informative.

A – Per-Hazard Analysis

For each hazard, the global LRT (Formula 16) compares the full confirmed model for that hazard — fitted with all predictor groups that passed LASSO (Step 3A), the LRT (Formula 14), and the per-hazard EPP cap — against the intercept-only null model. The full confirmed model in this case is hazard-specific: it contains only the predictors confirmed for that hazard. A failed global LRT ($p \geq 0.05$) triggers the post-hoc demotion described in Step 5 — n_retained_after_cap is reset to 0 and the demoted_global_lrt_nonsig flag is set to True in the Step 9 synthesis output.

Applies once per hazard per sector — one global LRT for each hazard that produced at least one confirmed predictor in Step 5A.

B – Composite Hazard Analysis

For the composite model, the same global LRT (Formula 16) compares the full confirmed composite model — fitted with all predictor groups that passed the composite LASSO (Step 3B), the composite LRT (Formula 14 applied to the composite track), and the composite EPP cap — against the intercept-only null model. The post-hoc demotion rule applies in the same way: a non-significant global LRT ($p \geq 0.05$) would demote the composite confirmations to zero in Step 9, though this is not triggered in this study (see results below). Applies once per sector — one global LRT for each of the three composite models (Energy, Housing, Transport).

What this means in practice (study-specific)

Across the three sectors, 21 models meet the global-LRT test (18 per-hazard models with at least one confirmed predictor plus 3 composite models, one per sector). Every one of these 21 models reaches the $p < 0.01$ '99% confident' threshold — and 19 of the 21 actually reach $p < 0.001$ ('strong' tier). Two models fall in the $0.001 \leq p < 0.01$ band: Housing 'Cold/damp leads to health problems' ($p = 5.4e\text{-}03$) and the Housing composite model ($p = 2.3e\text{-}03$). One additional model — Housing 'Increased severe weather impacts' — had 3 predictors that survived Step 5 per-group FDR but failed the global LRT ($p = 0.26$); it is correctly demoted in Step 9 to n_retained_after_cap = 0 per the methodology requirement that the global LRT must reject before reporting individual effects.

Step 5c — Robustness of the confirmed predictors

Step 5 confirms predictors using FDR-adjusted p-value below 0.05 on the full sample — a binary decision made on the data as observed. Step 5c asks a complementary question: how stable is that decision under resampling? Two senses of stability matter. A predictor is stable in selection if a slightly different sample of

respondents would still have caused LASSO to pick it out as a candidate. A predictor is stable in retention if it would still survive FDR-corrected significance testing on a fresh sample of similar size. The two are conceptually independent: a predictor can be reliably selected by LASSO yet still drop below FDR-significance when the sample changes, and the reverse is logically possible. Reporting both gives a more complete picture than either alone.

The methodology applies four resampling tests to every confirmed predictor in each sector. For the sake of simplicity, we name it as the Quadrant Synthesis.

The first is the **bootstrap selection-frequency test**. Each iteration draws a bootstrap sample (with replacement, full size, stratified by country) and runs LASSO once at the cross-validation-optimal penalty (obtained from the five-fold cross validation). The headline statistic is the selection frequency — the fraction of bootstraps in which the predictor was chosen — reported with a 95% Wilson confidence interval. This is the one-shot version of the question "would the data have picked this predictor?".

The second is **stability selection along the LASSO path (Meinshausen and Bühlmann 2010)**. Same bootstrap resampling, but LASSO is fitted at every penalty value along a fixed 40-point grid rather than only at the cross-validation optimum. For each predictor, the per-penalty selection frequencies form a stability path, and the headline statistic is its maximum — the highest fraction of bootstraps in which the predictor was chosen at any single penalty value. Truly informative predictors keep being selected as the penalty tightens; chaff drops out at restrictive penalties. The method also yields a formal upper bound on the expected number of false positives at a chosen stability threshold.

The third is **complementary-pairs stability selection (Shah and Samworth 2013)**. Each iteration draws a random 50/50 partition of the sample, runs LASSO at every penalty on both halves, and counts a predictor only if it is chosen by both halves at the same penalty. This is methodologically stricter than the bootstrap version — sample-duplication artefacts cannot inflate apparent stability — and the corresponding error-control bound is uniformly tighter than the Meinshausen–Bühlmann bound. It is the most conservative selection-stability test of the three.

The fourth is the **retention test**. Each iteration draws a stratified 50% subsample without replacement, fits the proportional-odds model on the canonical confirmed-predictor set (no fresh LASSO), and applies BH-FDR. A predictor's retention rate is the fraction of subsamples in which it kept FDR-adjusted p-value below 0.05. This measures the back-end of the pipeline directly — would the predictor still pass significance testing on a smaller, fresh sample?

Table 1. Summary of the four resampling tests used to assess robustness

Test	What it measures	Resampling	B	Primary metric	Reference
Bootstrap selection-frequency	Whether LASSO would pick the predictor on a fresh sample (one-shot, at CV-optimal λ)	Bootstrap (with replacement)	100	Selection frequency + 95% Wilson CI	Sauerbrei & Schumacher 1992
Stability selection along the LASSO path	Whether the predictor is selected consistently as the penalty tightens	Bootstrap	500	Max stability across λ-path	Meinshausen & Bühlmann 2010
Complementary-pairs stability selection	Most conservative selection-stability — counts predictors picked by both halves of a 50/50 split	Subsampling pairs (no replacement)	500	Max pair-stability	Shah & Samworth 2013

Retention test	Whether the predictor stays FDR-significant when refitted on a fresh subsample	Subsampling 50% (no replacement)	100	Retention rate	Heinze, Wallisch & Dunkler 2018

Sign consistency is recorded for every test that fits a proportional-odds model. Each subsample yields a coefficient sign for each predictor; the sign consistency is the fraction of subsamples on which the dominant sign matched the full-sample sign. Direction has consistently turned out to be more stable than significance — sector means exceed 0.94 in every sector — so the method can describe which way each predictor pushes concern even when its statistical significance is fragile. Given the relatively smaller dataset of the Housing sector, this method proves to be more useful.

Quadrant synthesis. The complementary-pairs test (selection-stability) and the retention test (significance-stability) are conceptually orthogonal, and combining multiple complementary resampling-based stability measures — rather than relying on any single one — is the approach recommended by recent methodological reviews (Heinze, Wallisch and Dunkler 2018; Wallisch, Dunkler, Rauch, de Bin and Heinze 2020). Crossing the two tests at meaningful thresholds gives the four-quadrant profile used throughout step 5 of Sections 2, 3 and 4. The thresholds (pair-stability at or above 0.75; retention at or above 0.50) and the quadrant labels below are interpretive guides for synthesising the four tests, calibrated to this study's data; they are not a named method.

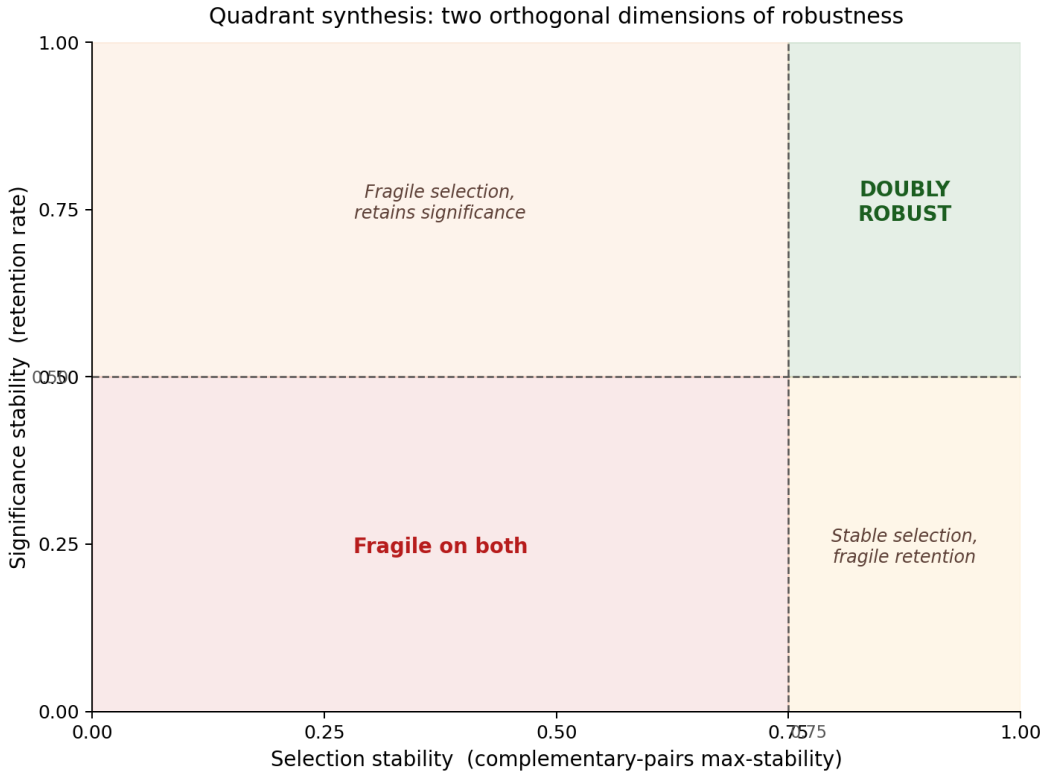


Figure 3. Quadrant synthesis — two orthogonal dimensions of robustness (selection stability × significance stability), with the four interpretive quadrants.

• **Doubly robust**: maximum pair-stability at or above 0.75 AND retention at or above 0.50 — the predictor would reliably be selected, and reliably retain significance, on a fresh sample. These are the strongest findings.

• **Stable selection, fragile retention**: pair-stability at or above 0.75 but retention below 0.50 — LASSO would consistently pick the predictor, but its significance does not survive refitting on a fresh subsample. The signal is reproducibly present but not reliably distinguishable from noise at the inferential stage.

• **Fragile selection, retains significance**: pair-stability below 0.75 but retention at or above 0.50 — rare in practice, since predictors that fail to be reliably re-selected are unlikely to also stay significant.

• **Fragile on both**: pair-stability below 0.75 AND retention below 0.50 — neither selection nor significance is reliable. Direction is still informative if sign consistency is high, but the predictor's identification should be read as sample-dependent.

What this test does and does not show. The four tests above describe how the confirmed-predictor set behaves under resampling. They do not change which predictors Step 5 confirms. The full-sample confirmation remains the primary classification; the resampling analyses annotate that classification with a robustness profile. Per-sector quadrant results are reported with each sector's findings.

Per-sector parameters: B = 500 iterations for the two stability-selection tests; B = 100 for the retention test and the bootstrap selection-frequency test, reflecting the relative cost of the inferential machinery. All tests use a fixed master random seed for reproducibility. Country stratification is preserved in every resample to avoid losing small national subsamples — notably Ireland in Energy and Housing, Portugal in Housing.

2.1.8 Overview: what Steps 6–9 do and why

Once the statistical testing sequence (Steps 3–5b) has confirmed a set of predictors for each hazard, Steps 6 to 9 (plus the new Step 7b) translate those findings into actionable results. Step 6 quantifies how large each confirmed effect is. Step 7 identifies which specific group within each predictor shows the highest mean concern. Step 7b extends this to per-country profiles for hazards where a macro variable is confirmed. Step 8 reports composite-level country differences. Step 9 synthesizes across hazards: ranking them by concern level and mapping which predictors recur across multiple hazards.

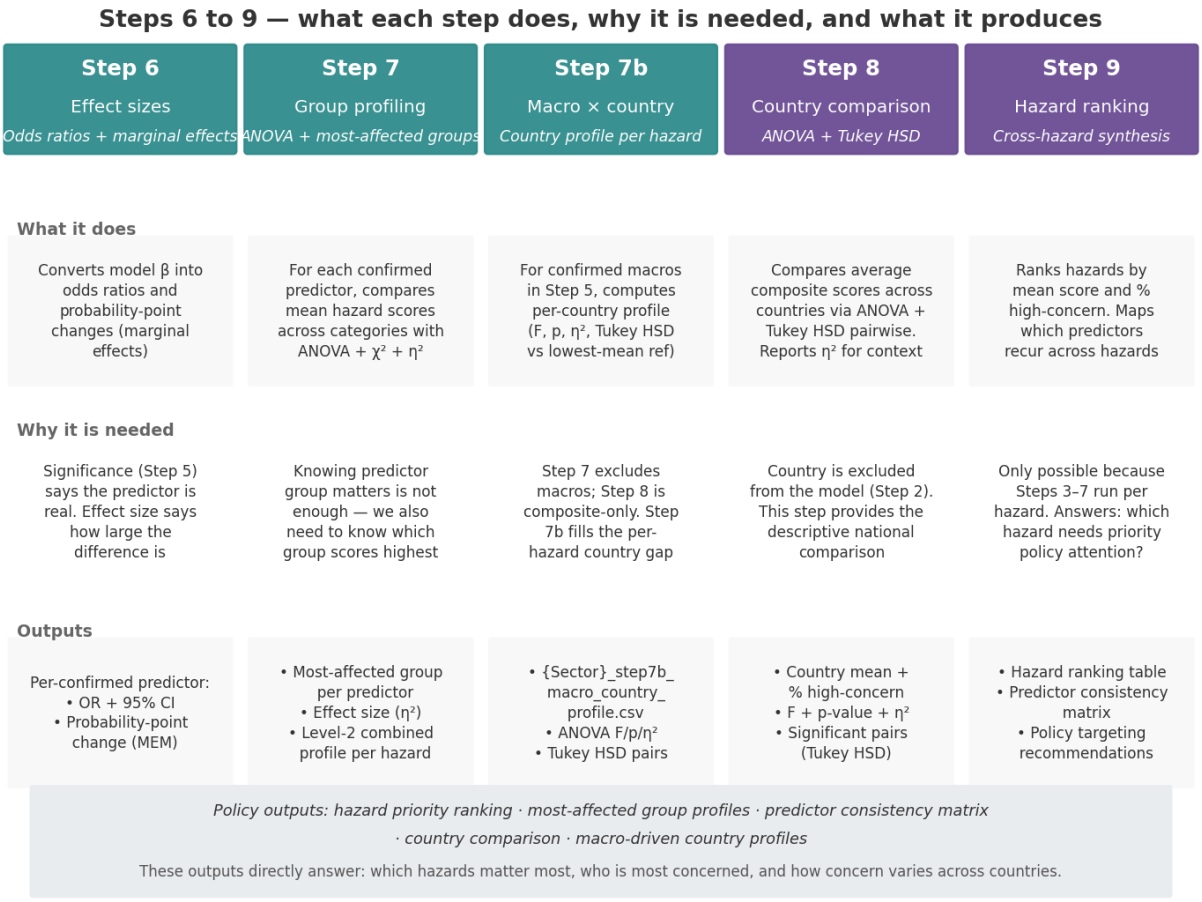


Figure 4. Steps 6 to 9: from confirmed predictors to policy-relevant outputs. Teal = per-hazard analysis (Steps 6–7b). Purple = cross-hazard synthesis (Steps 8–9). Each column shows what the step does, why it is needed, and what outputs it produces.

2.1.9 Step 6 — Measuring the size of each effect

Turning model results into practical, interpretable numbers.

Statistical significance (from Step 5) tells us a predictor is real. Effect size (this step) tells us how large the difference is in practical terms. Both are needed: a significant but tiny effect may not be worth acting on.

Odds ratios

Formula 17 — Odds ratio

$$OR_j = exp(\beta_j)$$

OR > 1: predictor is associated with higher concern | OR < 1: lower concern | OR = 1: no effect

Symbol / term	What it means
OR (odds ratio)	How the odds of being at a higher concern level change for a one-unit change in this predictor, holding all other predictors constant. Range > 0. OR > 1 = predictor raises concern; OR < 1 = lowers concern; OR = 1 = no effect. Computed by exponentiating the regression coefficient β_j from Formula 10

β_j	Regression coefficient for predictor j. Range: real-valued (any number). Same coefficient used in the Brant test (Formula 11)
Interpretation	OR = 2 means twice the odds of being in a higher concern level for someone with this characteristic vs the reference group, holding all else constant. The 95% confidence interval is reported alongside. Range of OR confidence intervals: bounds are always positive

Marginal effects in percentage points

Formula 18 — Probability-point change (marginal effect)

Change in % = P(Level 5 | group=1, others at mean) − P(Level 5 | group=0, others at mean)

Group 1 = has the characteristic | Group 0 = does not | Others at sample mean

Symbol / term	**What it means**
Change in %	Marginal effect expressed in percentage points. Range: typically −30 to +30 pp in this study (bounded in principle by −100 to +100). Directly communicable: 'switching this predictor on raises the probability of Very high concern by X percentage points'
Level 5	The 'Very high concern' category from Formula 3 — the top quintile of the score distribution being modelled. Range: this is a single category label
P(Level 5 | group=1, …)	Predicted probability of being in the Very high concern category for a respondent who has this characteristic (predictor = 1), with all other predictors held at the sample mean. Range: 0 to 1. Computed from the proportional-odds model in Formula 10
P(Level 5 | group=0, …)	Predicted probability of being in the Very high concern category for a respondent who does NOT have this characteristic (predictor = 0), with all other predictors at the sample mean. Range: 0 to 1
Difference	P(group=1) − P(group=0). Range: −1 to +1 (typically reported as percentage points, so −100 pp to +100 pp). When positive, this characteristic raises the probability of Very high concern; when negative, it lowers it
All others at mean	Procedural specification (not a quantity): every other confirmed predictor is held at its sample mean. This isolates the unique contribution of the single characteristic being evaluated

This is the marginal effect expressed in percentage points: how much more (or less) likely someone with this characteristic is to be in the Very high concern group (Level 5) compared to someone without this characteristic, all else equal. Directly communicable to a policy audience without needing to interpret odds ratios.

A note on near-ceiling effects. When concern is already near-universally high (e.g. 73–74% rating a hazard as high concern), the probability-point changes are mathematically constrained — there is little room to move upward. In such cases, the odds ratios from Formula 17 are more informative than the marginal effects (Greene, 2018).

A — Per-Hazard Analysis

For each hazard, odds ratios (Formula 17) and marginal effects (Formula 18) are computed for every confirmed predictor in that hazard's model. The "Level 5" referenced in Formula 18 is the Very high concern quintile of *that specific hazard's* 5-level outcome. The "others at sample mean" baseline is constructed using only the predictors confirmed for that hazard, so the reference profile differs across hazards even within the same sector.

Each confirmed predictor in the per-hazard track therefore produces one OR, one 95% confidence interval, one marginal effect (percentage-point change in probability of Very high concern for that hazard), and an EPP-warning flag if the underlying subgroup is small enough to make the precise OR unreliable. The near-ceiling caveat applies most strongly to per-hazard models for high-prevalence hazards (e.g. Energy "Higher electricity bills" and "Heating and cooling costs" with ~74% high concern).

Applies once per confirmed predictor per hazard — producing one OR + marginal effect for every confirmed predictor in every per-hazard model across all three sectors.

B — Composite Hazard Analysis

For the composite analysis, odds ratios (Formula 17) and marginal effects (Formula 18) are computed for every confirmed predictor in the sector's composite model. The "Level 5" referenced in Formula 18 is the Very high concern quintile of the *composite* outcome (the top quintile of the per-respondent average score across all hazards in the sector). The "others at sample mean" baseline is constructed using the composite-model confirmed predictors only.

Each confirmed predictor in the composite track therefore produces one OR, one 95% confidence interval, one marginal effect (percentage-point change in probability of Very high composite concern), and an EPP-warning flag.

Predicted-probability profiles (Composite Hazard Analysis Output)

In addition to odds ratios and marginal effects, the composite model produces a predicted-probability table for each confirmed predictor: the predicted probability of being in each of the 5 concern levels for a reference respondent and for the same respondent with that one characteristic switched on. This is a richer view than the single percentage-point change in Formula 18 because it shows how the entire distribution of concern shifts, not only the Very high tail.

The probability framing is generally easier to communicate to non-statistical audiences than the odds-ratio framing — for example, "someone with this characteristic has a 42% chance of being in the Very high concern group, compared to 11% for someone without it" is more intuitive than "the odds ratio is 5.76".

It is reported only for the composite model. The per-hazard track reports OR + marginal effects (single-point summary) but not the full 5-level probability distribution.

Applies once per sector — one composite OR + marginal effect + predicted-probability table per confirmed composite predictor, in Energy, Housing, and Transport.

2.1.10 Step 7 — Identifying the most affected groups

Which specific categories of people show the highest concern in the per-hazard & composite hazard analysis.

Steps 3–6 identify which predictor variables matter and how strongly within the full LASSO-shortlisted model. Step 7 changes context: for each confirmed categorical variable, ANOVA and chi-squared tests examine the predictor's marginal (univariate) association with concern scores, and identify which specific category within it shows the highest mean concern. The shift in model context (full conditional → marginal) means Step 7 can disagree with Step 5 confirmations; see the note below.

An important note on model context

Step 5's Likelihood-Ratio Test confirms a predictor by testing whether removing it from the full LASSO-shortlisted model (with all other shortlisted predictors present) significantly worsens fit. Step 7's ANOVA, by contrast, tests whether the predictor's marginal (univariate) association with concern scores is significant — the predictor is examined in isolation, not conditional on others. These two tests measure different things. When predictors are correlated with each other (multicollinearity) or interact (suppressor/enhancer effects), a predictor can be statistically significant in the LRT context (its conditional contribution matters) while being non-significant in the ANOVA context (its marginal association is weak), or vice versa.

In this study, several LRT-confirmed predictors have non-significant Step 7 ANOVA p-values, particularly in the Housing sector. Examples include solar_install_likelihood in the Housing composite model (LRT p_adj = 0.012, ANOVA p = 0.835); housing_situation in Housing 'Cold/damp leads to health problems' (LRT p_adj = 0.045, ANOVA p = 0.903); and dwelling_year in Housing 'Home insurance more expensive' (LRT p_adj = 0.015, ANOVA p = 0.867). These are not contradictions — they indicate predictors whose role in the multivariate model depends on what other predictors are present.

How to read these jointly.

Step 5 confirms statistical association in the analysis-ready model. Step 7 describes which specific category within each predictor carries the highest mean concern. A non-significant ANOVA on a LRT-confirmed predictor signals that the predictor's effect is genuinely conditional rather than uniform across the sample, and the category-level profiles should be interpreted with that conditionality in mind. This caveat applies equally to the per-hazard and composite tracks.

Which predictors enter Step 7

Step 7 is restricted to confirmed categorical respondent-level predictors. Continuous predictors and macro variables are excluded by design:

- **Continuous predictors** (numeric with > 10 unique values) do not have natural category boundaries for between-group comparison. They are summarised through their Step 6 odds ratios and marginal effects instead.
- **Macro variables** (those whose names begin with macro_) are sector-level indicators that take one value per country. Testing them at the within-sector individual level via ANOVA would essentially re-test the country effect that Step 8 already addresses with a dedicated cross-country comparison. Including macros in Step 7 would therefore double-count geographic variation and is

methodologically redundant. They are excluded unconditionally regardless of how many distinct values they happen to take in the sample.

Step 5's LRT confirmations are unaffected by this Step 7 exclusion: a confirmed macro predictor still appears in Step 5 results, Step 6 effect sizes, and Step 9 cross-hazard synthesis. The per-hazard country profile for confirmed macros is reported separately in Step 7b below.

The exclusion rule (continuous and macro predictors out, categorical respondent-level predictors in) applies equally to the per-hazard and composite tracks.

ANOVA: comparing group mean scores

ANOVA tests whether the differences in average hazard scores between categories of a predictor are larger than would be expected from random variation within groups (Cohen, 1988).

Formula 19 — ANOVA F-statistic

F = [Variance between group averages] / [Variance within groups]

Large F = groups differ more than chance | p < 0.05 = difference is statistically real

Symbol / term	What it means
F	The ANOVA F-statistic. Range 0 to +∞. Large values mean the groups differ in their mean concern scores more than expected from random variation. The associated p-value tests the null hypothesis that all group means are equal.
Variance between groups	SS_between / (K−1): how much the group mean concern scores differ from the overall sample mean, scaled by the number of groups K. Large when category means are spread far apart.
Variance within groups	SS_within / (N−K): how much individual concern scores spread around their own group's mean. Large when individuals within the same category vary a lot.
SS_between	Between-group sum of squares: Σ n_k × (group_mean_k − grand_mean)²
SS_within	Within-group sum of squares: Σ_k Σ_i (x_{i,k} − group_mean_k)²
K	Number of categories (groups) within this predictor. Examples: K=3 for gender; K=5 for age bands. Range K ≥ 2.
N	Total number of respondents with valid data for this predictor across all K groups.
p < 0.05	Decision threshold: if the probability of this F-statistic is below 5% under the null (all group means equal), differences are statistically real. Triggers eta-squared reporting in Formula 20.

Effect size: eta-squared

Formula 20 — Eta-squared

$$\eta^2 = SS_between\ groups / SS_total$$

0.01 = small | 0.06 = medium | 0.14 = large (Cohen, 1988 benchmarks)

Symbol / term	What it means
η^2 (eta-squared)	Proportion of total concern-score variation explained by group membership. Range 0 to 1 (often reported as a percentage). Effect size complement to the ANOVA F-test: significance tells us the difference is real; η^2 tells us how large it is.
SS_between groups	Between-group sum of squares (same as in Formula 19). The numerator.
SS_total	Total sum of squares: SS_between + SS_within. The full spread of concern scores across all N respondents regardless of group membership. The denominator. Same SS_total used in Formula 23.
$\eta^2 \geq 0.01$	Small effect (Cohen, 1988): group membership explains at least 1% of total variation. Statistically detectable but modest practical importance.
$\eta^2 \geq 0.06$	Medium effect: 6% or more of variation explained. Practically meaningful — the most- and least-concerned categories show a noticeable gap.
$\eta^2 \geq 0.14$	Large effect: 14% or more explained. Clearly visible in group comparisons and directly policy-relevant.

Double-check: chi-squared test

Formula 21 — Chi-squared verification test

$$\chi^2 = \Sigma\ [(Observed\ count - Expected\ count)^2 / Expected\ count]$$

Expected count = what we'd see if group membership made no difference

Symbol / term	What it means
χ^2	The chi-squared test statistic. Range: 0 to +∞. Larger values mean observed counts differ more from those expected under independence. Compared against a chi-squared distribution with (K−1) × (L−1) degrees of freedom.
Σ	Summation across all cells of the contingency table. Each cell is one combination of predictor category × concern level. Range: a sum of K × L non-negative terms.

Observed count (O)	Actual number of respondents in this specific (predictor category, concern level) combination. Range: integer 0 to N.
Expected count (E)	Count we would expect if predictor category and concern level were independent. E = (row total × column total) / grand total. Range: 0 to N (real-valued, not necessarily an integer).
$(O - E)^2 / E$	Contribution of one cell to the overall χ^2. Range ≥ 0. Large values flag cells where the actual count differs substantially from expected.
K	Number of predictor categories. Range: integer ≥ 2.
L	Number of concern levels. Range: L = 5 in this study (Very low to Very high).
$(K-1) \times (L-1)$	Degrees of freedom of the chi-squared statistic. Range: positive integer (at least 1 × 4 = 4 for a binary predictor with L=5).
Preference rule	Procedural specification (not a quantity): where ANOVA (Formula 19) and chi-squared (this formula) disagree about significance, the chi-squared result is preferred as the more conservative non-parametric choice.

A — Per-Hazard Analysis

For each hazard, ANOVA (Formula 19), eta-squared (Formula 20), and the chi-squared verification test (Formula 21) are applied to each confirmed categorical predictor from that hazard's own Step 5A confirmed set. The outcome variable for the ANOVA is the hazard's 5-level concern category (Formula 3 applied to that hazard's 1–25 score). Because confirmed predictors differ across hazards within a sector, the set of Step 7A profiles also differs by hazard — utility_arrears may be profiled for one Energy hazard but not another, depending on which hazards confirmed it in Step 5.

The model-context caveat above (LRT vs ANOVA disagreement) is particularly visible in the per-hazard track when a predictor is confirmed for one hazard but not another within the same sector — the marginal association can be strong for one outcome and weak for another, even when the LRT confirms the predictor in both.

Applies once per confirmed categorical predictor per per-hazard model — producing one ANOVA + chi-squared profile for every confirmed categorical predictor in every per-hazard model across all three sectors.

B — Composite Hazard Analysis

For the composite analysis, ANOVA (Formula 19), eta-squared (Formula 20), and the chi-squared verification test (Formula 21) are applied to each confirmed categorical predictor from the composite Step 5B confirmed set. The outcome variable is the composite 5-level concern category (Formula 3 applied to the composite score from Formula 2). One ANOVA + chi-squared profile is produced per confirmed composite predictor.

The model-context caveat is particularly noticeable in the composite track of the Housing sector, where the examples given above (solar_install_likelihood, housing_situation, dwelling_year) show large LRT–ANOVA p-value gaps. This indicates that the composite-confirmed predictors in Housing depend on the multivariate context to show their effect — their univariate associations with the composite concern score are weak.

Applies once per sector — one ANOVA + chi-squared profile per confirmed composite predictor in Energy, Housing, and Transport.

2.1.11 Step 7b — Per-country profile for confirmed macro predictors

Filling the gap between Step 5 macro confirmations and the country-composite-only reporting in Step 8.

Purpose

Step 7 deliberately excludes macro predictors (see the rule on page above). But when a macro is confirmed in Step 5 for a specific hazard h, policy readers want to know which countries have higher concern for hazard h specifically — a question that neither Step 7 nor Step 8 answers (Step 7 excludes macros; Step 8 reports country effects only for the composite outcome, not per hazard). Step 7b fills this gap.

Procedure

For each macro × hazard combination confirmed in Step 5:

- Group hazard scores by country.
- Run a one-way ANOVA across countries on that hazard's score (F, p, η^2).
- Run Tukey HSD post-hoc against a reference country (defined as the country with the lowest mean score for this hazard).
- Report the macro value attached to each country as a context column.
- Flag countries with n < 30 as small-sample (mirroring the events-per-predictor caveat used elsewhere).

Outputs

- {Sector}_step7b_macro_country_profile.csv — one row per (hazard, country) pair: n, macro value, mean score, % high concern, small-n flag.
- {Sector}_step7b_macro_anova.csv — one row per (hazard, macro) confirmed combination: F, p-value, η^2 and eta-squared label.
- {Sector}_step7b_macro_tukey.csv — pairwise post-hoc against the reference country.

Interpretation

Step 7b is descriptive, not inferential. It does not add an additional hypothesis test or re-test what Step 5 already confirmed. The ANOVA and Tukey results sometimes disagree with Step 5 — for example, a macro may be confirmed as a Step 5 predictor for a hazard, but the country-level ANOVA on that same hazard may not be significant. When this happens, it means the Step 5 macro effect operates through mechanisms that do not show up cleanly as country-level mean differences (typically because the macro is one of several correlated country-level factors and the multivariate regression isolates a partial effect that the univariate ANOVA cannot recover).

Step 7b is the empirical bridge between Step 5 ('this macro matters') and Step 8 ('countries differ on the composite score'). It produces per-hazard country profiles that are otherwise missing from the deliverable.

Empirical results from this study

Across the three sectors, 9 macro × hazard combinations were confirmed in Step 5 and profiled in Step 7b:

Energy (3 combinations, all with the electricity price band macro, macro_electricity_consumption). Heating and cooling costs × electricity price band: F = 3.69, p = 0.006, η^2 = 0.035 (statistically significant). Italy mean 17.16 vs Portugal mean 13.05 (gap +4.11 points), but no Tukey pair reaches significance individually because the Portugal reference has n = 22. Higher electricity bills × electricity price band: ANOVA p = 0.054 (marginal). House value decrease × electricity price band: ANOVA p = 0.243 (the macro effect from Step 5 operates through mechanisms not captured by country means).

Housing (3 combinations). Cold/damp leads to health problems × heating_degree_days (and also × cold_home_pct, same country gap): F = 6.99, p = 0.001, η^2 = 0.051. Tukey Portugal vs Germany p = 0.0007 (+4.09 points). Notably, Portugal has the LOWEST heating-degree-days of the three Housing countries — the cold-damp-health concern is therefore NOT climate-driven, more likely housing-stock quality. More frequent power cuts × asthma_death_rate: F = 15.54, p < 0.001, η^2 = 0.107 — the only 'medium' effect-size in any Step 7b output. Ireland vs Germany p = 0.002 (+4.51); Portugal vs Germany p < 0.001 (+3.95).

Transport (3 combinations, all with macro_modal_split_cars). Pollution exposure: F = 6.42, p = 0.012, η^2 = 0.018. Italy +1.97 vs Spain (p = 0.012). Restricted from city centres: F = 6.17, p = 0.013. Spain +2.00 vs Italy (p = 0.014). NOTE the direction flip — Italians worry more about pollution, Spaniards worry more about city-centre restrictions, despite both findings being driven by the same confirmed macro. Composite: F = 0.75, p = 0.39 — not significant (consistent with Step 8).

These Step 7b results are the empirical bridge between the macro confirmations in Step 5 and the composite-level country comparisons in Step 8. Without Step 7b, the per-hazard country picture would be missing from the deliverable.

2.1.12 Step 8 — Comparing results across countries

Describing national differences in hazard concern at the composite level.

Country is not included as a model predictor (Step 2 explains why). This step provides a descriptive country comparison alongside formal tests of whether national differences are statistically real. Step 8 covers composite concern only. For per-hazard country differences in the specific hazards where a macro is confirmed in Step 5, see Step 7b above.

Step 8a: are country differences real? One-way ANOVA

The same ANOVA method from Step 7 is applied with countries as the groups. The F-statistic tests whether between-country variation exceeds within-country variation (Snijders & Bosker, 2012).

Formula 22 — ANOVA F-statistic for country comparison

F = [Variation between country means] / [Variation within countries]

K = number of countries | N = total respondents | p < 0.05 = country effect is real

Symbol / term	What it means
F	ANOVA F-statistic for the country comparison. Range: 0 to +∞. Same formula as F-statistic in Formula 19, but the groups are countries and the outcome is the composite

	concern level (quintile band 1–5 from Formula 3) averaged across all hazards in the sector
Between-country variation	SS_between / (K−1). Range ≥ 0. How much country mean composite-concern scores differ from the overall sample mean. Large when countries differ substantially in average concern
Within-country variation	SS_within / (N−K). Range ≥ 0. How much individual respondents' composite concern scores spread around their own country's mean. Captures within-country heterogeneity that Steps 3–7 are designed to explain
K	Number of countries. Range: K = 5 for Energy (Germany, Italy, Spain, Portugal, Ireland), 3 for Housing (Germany, Portugal, Ireland), 2 for Transport (Italy, Spain). A significant overall F triggers the Tukey HSD follow-up in Formula 24
N	Total respondents in this sector. Range: positive integer. N = 407 for Energy, 263 for Housing, 358 for Transport
$p < 0.05$	Decision threshold (not a quantity): at least one country's mean differs significantly from the others when the F-statistic's p-value is below this level. Triggers Tukey HSD (Formula 24)

Formula 23 — Eta-squared for country effect

$$\eta^2 = SS_between\ countries / SS_total$$

Small η^2 means individual factors explain far more than nationality

Symbol / term	**What it means**
η^2	Eta-squared for the country effect. Same formula as Formula 20 but applied to countries as groups. Range 0 to 1. Quantifies the practical importance of the country ANOVA result from Formula 22: a significant F-test ($p < 0.05$) tells us country differences are real; η^2 tells us how large they are relative to all variation in the data
SS_between countries	Between-country sum of squares (same definition as in Formula 22). The numerator
SS_total	Total sum of squared deviations across all respondents: SS_between + SS_within from Formula 22. Same denominator used in Formulas 16 and 18
Small η^2	When η^2 is small (e.g., 0.01–0.03), country membership explains only 1–3% of total variation in concern scores. Predictors confirmed in Steps 3–7 typically explain substantially more

Small η² across sectors (study-specific)

Country membership consistently explains a small fraction of total variation in composite concern scores across all three sectors: Energy $\eta^2 = 0.016$ (1.6%), Housing $\eta^2 = 0.041$ (4.1%), Transport $\eta^2 = 0.002$ (0.2%). In every sector, the predictor variables confirmed in Steps 3–7 — home conditions, financial vulnerability, EV attitudes, and macro indicators — explain substantially more variation than which country a respondent lives in. Housing's slightly larger η^2 (4.1%) reflects the Germany-Portugal mean difference confirmed by Tukey HSD; the substantive conclusion that country is a minor driver compared to socio-demographic and behavioural factors is consistent across all three sectors.

Country ANOVA results in this study

The triggers row for $p < 0.05$ should be read as follows. At least one country's mean concern level differs significantly from the others when the overall ANOVA p-value is below 0.05; this triggers the Tukey HSD post-hoc test (Formula 24). Not significant in Energy ($p = 0.168$, $K = 5$ countries) or Transport ($p = 0.386$, $K = 2$ countries); significant in Housing ($p = 0.005$, $K = 3$ countries) — the lowest country-ANOVA p-value across the three sectors.

Step 8b: identifying which specific countries differ significantly

When the overall country ANOVA is significant ($p < 0.05$), we know that at least one country shows a meaningfully different level of hazard concern from the others. ANOVA does not tell us which country or countries are responsible. Any comparison that asks whether Country A and Country B differ is a pairwise comparison. With small sample sizes (Ireland n = 11 and Portugal n = 22 in Energy, for example), a large apparent gap can easily be the result of random sampling variation rather than a genuine national difference. The test is what separates real differences from noise.

Formula 24 — Tukey HSD threshold

$$HSD = q \times \sqrt{(MS_within / n_harmonic)}$$

q = Studentised range critical value | MS_within = mean squared error within countries | n_harmonic = harmonic mean of country sample sizes

Symbol / term	What it means
HSD	Honestly Significant Difference threshold. Range > 0. Two countries are declared significantly different if the absolute difference in their mean composite concern scores exceeds this HSD
q	Studentised range critical value from a q-distribution table, indexed by K (number of countries) and the within-group degrees of freedom (N − K). Range > 0. At α = 0.05, q is roughly 3.0 to 3.5 for the K and N values in this study
MS_within	Mean squared error within countries: SS_within / (N − K). Range > 0. Same denominator as the within-country variation in Formula 22. Measures how much concern scores spread around each country's own mean. Larger spread → larger HSD → harder to declare countries different

n_harmonic	Harmonic mean of country sample sizes: K / Σ(1/n_k). Range > 0. Corrects for unequal group sizes (e.g., Ireland n=11 vs Germany n=149 in Energy). Countries with smaller samples contribute less certainty, so HSD is widened to account for this
√ (square root)	Square-root operator. Converts the variance-scaled quantity back to the original score scale (1 to 25) so the HSD can be compared directly to raw mean differences between countries
|mean_A − mean_B|	Absolute difference in mean composite concern scores between Country A and Country B. Range ≥ 0 (typically 0 to 5 on the 1–25 composite scale in this study). If this exceeds HSD, the pair is declared significantly different at the family-wise α level
Decision rule	Procedural specification (not a quantity): if |mean_A − mean_B| > HSD then declare Country A and Country B significantly different. The p-value reported in the posthoc CSV file is the q-distribution probability of observing this difference under the null

Application in this study

In the Energy sector, the overall country ANOVA is not significant (p = 0.168), so the Tukey HSD step is not reached — there is nothing to locate. In the Transport sector there are only two countries (Italy and Spain), so there is only one comparison to make (p = 0.386, also not significant) and no multiple-comparison problem arises. The Tukey HSD is most relevant in the Housing sector, where the overall ANOVA is significant (p = 0.005) and there are 3 countries giving 3 pairwise comparisons. After Tukey HSD adjustment, the only confirmed significant pair is Germany vs Portugal (p = 0.005, mean difference = −1.79 on the 1–25 composite scale; Germany mean below Portugal mean). Germany-Ireland and Portugal-Ireland pairs are not significant.

2.1.13 Step 9 — Comparing results across hazards

Ranking hazards and mapping predictor consistency — only possible because the analysis is run per hazard.

Multiple testing note. The BH-FDR correction was applied independently within each hazard's own set of predictor tests. The family-wise error rate across all hazards within a sector — up to 130 simultaneous tests in the Housing sector — is not controlled at the sector level. A predictor confirmed for a single hazard should therefore be interpreted with appropriate caution. Predictors confirmed across multiple hazards carry substantially stronger evidential weight, as independent confirmation across separate models acts as an informal cross-hazard validation.

Step 9 is the synthesis step that compares the seven (or more) separate hazard analyses produced by the per-hazard analysis. It has no composite counterpart — the composite model produces a single set of predictors for the sector, not a hazard-by-hazard comparison. This comparison is one of the primary justifications for running the analysis per hazard rather than on the composite.

Ranking the hazards

We rank hazards by their average raw score on the 1–25 scale and by the percentage of respondents rating them as high concern (score ≥ 12 out of 25). This directly answers the policy question: which hazard should be addressed first? (Slovic, 1987; Wachinger et al., 2013).

Mapping predictor consistency

We produce a matrix showing which predictors were confirmed for which hazards. A predictor confirmed for many hazards is a robust cross-cutting finding; one confirmed for only one hazard is hazard-specific. This mapping would be impossible with a composite model — it requires having run a separate confirmed predictor set for each hazard.

Sign consistency check (Step 9b A/B convergence analysis)

Predictors that appear in multiple confirmed models (across hazards within a sector, or between the per-hazard and composite analyses) are checked for sign consistency: if the effect is positive in some appearances and negative in others, the predictor is flagged with sign_consistency = 'MIXED (k+ / m−)' in the step9b_AB_convergence.csv output. Where this occurs, the predictor's reported convergence classification (Convergent / Partially convergent / A-cross-cutting / A-specific / B-specific) should be read alongside the sign-consistency flag. In this study, the most notable mixed-sign predictors are gender__Woman or Non-binary (positive in Energy, negative in Housing 'Laws forbid selling/renting'), household_type__Multi-generational (positive in Energy 'struggling_to_pay', negative in Housing insurance/composite), and dwelling_year__Before 1945 (varying sign across hazards). These should be reported as evidence of context-specific rather than uniform effects.

What it means when no predictors are confirmed

If a hazard has no confirmed predictors, concern about it is distributed almost equally across all respondent types — regardless of age, income, gender, or any other measured characteristic. This is not a failure; it is a substantive finding indicating that this hazard is a universal concern requiring no targeted group-specific response (Wachinger et al., 2013).

3. Results

3.1 Introduction

This section reports the outcomes of the twin-transition hazard perception analysis for the three sectors studied: Energy, Housing, and Transport. The structure mirrors the nine-step analytical pipeline described in Section 2. For each sector, every methodological step has its own subsection containing four elements: the purpose of the step, the sector-specific results (tables and charts), a plain-English explanation of what the numbers mean and how they should be interpreted, and a key takeaway for this sector.

Where a technical term first appears we explain it once here:

- Hazard score — each hazard was rated on a 1–25 scale by multiplying perceived Likelihood (1–5) by perceived Impact (1–5). A score of 12 or higher is classified as high concern.
- Confirmed predictor — a respondent characteristic that survived three filters: LASSO selection (Step 3), Likelihood-Ratio Test with Benjamini-Hochberg FDR correction (Step 5), and the per-hazard events-per-predictor cap (also Step 5).

- Odds ratio (OR) — how many times higher (OR > 1) or lower (OR < 1) the odds of being in a higher concern category are.
- Marginal effect in percentage points (MEM) — change in predicted probability of high concern when predictor switches on.
- McFadden's R^2 — how well the model explains variation. 0.10+ is a good fit for ordinal survey data.
- EPP warning — flag for estimates based on small subgroups where the standard error is unreliable.

3.1.1 Overview across the three sectors

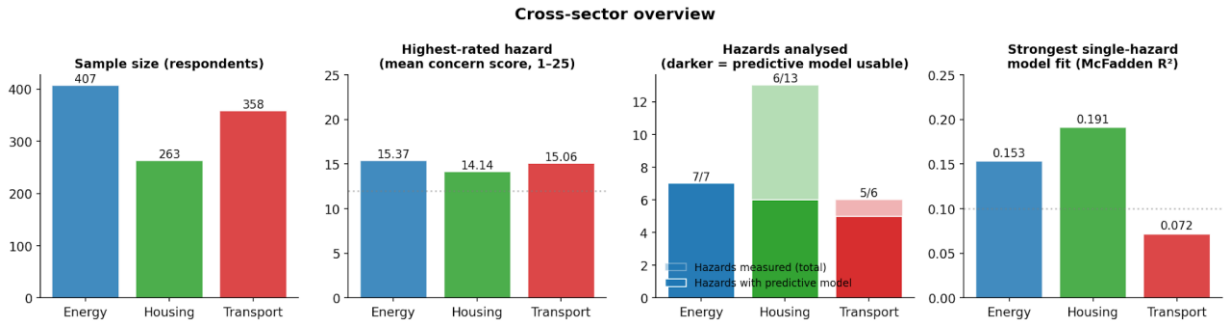


Figure 5. Cross-sector headline measures.

The three sectors are summarised in the table below. Cronbach's α measures internal consistency — values above 0.70 are acceptable; above 0.80 are good.

Table 2. Results overview across three sectors

Sector	Sample	Countries	Hazards	Cronbach's α	Brant violations
Energy	407	Germany, Italy, Spain, Portugal, Ireland	7	0.8157 (good)	0 of 153
Housing	263	Germany, Portugal, Ireland	13	0.8383 (good)	0 of 186
Transport	358	Italy, Spain	6	0.7921 (acceptable)	0 of 69

3.1.2 One pattern that runs through all three sectors

The most universally-shared concerns are the hardest to predict from individual characteristics. The hazards rated most-concerning by the largest share of respondents — heating and cooling costs in Energy and Housing, fuel and repair costs in Transport — have the weakest predictive models. Hazards where concern is split more evenly across the sample produce the strongest models.

This is a statistical consequence: when almost everyone reports the same concern level, there is little variation between respondents for a model to explain. Where the population is genuinely divided, individual characteristics matter more.

Practical implication. Hazards with high mean concern are universal concerns that policy needs to address broadly. Hazards with moderate mean concern are where targeted policy can find traction, because individual characteristics distinguish who is concerned from who is not.

3.2 Energy sector results (n = 407)

This section presents the Energy sector results step by step, matching the methodology document. Each subsection corresponds to one step in the analytical pipeline and contains the purpose of the step, the Energy-specific results, a plain-English explanation of what the numbers mean, and a key takeaway for this sector.

3.2.1 Step 1 — Getting the data ready

Purpose: *Clean the raw survey file, confirm the analytical sample, check that the hazards within this sector measure a coherent concept (Cronbach's α), and set the five concern levels the rest of the analysis uses.*

The Energy analytical sample is 407 respondents (across 159 raw survey columns at the start of cleaning). The country breakdown is:

Table 3. Energy sector - Data distribution

Country	n	% of sample
Germany	149	36.6%
Italy	129	31.7%
Spain	96	23.6%
Portugal	22	5.4%
Ireland	11	2.7%

The raw hazard scores (Likelihood × Impact, 1–25 scale) before any modelling are:

Table 4. Energy sector - Raw hazard scores

Hazard	Variable name	Mean (1–25)	% high concern (≥12)
Higher electricity bills	hazard_elec_bill	14.96	74.0%
Heating and cooling costs increase	hazard_heat_cool	15.37	73.0%
Struggling to pay bills each month	hazard_bill_struggle	9.92	36.1%
House value decrease (no solar)	hazard_house_value	8.88	31.7%
Missing out on solar savings	hazard_miss_savings	11.10	43.5%
New taxes or fines for inefficiency	hazard_tax_fines	9.77	36.6%

More frequent power cuts	hazard_power_cuts	7.67	23.1%

Internal consistency. Cronbach's α = 0.8157 — good (≥ 0.80). This confirms the hazards within the Energy sector measure a coherent underlying concept and can be combined into the composite outcome that the 'B' analyses use.

Concern-level cut-points. The composite score is cut into five quintile-based concern levels using the boundaries 7.17 | 9.57 | 11.57 | 14.86. This produces approximately equal-sized groups (the actual sizes are 20.1 / 19.9 / 19.9 / 19.9 / 20.1 per cent of the sample).

What this means. The Energy sample is large enough (n = 407) and well-spread across five EU countries to support the planned modelling. The Cronbach's α of 0.82 is important: it tells us that respondents who were worried about one Energy hazard tended to be worried about other Energy hazards too. In other words, the seven Energy hazards measure a single coherent dimension of concern — there is such a thing as 'overall Energy concern', not seven unrelated worries. The composite cut-points (7.2 / 9.6 / 11.6 / 14.9) show that the median respondent sits just below the high-concern threshold of 12, meaning the population is genuinely split: roughly half are in the lower three concern quintiles and half are in the upper two. This split is exactly what the modelling needs in order to identify drivers of concern.

► **Key takeaway: The Energy sample is sufficient, internally consistent, and well-distributed across concern levels — the modelling can proceed with confidence.**

Limitations of the dataset

The Energy sample (n = 407) spans five countries with Germany the plurality (149, 37%), followed by Italy (129, 32%) and Spain (96, 24%); Portugal (22, 5%) and Ireland (11, 3%) are too small to anchor country-specific claims. This composition has two analytical consequences. First, because the LASSO selection step pools observations without country weights, predictor selection is weighted toward the larger national samples — Germany, Italy and Spain — and country-level macro variables enter the models with variance dominated by these three. Second, the Step 5c robustness tests stratify by country in every resample, so sign-consistency and stability statistics describe robustness within the observed country mix rather than across structurally different ones. Findings are most defensible for the populations sampled and should not be read as estimates of Energy concern in countries that are under-sampled or unrepresented.

3.2.2 Step 2 — Preparing the data for the statistical model

Purpose: *Turn the cleaned survey answers into numeric columns the ordinal regression can use: one-hot-encode categorical variables, bin age into groups, exclude country labels (replaced by national macro variables), and fill remaining missing values with iterative kNN imputation.*

After encoding and feature engineering, the Energy dataset moves from 159 raw survey columns to 101 candidate predictor variables (each categorical answer becomes one yes/no column per category; age is grouped into bins; macro variables for the respondent's country are added).

Country labels are deliberately excluded from the predictor set. Section 2 explains why: country and the national macro variables carry overlapping information, and including both would cause a mathematical conflict in the model. Country differences are reported descriptively in Step 8 instead.

Missing values in the few remaining cells are filled using iterative kNN imputation based on the five most similar respondents.

The Energy sector adds nine macro variables: **percentage of households in arrears on utility bills, percentage unable to keep the home adequately warm, housing-cost overburden rate, household final consumption expenditure, regional household disposable income, heating degree-days, cooling degree-days, total final energy consumption by households, and household electricity price band**.

What this means. Moving from 159 raw columns to 101 candidate predictors is a careful expansion: each categorical answer becomes one numeric column per category, so 'income band' (one survey question with seven answer categories) becomes seven yes/no columns. The deliberate exclusion of country labels is the most consequential choice at this step. By using national macro variables (electricity price band, household expenditure, utility arrears percentages, cooling degree-days) instead of country dummies, the analysis can attribute concern to specific national circumstances rather than just labelling people by passport. This is what makes findings transferable: when 'higher national utility-arrears percentage' predicts concern, that finding applies to any country with high arrears, not just to Italy or Spain.

► **Key takeaway: Country effects are carried by specific national macro variables, not by country labels — making findings more meaningful and more transferable to other countries.**

3.2.3 Step 3 — Choosing which variables to test (LASSO)

Purpose: *Use LASSO regression with 5-fold cross-validation to shortlist the predictors most worth testing for each hazard. LASSO is a practical filter, not a formal test.*

LASSO is run separately for each hazard and for the composite outcome.

A — Per-Hazard Analysis

Across all per-hazard models, LASSO selected a total of 149 predictor groups. Selection counts for Energy range from 11 (new taxes / fines) to 33 (missing solar savings). High counts mean many candidate predictors correlate weakly with the outcome — they are worth checking but most will fail formal testing in Step 5. Low counts (such as new taxes) tell us that few respondent characteristics correlate even weakly with that hazard.

Table 5. Energy sector – per-hazard predictor count

Hazard	LASSO selections
Higher electricity bills	18
Heating and cooling costs increase	15
Struggling to pay bills each month	31
House value decrease (no solar)	27
Missing out on solar savings	33
New taxes or fines for inefficiency	11
More frequent power cuts	14

B — Composite Hazard Analysis

For the Energy sector composite outcome, LASSO selected 51 predictors out of 101 candidates. This is a reasonable shortlist — not so few that we miss real signals, not so many that the multiple-testing burden becomes unmanageable.

► **Key takeaway: Energy has plenty of candidate predictors to test, especially for the solar-savings and arrears-related hazards — but most will be filtered out by formal testing in the next step.**

3.2.4 Step 4 — The main statistical model (proportional odds)

Purpose: *Fit an ordinal (proportional odds) regression for each hazard, and test the proportional-odds assumption with the Brant test. McFadden's R^2 describes how well each model explains variation in concern.*

Proportional-odds assumption (Brant test). The proportional-odds assumption was tested at the feature level (0 violations in 258 tests) and the group level (0 violations in 179 tests). The model is appropriate for the Energy data.

A — Per-Hazard Analysis

Per-hazard model fits. The figure below shows the full-LASSO McFadden R^2 for each hazard alongside the number of confirmed predictor groups.

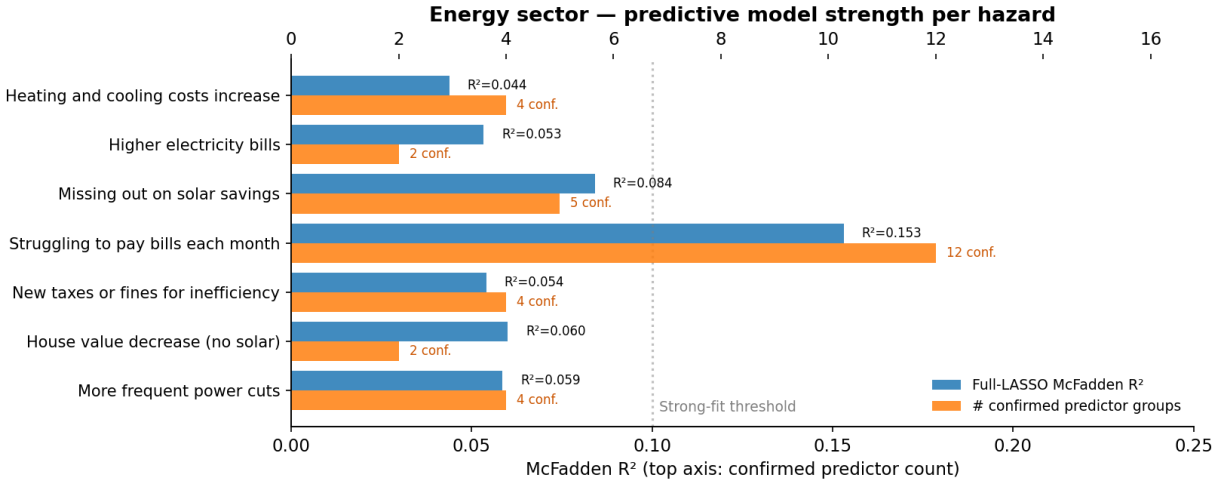


Figure 6. Predictive model strength per Energy hazard.

The per-hazard story is more varied: 'Struggling to pay bills' has $R^2 = 0.15$ (strong fit), reflecting that financial-pressure characteristics clearly distinguish concerned from unconcerned respondents. At the other end, 'Heating and cooling costs' has $R^2 = 0.04$ — not because the model is broken, but because 73 per cent of respondents are already in the high-concern group, leaving little variation for any model to explain.

B — Composite Hazard Analysis

Composite model fit. For the Energy composite outcome, the full-LASSO model achieves McFadden $R^2 = 0.1138$ (Nagelkerke $R^2 = 0.3194$) on n = 407.

The composite-model McFadden R^2 of 0.11 means the confirmed predictors together explain a meaningful share of variation in who is highly concerned. For ordinal survey models, 0.10 or above is the 'good' threshold, and the composite reaches it.

► **Key takeaway: Energy models are statistically valid and the composite explains a meaningful share of concern variation, particularly for the financial-pressure hazards.**

3.2.5 Step 5 — Confirming which predictors are real (LRT + BH-FDR + EPP cap)

Purpose: *Test each LASSO-selected predictor with a Likelihood-Ratio Test, correct for multiple testing using Benjamini-Hochberg FDR control at $\alpha = 0.05$, then apply an events-per-predictor (EPP) cap.*

A — Per-Hazard Analysis

Per-hazard FDR results — how many of the LASSO-selected predictors survived BH-FDR correction and the EPP cap:

Table 6. Energy sector – per-hazard predictor confirmation dimensions

Hazard	LASSO selected	FDR significant	Retained after EPP cap	EPP warnings
Higher electricity bills	18	2	2	0
Heating and cooling costs increase	15	4	4	0
Struggling to pay bills each month	31	12	12	0
House value decrease (no solar)	27	2	2	0
Missing out on solar savings	33	5	5	1
New taxes or fines for inefficiency	11	4	4	0
More frequent power cuts	14	4	4	0

The per-hazard thinning is severe and intentional: 'House value decrease' goes from 27 LASSO selections to 2 confirmed, meaning 25 candidates that LASSO thought might matter did not survive multiple-testing correction. This is not a flaw — it is the safeguard that prevents over-claiming.

B — Composite Hazard Analysis

For the Energy composite outcome, 30 predictor groups were tested. After BH-FDR and EPP cap, 8 are retained as confirmed. The evidence-strength breakdown is:

Table 7. Energy sector - Predictors strength distribution

Evidence strength	Count
strong	4
moderate	2
borderline (interpret with caution — post-LASSO bias)	2
not significant	22

The evidence-strength breakdown for the composite (4 strong, 2 moderate, 2 borderline, 22 not significant) is exactly the pattern good methodology produces: most LASSO candidates do not survive formal testing,

but the ones that do are robust. The 'borderline' label is important — those predictors pass the threshold but only just, and the methodology notes they may partly reflect post-LASSO selection bias.

► **Key takeaway: Heavy filtering at Step 5 is by design — only robust, multiply-corrected predictors survive, so Energy findings can be reported with statistical confidence.**

3.2.6 Step 5b — Validating the overall predictive model (global LRT)

Purpose: *Test whether the entire confirmed predictor set together is significantly better than a null (intercept-only) model.*

A — Per-Hazard Analysis

All seven Energy hazard models pass the global Likelihood-Ratio Test. 'Struggling to pay bills' reaches 'high' tier. The two weakest hazard models (heating/cooling, electricity bills) are still globally significant but reach only 'low (weak fit)' tier — meaning we trust the set is real, but each individual predictor inside it carries less explanatory power.

Table 8. Energy Sector - per-hazard global likelihood ratio outcome

Hazard	Global LRT p	Significant?	Full R^2	Model strength tier
Higher electricity bills	0.0007986	Yes	0.0533	moderate
Heating and cooling costs increase	3.287e-08	Yes	0.0438	low (weak fit)
Struggling to pay bills each month	5.932e-21	Yes	0.1531	high
House value decrease (no solar)	0.0007757	Yes	0.06	moderate
Missing out on solar savings	2.304e-05	Yes	0.0841	moderate
New taxes or fines for inefficiency	1.368e-10	Yes	0.054	moderate
More frequent power cuts	8.993e-10	Yes	0.0585	moderate

B — Composite Hazard Analysis

The Energy composite model achieves the 'high' tier on the global LRT.

► **Key takeaway: The Energy confirmed predictors are not just individually significant — they are jointly explanatory, which means the set of findings can be treated as a coherent model of concern.**

Robustness of the Energy findings

The four resampling tests of Step 5c were applied to the 41 confirmed Energy predictors. Selection stability along the LASSO path under bootstrap resampling (Meinshausen and Bühlmann, B = 500) shows the confirmed set is strongly preferred by LASSO: mean maximum stability is 0.97, and 37 of 41 predictors clear the strict 0.90 threshold. Under the more conservative complementary-pairs subsampling (Shah and Samworth, B = 500), mean stability is 0.90 — a small reduction indicating that the bootstrap result is not appreciably inflated by sample duplication. The retention test (B = 100, 50% subsamples) shows weaker back-end behaviour: mean retention is 0.38, median 0.28, and only 11 confirmed predictors retain FDR-significance in at least half of subsamples.

Crossing complementary-pairs stability at or above 0.75 with retention at or above 0.50 gives the four-quadrant profile defined in Step 5c.

Doubly robust (11 of 41) — reliably selected and reliably significant under refitting. The anchoring findings are: utility-bill arrears (confirmed in three hazards plus the composite, retention 0.64 to 0.92); gender (composite, new taxes, more frequent power cuts; retention 0.73 to 0.92); bill confidence (composite, retention 0.66); income band (struggling to pay, retention 0.76); dwelling year (struggling to pay, retention 0.51); and ethnic minority (new taxes, retention 0.55).

Stable selection, fragile retention (25 of 41) — LASSO would consistently pick these predictors but their FDR significance does not survive subsample refitting. This category contains the three macro-predictor instances (macro_electricity_consumption in heating and cooling, electricity bills, and house value) along with most country-confirmed and lower-frequency predictors.

Fragile on both (4 of 41) — neither selection nor significance is reliable.

Fragile selection, retains significance (1 of 41) — a single edge case in which retention happens to hold despite weak selection.

Direction is universally robust. Sign consistency averages 0.99 across the 41 confirmed predictors, and every one of them keeps its full-sample sign as the dominant direction across subsamples. Where significance is fragile, the direction of association remains a reliable description of who is more concerned.

Key takeaway: The 11 doubly robust predictors constitute the strongest Energy findings — they reproduce on every robustness criterion and should be cited at the predictor level. The 25 stable-selection-fragile-retention predictors should be read at the construct level (financial pressure, country effects, household composition) rather than as individual point estimates. The macro effects in particular are consistently selected but consistently fragile in retention — the data prefers them as candidates, but their significance is sensitive to which respondents are observed.

3.2.7 Step 6 — Measuring the size of each effect (odds ratios and predicted probabilities)

Purpose: Purpose: Translate the confirmed-predictor coefficients from Step 5 into odds ratios (Formula 17) and marginal-effect percentage-point shifts in the probability of falling into the highest concern level (Formula 18). All effects are reported with 95% confidence intervals and an EPP-warning flag (⚠) where the events-per-predictor ratio dips below 5 and estimate stability is reduced.

A — Per-Hazard Analysis

For each of the 7 Energy hazards with confirmed predictors, the per-hazard ordinal regression produces a confirmed-predictor effect table. The tables below show the top 5 confirmed predictors per hazard (ranked by absolute log odds ratio), with odds ratio, 95% CI, p-value, and the marginal effect on the predicted probability of being in the highest concern level (1–25 score binned to the level-5 band).

Hazard 1: Struggling to pay bills each month — 21 confirmed predictors.

Table 9. Per-hazard Step 6 effects: Struggling to pay bills each month (top 5 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
utility_arrears: Yes, once	4.03 (1.84–8.83)	0.0005	+25.5	
utility_arrears: Yes, twice or more	3.40 (1.46–7.89)	0.0045	+21.7	
dwelling_year: 1961-1980	0.37 (0.23–0.59)	<0.0001	-10.4	
bill_confidence: Not very confident — I often need help or support	2.56 (1.45–4.52)	0.0012	+15.1	
household_type: Multi-generational / extended	2.41 (1.43–4.07)	0.0009	+13.6	

Hazard 2: Missing out on solar savings — 17 confirmed predictors.

Table 10. Per-hazard Step 6 effects: Missing out on solar savings (top 5 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
solar_install_likelihood: Very unlikely	0.13 (0.04–0.41)	0.0005	-16.3	
dwelling_year: 1945-1960	3.70 (1.58–8.65)	0.0026	+26.4	
political_group: Yes	0.33 (0.15–0.69)	0.0034	-12.1	
solar_install_likelihood: Unlikely	0.39 (0.16–0.92)	0.0308	-10.7	
solar_install_likelihood: Not sure	0.48 (0.14–1.59)	0.2290	-8.7	⚠

More frequent power cuts — 6 confirmed predictors.

Table 11. Per-hazard Step 6 effects: More frequent power cuts (top 5 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
bill_confidence: Not at all confident — I find it very difficult to understand my energy bill.	6.05 (1.89–19.39)	0.0025	+39.0	
utility_arrears: Yes, twice or more	3.33 (1.46–7.59)	0.0042	+24.0	
political_group: Yes	0.44 (0.22–0.89)	0.0226	-9.8	
utility_arrears: Yes, once	2.15 (1.03–4.51)	0.0417	+14.1	
gender: Woman or Non-binary	2.11 (1.48–3.02)	<0.0001	+11.5	

Heating and cooling costs increase — 4 confirmed predictors.

Table 12. Per-hazard Step 6 effects: Heating and cooling costs increase (top 4 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
utility_arrears: Yes, twice or more	3.00 (1.38–6.51)	0.0056	+21.5	
religious_minority: Yes	0.38 (0.17–0.88)	0.0239	-10.9	
home_problems_count	1.29 (1.13–1.47)	0.0002	+4.1	
macro_electricity_consumption	1.23 (1.08–1.40)	0.0018	+3.4	

New taxes or fines for inefficiency — 5 confirmed predictors.

Table 13. Per-hazard Step 6 effects: New taxes or fines for inefficiency (top 5 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
utility_arrears: Yes, twice or more	4.11 (1.89–8.96)	0.0004	+28.9	
ethnic_minority: Yes	3.04 (1.62–5.70)	0.0005	+21.3	

political_group: Yes	0.36 (0.18–0.75)	0.0058	-11.3	
utility_arrears: Yes, once	2.60 (1.29–5.28)	0.0079	+18.0	
gender: Woman or Non-binary	2.13 (1.49–3.03)	<0.0001	+11.4	

Higher electricity bills — 2 confirmed predictors.

Table 14. Per-hazard Step 6 effects: Higher electricity bills (top 2 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
political_group: Yes	0.34 (0.17–0.70)	0.0035	-12.5	
macro_electricity_consumption	1.16 (1.02–1.32)	0.0249	+2.4	

House value decrease (no solar) — 2 confirmed predictors.

Table 15. Per-hazard Step 6 effects: House value decrease (no solar) (top 2 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM	EPP Warning
political_group: Yes	0.28 (0.13–0.60)	0.0010	-13.9	
macro_electricity_consumption	1.12 (0.99–1.28)	0.0741	+1.9	

What the per-hazard effects tell us

Three patterns run through the Energy per-hazard ORs:

1. **utility_arrears is the single most consistent predictor.** "Yes, twice or more" appears at the top of the OR ranking for four of the seven Energy hazards (heating/cooling costs, more frequent power cuts, new taxes/fines, struggling to pay bills), with ORs between 3.0 and 4.1. Respondents with repeated utility arrears carry 3–4 times the odds of high concern on these hazards relative to those with no arrears history.
2. **political_group__Yes is the only consistent protective predictor.** Across five Energy hazards, respondents who identify with a political group show OR between 0.28 and 0.44 — a 2.3- to 3.5-fold lower odds of high concern. This is interpretable as institutional trust mediating concern, not as the predictor itself being protective.
3. **The two most-confirmed hazards have very different drivers.** struggling_to_pay_bills_each_month (21 confirmed predictors) is dominated by direct financial-pressure variables; missing_out_on_solar_savings (17 confirmed predictors) is dominated by solar_install_likelihood categories and dwelling-year. The per-hazard tables surface these distinctive driver sets — visible nowhere else in the analysis.

B — Composite Hazard Analysis

For the composite analysis, Step 6 reports the odds ratios for the confirmed predictors in the Energy composite ordinal regression.

The composite Energy model retains **20 confirmed predictors** (12 significant at p < 0.05). The top 5 by absolute log odds ratio:

Table 16. Composite Step 6 effects: Energy composite model (top 5 by absolute log odds ratio).

Sub-group	OR	p-value	MEM (PCT PTS)	EPP Warning
utility_arrears: Yes, twice or more	5.76 (2.59–12.78)	<0.0001	+35.4	
bill_confidence: Not at all confident — I find it very difficult to understand my energy bill.	3.85 (1.37–10.79)	0.0105	+26.1	
dwelling_year: 1945-1960	3.59 (1.41–9.10)	0.0072	+24.2	
political_group: Yes	0.28 (0.13–0.61)	0.0013	-11.8	
bill_confidence: Not very confident — I often need help or support	3.20 (1.81–5.66)	0.0001	+20.7	

What this means (Composite Analysis).

The five strongest predictors of overall Energy concern (by absolute log odds ratio) reproduce the per-hazard pattern at the aggregate level. utility_arrears 'Yes, twice or more' has OR = 5.76 in the composite model, meaning respondents with repeated arrears have nearly six times the odds of being highly concerned about Energy hazards. That is not a marginal effect; it is one of the largest effect sizes in the entire study. The pattern is consistent: every top predictor is some form of financial-pressure indicator — arrears history, low bill confidence, mid-century post-war housing. The single negative effect comes from political affiliation: respondents who identify with a political group are systematically less concerned, possibly reflecting trust in institutional handling of the transition or different lived experience of energy costs.

► **Key takeaway: Energy concern is driven by financial pressure — respondents already behind on bills are nearly six times more likely to be highly concerned about Energy hazards. Per-hazard ORs add the targeting precision (different hazards pick out different driver subgroups), while the composite ORs and predicted-probability shifts make the aggregate magnitude of effect concrete for policy.**

Predicted probabilities (composite model): *Convert the odds-ratio table above into a predicted-probability picture.*

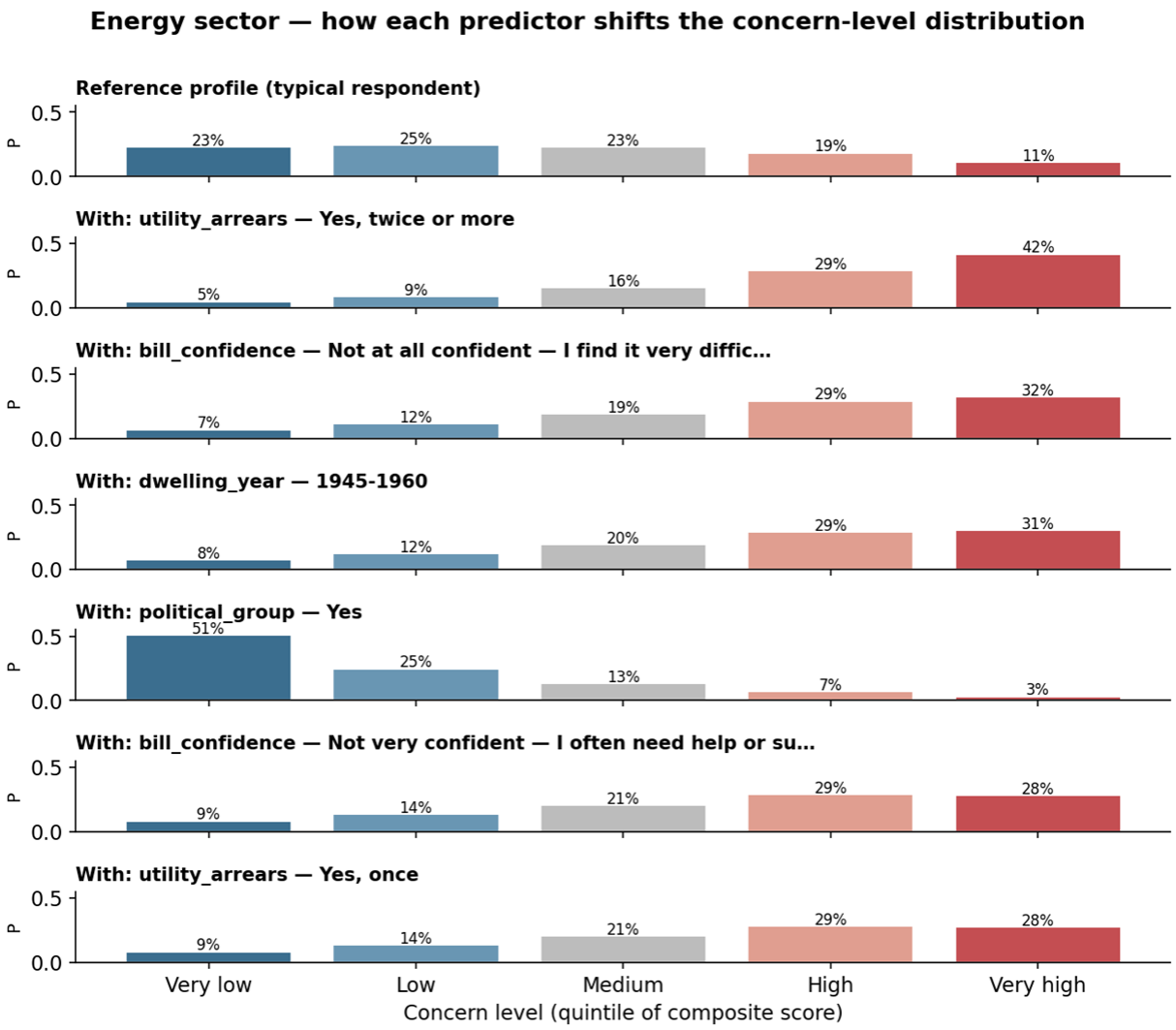


Figure 7. How each of the top predictors shifts the predicted distribution of concern levels.

How to read this chart. The reference profile has a 11% probability of falling in the 'very high concern' group. The rows below show how that probability changes when each predictor is 'switched on'.

What this means. The probability picture is more intuitive than odds ratios. A reference respondent — someone with none of the predictor characteristics 'switched on' — has an 11 per cent chance of falling in the 'very high concern' group. If that same person has been in utility arrears twice or more, the probability jumps to 42 per cent: almost a four-fold increase. The political-affiliation effect goes the other way: identifying with a political group drops the very-high-concern probability from 11 per cent to 3 per cent. These are not subtle shifts — they describe genuinely different concern populations within the sample.

3.2.8 Step 7 — Identifying the most affected groups (ANOVA)

Purpose: *For each confirmed categorical predictor, run a one-way ANOVA on the composite concern score and identify which categories show highest concern. Descriptive complement to the regression-based Step 6.*

A — Per-Hazard Analysis

For each of the 7 Energy hazards with confirmed predictors, a one-way ANOVA was run on every confirmed categorical predictor in that hazard's own Step 5A confirmed set, using the hazard's 1–25 score as the outcome. Macros (sector-level indicators) and continuous predictors are excluded by methodology from the main ANOVA — macros because their within-sector variation is purely country-level (covered separately in

Step 7b for any macro that is FDR-confirmed), continuous predictors because they have no natural category boundaries for between-group comparison.

Across all 7 per-hazard models, **29 categorical predictor ANOVAs were performed; 19 (66%) showed a statistically significant univariate effect at α = 0.05**. The chi-squared verification test (Formula 21) was run alongside each ANOVA and agreed with the ANOVA decision in all 29 cases.

Hazard 1: Higher electricity bills — 1 confirmed categorical predictor tested, 1 significant.

Table 17. Per-hazard Step 7 ANOVA: Higher electricity bills.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
political_group	No (does not identify)	Yes (identifies with group)	+3.79	0.0115	0.020

Respondents who do not identify with any political group carry higher mean concern about higher electricity bills than those who do — a 3.79-point gap on the 1–25 scale. Those who prefer not to say sit between the two but closer to "No". This matches the protective direction of political_group reported in the Step 6 odds ratios.

Note: *The Step 5 LRT confirms exactly 2 predictors for this hazard: macro_electricity_consumption (excluded from ANOVA per the macro rule, profiled in Step 7b instead) and political_group (tested in ANOVA above). An additional post-hoc file for utility_arrears exists in the output folder, but utility_arrears did not pass FDR for this hazard (p_adj = 0.054) and is correctly absent from the ANOVA results.*

Heating and cooling costs increase — 3 confirmed categorical predictors tested, 2 significant.

Table 18. Per-hazard Step 7 ANOVA: Heating and cooling costs increase.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
utility_arrears	Yes, twice or more	No	+5.07	0.0013	0.034
home_problems_count	6 home problems	2 home problems	+8.75	0.0182	0.052

Respondents with **repeated utility arrears (twice or more)** carry the highest mean concern, +5.07 points above those with no arrears history. The home-problems gradient is even steeper: respondents reporting 6 dwelling problems have +8.75-point higher mean concern than those reporting 2 — the steepest categorical gradient in this hazard's profile. Religious-minority status is confirmed in Step 5 but shows no significant pairwise differences in Step 7, indicating its effect operates conditionally on the other predictors rather than as a stand-alone group-mean shift.

Struggling to pay bills each month — 11 confirmed categorical predictors tested, 8 significant.

Table 19. Highest- and lowest-concern categories: Struggling to pay bills each month (Energy per-hazard Step 7).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
utility_arrears	Yes, twice or more	No	+6.44	<0.0001	0.078

can_cover_solar_cost	No (cannot cover)	Yes (can cover)	+5.12	<0.0001	0.110
income_band	Band 1 (lowest income)	Band 5 (mid)	+5.92	0.0005	0.096
dwelling_year	1945–1960	1961–1980	+6.38	0.0054	0.054
bill_confidence	Not very confident	Very confident	+3.53	0.0185	0.029
aware_gov_programmes	No (not aware)	Yes (aware)	+4.36	0.0117	0.024
energy_improvements	"I don't know"	Yes (improvements made)	+3.30	0.0221	0.018
household_type	Other shared / Single parent	Couple, no children	(range ~+4 pts)	(overall ANOVA p=0.022)	0.032

Five of the eight significant predictors point to the same vulnerable population: respondents with repeated utility arrears, no ability to cover solar costs, low income (band 1), low bill confidence, and no awareness of government programmes all carry 3.3–6.4 points higher mean concern than the corresponding "safer" category. The dwelling-year finding (1945–1960 housing stock highest, 1961–1980 lowest) reflects the older-housing-stock penalty already visible in the per-hazard Step 6 odds ratios. Income band shows a stair-step pattern: bands 1–4 (lower-income) cluster high, bands 5–9 (mid-to-high) cluster low; the largest gap is band 1 vs band 5 (+5.92 points).

Note: *The Step 5 LRT confirms exactly 12 predictors. Eleven are tested in the ANOVA above; the 12th — age — is continuous (no categorical groups) and is correctly excluded from Step 7 per the methodology rule that continuous predictors are summarised through their Step 6 odds ratios rather than ANOVA.*

House value decrease (no solar) — 1 predictor tested, 0 significant.

The single confirmed categorical predictor political_group does not reach univariate significance for this hazard (F=3.01, p=0.0504 — exactly at the threshold). Step 6 odds ratios are the appropriate effect-size summary; no per-category finding can be reported from Step 7.

Missing out on solar savings — 5 predictors tested, 1 significant.

Table 20. Highest- and lowest-concern categories: Missing out on solar savings (Energy per-hazard Step 7).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
solar_install_likelihood	Likely to install	Very unlikely	+7.17	0.0001	0.054

The most striking — and at first counter-intuitive — finding in the Energy per-hazard ANOVAs: respondents who say they are **likely** or **very likely** to install solar carry the highest concern about missing out on savings, +7.17 points above those who are very unlikely to install. The reading is that prospective solar adopters are most acutely aware of what they would lose if the opportunity disappeared, whereas those who have already ruled out solar feel no exposure to the hazard. The other four confirmed predictors (political_group, dwelling_year, home_type, english_proficiency) do not show significant univariate effects for this hazard; their roles in the Step 5 model are conditional on the other shortlisted predictors.

Note: *The Step 5 LRT confirms exactly 5 categorical predictors, all five tested in the ANOVA above. Additional post-hoc files exist in the output folder for age_group, utility_arrears, macro_consumption_expenditure, macro_electricity_consumption, and macro_utility_arrears_pct — these correspond to LASSO-selected predictors that did not pass FDR (p_adj between 0.05 and 0.52) and are not part of the confirmed-predictor set.*

New taxes or fines for inefficiency — 4 predictors tested, 3 significant.

Table 21. Highest- and lowest-concern categories: New taxes or fines for inefficiency (Energy per-hazard Step 7).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
gender	Woman or Non-binary	Man	+3.27	<0.0001	0.055
utility_arrears	Yes, twice or more	No	+5.44	0.0005	0.055
ethnic_minority	Yes (ethnic minority)	No	+3.52	0.0098	0.021

The distributive-impact pattern is clear: **women and non-binary respondents**, **repeat-arrears households**, and **ethnic-minority respondents** all carry significantly higher concern about who carries the cost of inefficiency penalties than their reference categories. The arrears gap (+5.44) is the largest of the three, and the gender and ethnic-minority gaps (+3.27 and +3.52) are remarkably similar in magnitude — pointing to a broad equity dimension rather than a single demographic driver. Political-group identification is confirmed in Step 5 but not univariately significant in Step 7 for this hazard.

More frequent power cuts — 4 predictors tested, 4 significant.

Table 22. Highest- and lowest-concern categories: More frequent power cuts (Energy per-hazard Step 7).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
bill_confidence	Not at all confident	Very confident	+8.65	0.0021	0.066
utility_arrears	Yes, twice or more	No	+4.15	0.0124	0.037
gender	Woman or Non-binary	Man	+3.01	<0.0001	0.060
political_group	Prefer not to say / No	Yes (identifies)	+3.39	<0.0001	0.023

This is the only Energy hazard where every confirmed categorical predictor is also univariately significant. The largest gap in the entire Energy per-hazard ANOVA set appears here: respondents **"not at all confident" about understanding their bill** carry mean concern +8.65 points above the "very confident" group — a near-third-of-the-scale gap on the 1–25 measure. The gender, arrears, and political-affiliation effects all follow the patterns already observed in other hazards.

What the per-hazard ANOVAs tell us

Three patterns emerge across the 7 hazards once the focus shifts from "which predictor is significant" to "which category is most concerned":

1. **The highest-concern category is almost always the most vulnerable subgroup.** Utility arrears "twice or more", income band 1, "cannot cover solar cost", "not at all confident" about bills, "not aware of government programmes", "ethnic minority", and "woman or non-binary" all consistently sit at the

top of their respective category rankings. This is methodologically reassuring — it shows the survey is capturing real vulnerability gradients rather than noise.

2. **The largest single category gaps are concentrated in the financial-pressure and bill-comprehension predictors.** The two largest within-predictor gaps are bill_confidence in power cuts (+8.65 points) and home_problems_count in heating/cooling (+8.75 points). Both represent more than a third of the 1–25 scale — they are the most actionable findings in Step 7 because they identify populations where targeted intervention could move concern by a substantial margin.
3. **Predictor types behave differently in Step 7.** Binary predictors (gender, ethnic_minority) show one clean gap; ordered categorical predictors (utility_arrears, income_band) show a graded staircase; multi-level unordered predictors (household_type, dwelling_year) show a more diffuse pattern where multiple mid-categories cluster around the mean and only one or two outlier categories carry distinctive concern. The table format above emphasises the largest gap, but the underlying post-hoc files contain the full pairwise comparison set for any predictor where finer category-level analysis is needed.

B — Composite Hazard Analysis

For the composite analysis, ANOVA was run on the composite concern score using each confirmed categorical predictor from the Step 5B composite model. Across the 8 predictors retained in the composite model, 6 show a statistically significant univariate ANOVA effect at $\alpha = 0.05$. The category-level pattern reproduces what the per-hazard analyses show:

Table 23. Highest- and lowest-concern categories: Energy composite model (Step 7B). Top 3 by effect size.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
utility_arrears	Yes, twice or more	No	14.47	<0.0001	0.067
gender	Woman or Non-binary	Man	24.18	<0.0001	0.056
bill_confidence	Lower-confidence categories	Very confident	3.85	0.0044	0.037

Why Step 5 and Step 7 can disagree. Step 5 measures a predictor with all others held constant (multivariate); Step 7 tests it on its own (univariate). Disagreements signal effects that only become visible when other variables are controlled for.

What this means. The Step 7 ANOVA tests each predictor in isolation, without controlling for the others. The top three results (utility arrears, gender, bill confidence) match the Step 6 odds-ratio ranking, which means the Energy findings are robust to whether they are examined one-at-a-time or in the full multivariate model. The η^2 (eta-squared) numbers describe how much of the variation in composite concern each predictor explains on its own: utility arrears at $\eta^2 = 0.067$ reaches the 'medium' effect-size tier — respectable for survey data and consistent with its prominence in the regression.

► **Key takeaway: Energy's highest-concern categories — repeat-arrears households, lowest-income bands, women and non-binary respondents, those with low bill confidence, those unable to cover solar costs, and ethnic-minority respondents — line up consistently across the per-hazard and composite tracks. The largest single category gap in the entire Energy Step 7 result set is bill confidence on the More frequent power cuts hazard, where respondents not at all confident about understanding their bill carry**

mean concern +8.65 points above the most confident group — over a third of the 1–25 scale. This is the most policy-actionable per-category finding in the Energy sector.

3.2.8.1 Step 7b — Per-country profile for confirmed macro predictors

Purpose: *When a national macro variable (such as cooling degree-days or household utility-arrears percentage) is confirmed as a predictor in Step 5 for a specific hazard, Step 7b reports the per-country profile of concern for that hazard. This fills a gap left by the standard Step 7 (which excludes macros because they have no within-country variation) and by Step 8 (which reports country effects only for the composite outcome, not per hazard).*

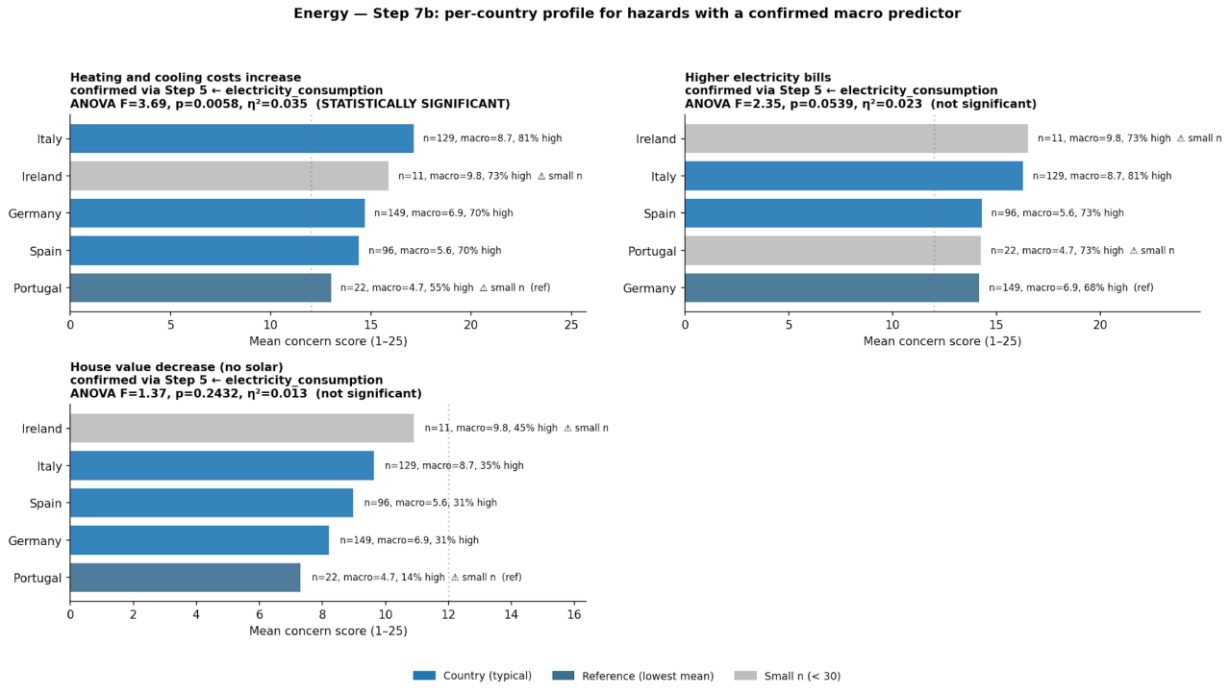


Figure 8. Per-country mean concern for Energy hazards where a macro predictor was confirmed in Step 5. Each panel shows the macro value attached to each country and flags countries with n < 30.

Across Energy, 3 macro × hazard combinations were confirmed in Step 5 and profiled per country here. The ANOVA results are summarised below.

Table 24. Energy sector - Effect of macro variables

Hazard	Macro predictor	F	p-value	η^2	Effect size	Sig
Heating and cooling costs increase	electricity_consumption	3.69	0.0058	0.035	small	**
Higher electricity bills	electricity_consumption	2.35	0.0539	0.023	small	n.s.
House value decrease (no solar)	electricity_consumption	1.37	0.2432	0.013	small	n.s.

No country pairs reach Tukey HSD significance at α = 0.05 for these combinations. The ANOVA's overall pattern suggests a gradient across countries rather than one country being extreme.

What this means. Italy carries the highest Energy concern in heating and cooling costs, where the mean Italian concern score (17.16) is more than four points above Portugal's (13.05); it is also among the highest in the other two macro-confirmed hazards. Ireland's mean nominally exceeds Italy's in higher electricity bills (16.55 vs 16.30) and house value decrease (10.91 vs 9.64), but with n=11 these point estimates are too noisy to interpret as country-level differences.The ANOVA across all five countries is statistically significant for heating/cooling (p=0.006). For higher electricity bills the cross-country ANOVA is marginal (p=0.054), and for house value decrease it is not significant (p=0.243) — meaning the macro effect captured in Step 5 for that hazard works through mechanisms that do not show up cleanly in country-level means. Pairwise Tukey tests do not reach significance individually, consistent with both Ireland (n=11) and Portugal (n=22) being too small to anchor reliable country-level comparisons.

► **Key takeaway:** Italy carries the highest Energy concern among the larger national samples in the macro-confirmed hazards, but Tukey post-hoc cannot isolate significant country pairs — Ireland (n=11) and Portugal (n=22) are too small to anchor reliable comparisons.

3.2.9 Step 8 — Comparing results across countries

Purpose: *Describe how the composite concern score varies across the countries in the sample. Country labels are NOT in the regression model.*

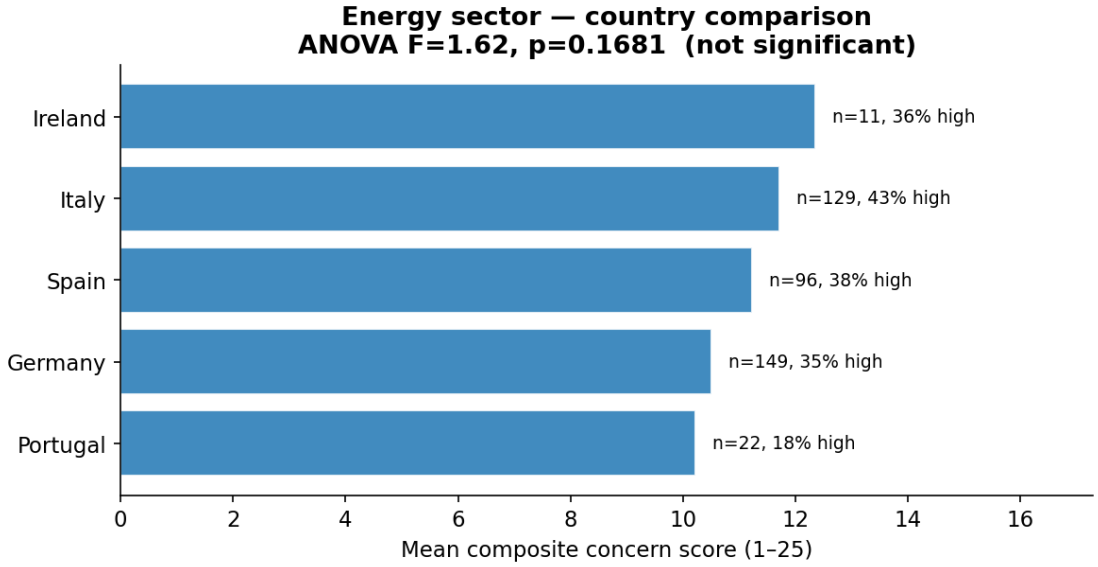


Figure 9. Mean composite Energy concern by country with ANOVA result.

The ANOVA across countries is not statistically significant (F = 1.62, p = 0.1681, η^2 = 0.016). No country pair shows significantly different overall Energy concern in this sample.

What this means. Country differences in Energy concern are not statistically significant (F = 1.62, p = 0.17). The mean composite scores by country span only 10.21 (Portugal) to 12.35 (Ireland), and the differences are within noise. This is a substantively important non-finding: it means the patterns of who is concerned and why — financial stress, dwelling age, bill confidence — recur across all five EU countries in the sample, rather than being driven by one country's particular politics or economic situation. Policy informed by these findings is therefore likely to apply broadly rather than needing to be country-specific.

► **Key takeaway:** **Energy concern is statistically uniform across the five EU countries in the sample — the financial-pressure pattern is not a German story or an Italian story, it is a European story.**

3.2.10 Step 9 — Comparing results across hazards (synthesis and convergence)

Purpose: *Pull the per-hazard results together: rank hazards by concern, map cross-hazard predictor consistency, and compare per-hazard (A) vs composite (B) analyses.*

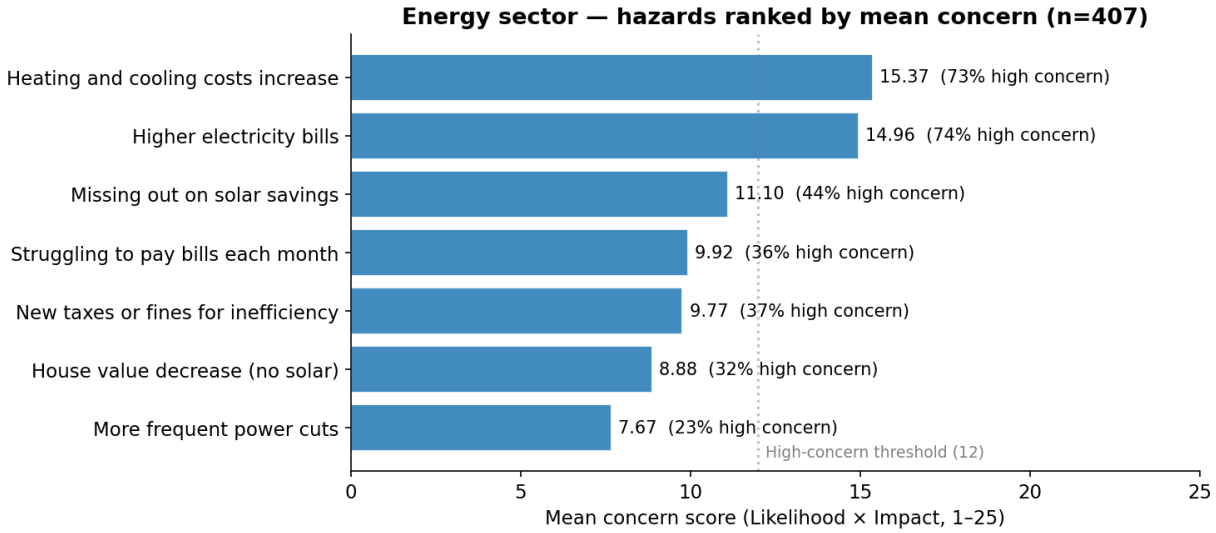


Figure 10. Energy hazards ranked by mean concern score.

Cross-hazard predictor consistency. 7 predictors are confirmed in two or more hazards; the most cross-cutting predictor, 'political_group', is confirmed in 5 hazards.

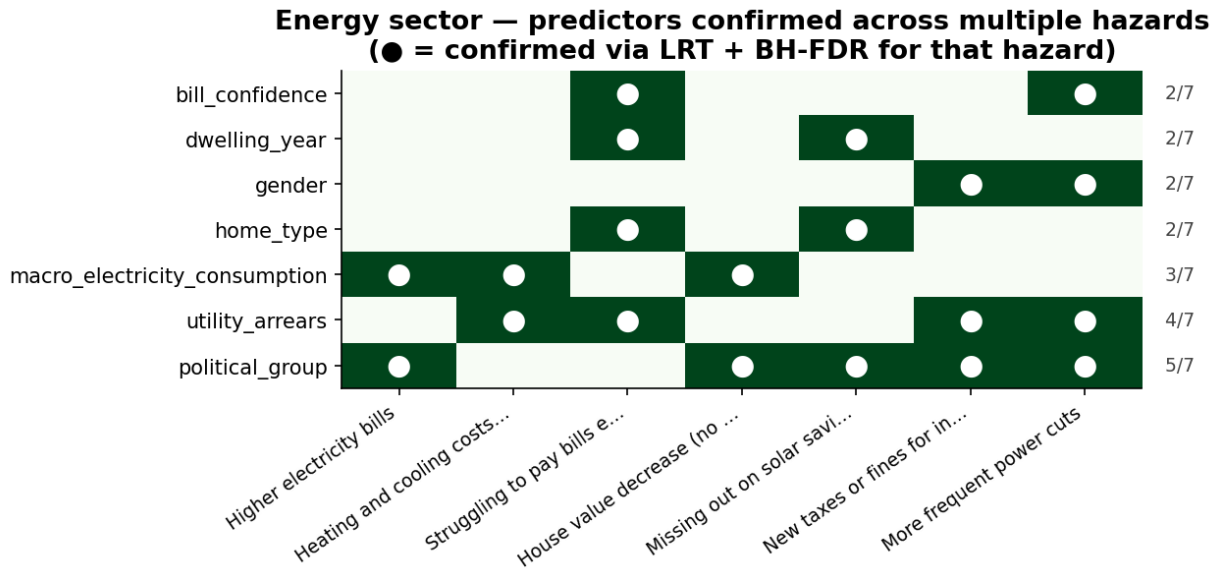


Figure 11. Predictors that are confirmed across more than one Energy hazard.

3.2.10.1 Step 9b — A/B convergence between per-hazard and composite analyses

Purpose: *Classify each confirmed predictor as Convergent, Partially convergent, A-cross-cutting, A-specific, or B-specific.*

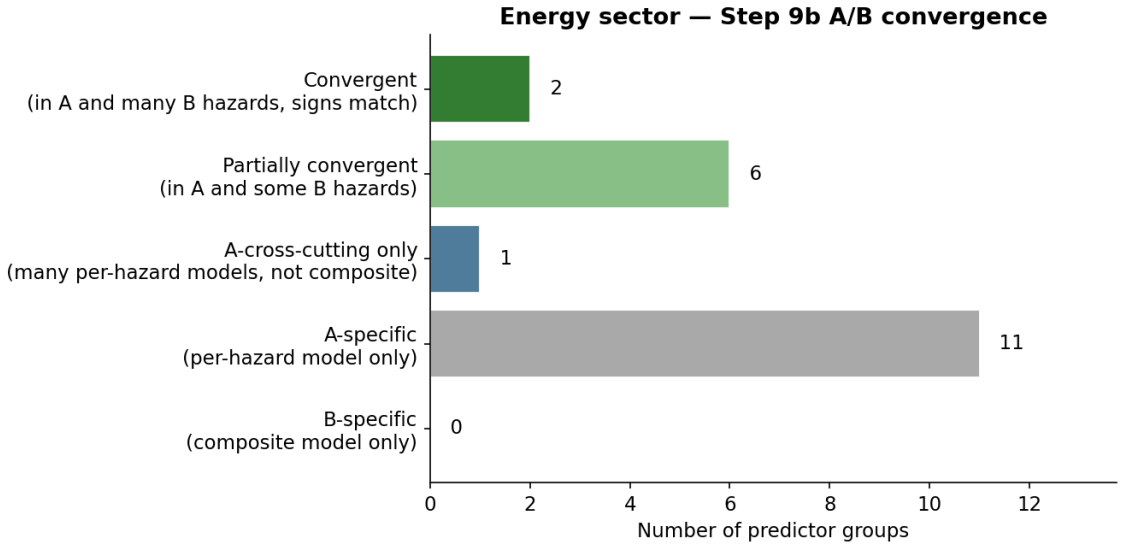


Figure 12. Step 9b classification of confirmed predictors by A/B convergence.

In Energy, 2 predictor group(s) are classified as Convergent, 6 as Partially convergent, 11 as A-specific, and 1 as A-cross-cutting.

What this means. The hazard ranking shows people are most concerned about heating/cooling costs and electricity bills (around 74 per cent in high concern), and least concerned about power cuts (23 per cent). The cross-hazard predictor map shows political identity is confirmed in five of the seven hazards — by far the most cross-cutting predictor. Utility arrears is confirmed in four hazards. These two characteristics are the through-lines of Energy concern. The A/B convergence analysis flags political_group and utility_arrears as 'Convergent' — they show up in the composite model and in multiple per-hazard models with the same direction of effect, which is the most stringent test of robustness this study applies.

► **Key takeaway: Two characteristics structure Energy concern across the board: political identity (associated with lower concern) and utility-bill arrears history (associated with higher concern). These are the most robust findings.**

3.2.11 Headline for Energy sector

Financial pressure indicators (utility-bill arrears, low bill confidence, mid-century housing) are the strongest predictors of Energy concern in this sample.

3.3 Housing sector results (n = 263)

This section presents the Housing sector results step by step, matching the methodology document. Each subsection corresponds to one step in the analytical pipeline and contains the purpose of the step, the Housing-specific results, a plain-English explanation of what the numbers mean, and a key takeaway for this sector.

3.3.1 Step 1 — Getting the data ready

Purpose: *Clean the raw survey file, confirm the analytical sample, check that the hazards within this sector measure a coherent concept (Cronbach's α), and set the five concern levels the rest of the analysis uses.*

The Housing analytical sample is 263 respondents (across 185 raw survey columns at the start of cleaning). The country breakdown is:

Table 25. Housing sector - Data distribution

Country	n	% of sample
Germany	197	74.9%
Portugal	49	18.6%
Ireland	17	6.5%

The raw hazard scores (Likelihood × Impact, 1–25 scale) before any modelling are:

Table 26. Housing sector - Raw hazard scores

Hazard	Variable name	Mean (1–25)	% high concern (≥12)
Higher electricity bills	hazard_elec_bill	13.98	67.0%
Heating and cooling costs increase	hazard_heat_cool	14.14	68.2%
Struggling to pay bills each month	hazard_bill_struggle	9.05	32.6%
House value decrease (no solar)	hazard_house_value	7.20	21.2%
Missing out on solar savings	hazard_miss_savings	10.40	44.3%
New taxes or fines for inefficiency	hazard_tax_fines	9.40	37.1%
More frequent power cuts	hazard_power_cuts	6.99	18.6%
Home insurance more expensive	hazard_insurance_cost	8.62	32.6%
Laws forbid selling/renting non-renovated	hazard_rent_ban	6.59	18.2%
Home more damp or mould	hazard_damp_mould	8.29	29.2%
Insurers classify home as high risk	hazard_insurer_risk	6.36	18.6%
Increased severe weather impacts	hazard_severe_weather	9.47	36.7%
Cold/damp leads to health problems	hazard_cold_damp_health	9.36	36.4%

Internal consistency. Cronbach's α = 0.8383 — good (≥ 0.80). This confirms the hazards within the Housing sector measure a coherent underlying concept and can be combined into the composite outcome that the 'B' analyses use.

Concern-level cut-points. The composite score is cut into five quintile-based concern levels using the boundaries 6.11 | 8.45 | 10.08 | 11.92. This produces approximately equal-sized groups (the actual sizes are 20.1 / 19.9 / 19.9 / 19.9 / 20.1 per cent of the sample).

What this means. The Housing sample of 263 respondents is smaller than Energy but covers more hazards (13 instead of 7). The Cronbach's α of 0.84 is the highest of the three sectors, telling us the 13 Housing hazards form a particularly coherent concept — respondents have a consistent view of what 'Housing concern' means. The country distribution is heavily weighted toward Germany (75 per cent), which is the most important structural feature of the Housing sample to bear in mind throughout the rest of the analysis. The quintile cut-points (6.1 / 8.5 / 10.1 / 11.9) show that the median respondent sits in 'moderate' concern, lower than in Energy or Transport, reflecting that several Housing hazards (severe weather, insurers reclassifying homes, laws restricting non-renovated homes) are not yet as widely experienced.

► **Key takeaway: The Housing hazards measure a particularly coherent concept (α = 0.84), but the sample is heavily weighted toward Germany, which shapes how country-level findings should be read.**

Limitations of the dataset

The Housing sample (n = 263) is heavily concentrated in Germany (197, 75%), with smaller subsamples from Portugal (49, 19%) and Ireland (17, 7%). This is functionally a study of Housing concern in a predominantly German sample, with limited representation of two further national contexts. Three analytical consequences follow.

First, the LASSO selection step pools observations without country weights, so three-quarters of the gradient signal comes from German respondents — confirmed predictors are those most discriminative within the German subsample. Predictors that might separate concern groups in Portuguese or Irish housing markets but not in the German one tend to be filtered out at this step.

Second, country-level macro variables become near-binary "Germany versus rest" contrasts. The macro-confirmed findings reflect this asymmetry: they describe how Germany differs from the small non-German subsample on each dimension, not a richer cross-national gradient.

Third, the Step 5c robustness tests stratify by country in every resample, so each subsample reproduces the 75-percent-German composition. The high sign-consistency (mean 0.94) and complementary-pairs stability statistics describe robustness within this composition; they cannot speak to whether the same predictors would emerge in a more nationally balanced sample.

The headline findings — the housing-cost-band cluster, the dampness-related predictors, the BER and dwelling-age predictors, and the single doubly-robust heat_fuel_wood predictor — are best read as describing a predominantly German Housing perception space. They remain valid descriptions of the data observed. Generalisation to the Portuguese or Irish national populations would require dedicated samples; generalisation to EU countries not represented is outside the study's scope.

3.3.2 Step 2 — Preparing the data for the statistical model

Purpose: *Turn the cleaned survey answers into numeric columns the ordinal regression can use: one-hot-encode categorical variables, bin age into groups, exclude country labels (replaced by national macro variables), and fill remaining missing values with iterative kNN imputation.*

After encoding and feature engineering, the Housing dataset moves from 185 raw survey columns to 111 candidate predictor variables (each categorical answer becomes one yes/no column per category; age is grouped into bins; macro variables for the respondent's country are added).

Country labels are deliberately excluded from the predictor set. Section 2 explains why: country and the national macro variables carry overlapping information, and including both would cause a mathematical conflict in the model. Country differences are reported descriptively in Step 8 instead.

Missing values in the few remaining cells are filled using iterative kNN imputation based on the five most similar respondents.

The Housing sector adds nine macro variables: **percentage reporting the dwelling is too dark, respiratory/circulatory death rate per 100,000, percentage reporting leaks/damp/rot, excess winter mortality, percentage unable to keep the home adequately warm, housing-cost overburden rate, heating degree-days, cooling degree-days, and severe housing deprivation rate**.

What this means. Housing has 185 raw survey columns, more than Energy or Transport, reflecting the breadth of the 13 Housing hazards (financial, structural, regulatory, and health-related). After encoding and feature engineering, there are 111 candidate predictor variables. The macro variables added (cold-home percentage, heating degree-days, solar dwelling percentage, housing arrears percentage) are particularly relevant to the Housing context — they capture climate exposure and housing-market characteristics of each respondent's country. The kNN imputation step matters more in Housing because some questions (e.g. about energy ratings, dwelling year) have higher missing-data rates than core demographic questions.

► **Key takeaway: Housing's broader hazard set translates into a wider predictor universe, and the macro variables capture climate and housing-market context directly relevant to perceived risk.**

3.3.3 Step 3 — Choosing which variables to test (LASSO)

Purpose: *Use LASSO regression with 5-fold cross-validation to shortlist the predictors most worth testing for each hazard. LASSO is a practical filter, not a formal test.*

LASSO is run separately for each hazard and for the composite outcome.

A — Per-Hazard Analysis

Across all per-hazard models, LASSO selected a total of 236 predictor groups. The Housing LASSO results are dramatic: 'Cold/damp leading to health problems' has 61 LASSO-selected predictors, 'Damp or mould' has 55, while 'Higher electricity bills' has zero. This wide variation tells us something substantive about Housing perception: the health-related hazards have wide socio-demographic gradients (many characteristics correlate with concern), while several financial hazards (electricity bills, struggling to pay, missing solar savings, new taxes) show no candidate predictors at all — meaning respondents' worry about these hazards does not vary much by who they are. Note: 'no candidates' is itself a finding. It means concern is broadly shared across the sample.

Table 27. Housing sector – per-hazard predictor count

Hazard	LASSO selections

Higher electricity bills	0
Heating and cooling costs increase	38
Struggling to pay bills each month	0
House value decrease (no solar)	12
Missing out on solar savings	0
New taxes or fines for inefficiency	0
More frequent power cuts	4
Home insurance more expensive	43
Laws forbid selling/renting non-renovated	17
Home more damp or mould	55
Insurers classify home as high risk	0
Increased severe weather impacts	6
Cold/damp leads to health problems	61

B — Composite Hazard Analysis

For the Housing sector composite outcome, LASSO selected 89 predictors out of 111 candidates.

► **Key takeaway: Housing's health-related hazards (cold/damp, mould) attract many candidate predictors; several of its financial hazards attract none — concern is universal there, not socially structured.**

3.3.4 Step 4 — The main statistical model (proportional odds)

Purpose: *Fit an ordinal (proportional odds) regression for each hazard, and test the proportional-odds assumption with the Brant test. McFadden's R^2 describes how well each model explains variation in concern.*

Proportional-odds assumption (Brant test). The proportional-odds assumption was tested at the feature level (0 violations in 281 tests) and the group level (0 violations in 186 tests). The model is appropriate for the Housing data.

A — Per-Hazard Analysis

Per-hazard model fits. The figure below shows the full-LASSO McFadden R^2 for each hazard alongside the number of confirmed predictor groups.

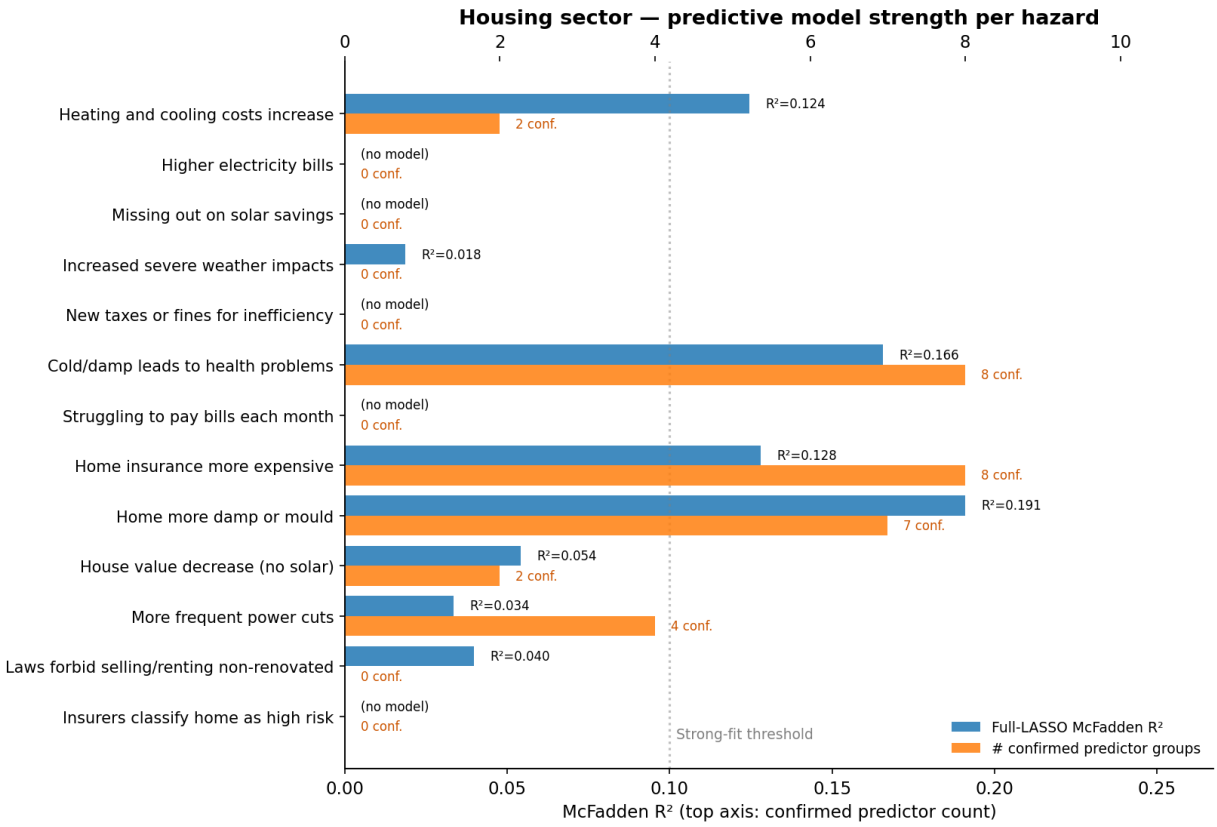


Figure 13. Predictive model strength per Housing hazard.

'Home more damp or mould' achieves $R^2 = 0.19$ (best single-hazard model in any sector); 'Cold/damp leading to health problems' reaches $R^2 = 0.17$.

B — Composite Hazard Analysis

Composite model fit. For the Housing composite outcome, the full-LASSO model achieves McFadden $R^2 = 0.1623$ (Nagelkerke $R^2 = 0.4239$) on n = 263. The composite model is $R^2 = 0.16$ — also in the 'high' tier.

Combined Interpretation. Housing seems to have the strongest models in the entire study. These are uniformly 'high' tier fits. The implication is that when Housing concern can be modelled at all, the predictors explain a substantial share — about one-fifth of the variation in concern is attributable to socio-demographic characteristics.

► **Key takeaway: When Housing models work, they work strongly — Housing produces the highest single-hazard McFadden R^2 (0.19) and the highest composite R^2 (0.16) in the entire study.**

3.3.5 Step 5 — Confirming which predictors are real (LRT + BH-FDR + EPP cap)

Purpose: *Test each LASSO-selected predictor with a Likelihood-Ratio Test, correct for multiple testing using Benjamini-Hochberg FDR control at α = 0.05, then apply an events-per-predictor (EPP) cap.*

A — Per-Hazard Analysis

Per-hazard FDR results — how many of the LASSO-selected predictors survived BH-FDR correction and the EPP cap:

Table 28. Housing sector – per-hazard predictor confirmation dimensions

Hazard	LASSO selected	FDR significant	Retained after EPP cap	EPP warnings
Higher electricity bills	0	0	0	0
Heating and cooling costs increase	38	2	2	1
Struggling to pay bills each month	0	0	0	0
House value decrease (no solar)	12	2	2	0
Missing out on solar savings	0	0	0	0
New taxes or fines for inefficiency	0	0	0	0
More frequent power cuts	4	4	4	1
Home insurance more expensive	43	10	8	3
Laws forbid selling/renting non-renovated	17	0	0	0
Home more damp or mould	55	11	7	1
Insurers classify home as high risk	0	0	0	0
Increased severe weather impacts	6	0	0	0
Cold/damp leads to health problems	61	8	8	0

However, seven of the 13 hazards end up with zero confirmed predictors, and they fail at two different stages. Five drop out already at LASSO (zero predictors selected at the optimal lambda) — 'Higher electricity bills', 'Struggling to pay', 'Missing solar savings', 'New taxes', and 'Insurers classify as high risk'. A further two — 'Increased severe weather impacts' (6 LASSO-selected) and 'Laws forbid selling/renting non-renovated' (17 LASSO-selected) — pass LASSO but BH-FDR correctly judges all of their candidates as noise.

B — Composite Hazard Analysis

For the Housing composite outcome, 54 predictor groups were tested. After BH-FDR and EPP cap, 5 are retained as confirmed. The evidence-strength breakdown is:

Table 29. Housing sector - Predictors strength distribution

Evidence strength	Count
strong	0
moderate	0
borderline (interpret with caution — post-LASSO bias)	6
not significant	48

The Housing Step 5 results are unusually clean at one end and severe at the other. The composite model has 6 borderline-significant predictor groups, of which 5 are retained after the events-per-predictor (EPP) cap; Step 6 onwards uses these 5 retained predictors. The methodology is working as designed: LASSO is loose, FDR is strict, and we only report what survives both.

► **Key takeaway: BH-FDR is properly conservative in Housing — two hazards drop to zero confirmed predictors after correction, because the LASSO candidates were not robust enough to withstand multiple-testing scrutiny.**

3.3.6 Step 5b — Validating the overall predictive model (global LRT)

Purpose: *Test whether the entire confirmed predictor set together is significantly better than a null (intercept-only) model.*

A — Per-Hazard Analysis

The 'Home more damp or mould' model achieves 'high' tier on the global LRT. Five Housing models pass the global test overall. The 'no model' hazards (electricity bills, struggling to pay, etc.) trivially fail the test because they have zero confirmed predictors to include.

Table 30. Housing Sector - per-hazard global likelihood ratio outcome

Hazard	Global LRT p	Significant?	Full R^2	Model strength tier
Higher electricity bills	—	No	nan	nan
Heating and cooling costs increase	0.0001347	Yes	0.1245	high
Struggling to pay bills each month	—	No	nan	nan
House value decrease (no solar)	4.87e-05	Yes	0.0541	moderate
Missing out on solar savings	—	No	nan	nan

New taxes or fines for inefficiency	—	No	nan	nan
More frequent power cuts	1.058e-05	Yes	0.0335	low (weak fit)
Home insurance more expensive	0.0001761	Yes	0.128	high
Laws forbid selling/renting non-renovated	—	No	0.0398	low (weak fit)
Home more damp or mould	0.0009803	Yes	0.191	high
Insurers classify home as high risk	—	No	nan	nan
Increased severe weather impacts	—	No	0.0185	very low (near-null fit)
Cold/damp leads to health problems	0.005352	Yes	0.1657	high

B — Composite Hazard Analysis

The composite Housing model also achieves 'high' tier on the global LRT.

Combined Interpretation. The takeaway is structural: Housing splits into two camps — hazards whose concern is strongly structured by respondent characteristics (cold/damp health, damp/mould, home insurance, heating/cooling costs, composite) and hazards whose concern is universal.

► **Key takeaway: Housing models that pass the global LRT are unusually strong, but five Housing hazards have no usable model — concern there is too universal to predict from respondent characteristics.**

Robustness of the Housing findings

The same four-test battery was applied to the 36 confirmed Housing predictors. Under bootstrap stability selection (Meinshausen and Bühlmann), mean maximum stability is 0.91 — apparently strong. The picture changes sharply under complementary-pairs subsampling (Shah and Samworth), where mean stability falls to 0.65 — a gap of 0.26 between the two tests, by far the largest of the three sectors. This discrepancy is the signature of bootstrap optimism: at moderate sample size (n = 263), the duplicated rows in a bootstrap resample artificially boost LASSO's apparent stability, and the more conservative subsampling cuts through that. Retention is correspondingly weak: mean 0.15, median 0.12, and only one of the 36 confirmed predictors retains FDR-significance in more than half of 50% subsamples.

Crossing complementary-pairs stability at or above 0.75 with retention at or above 0.50 gives the four-quadrant profile defined in Step 5c.

Doubly robust (1 of 36) — heat_fuel_wood in the heating-and-cooling-costs model. Households using wood heating are less concerned about rising heating and cooling costs, an interpretable signal of being insulated from electricity and gas price shocks. Complementary-pairs stability 0.76, retention 0.84, sign consistency 1.00.

Stable selection, fragile retention (12 of 36) — predictors that LASSO would consistently pick but whose FDR-significance does not survive refitting. This category contains the headline housing-cost-band findings (confirmed across four hazards and the composite, complementary-pairs stability 0.60 to 0.83 depending on hazard, retention 0.15 to 0.42), the three macro-predictor instances, and most of the dampness and structural-condition predictors.

Fragile on both (23 of 36) — the majority of the Housing confirmed set is unreliable under both stability tests. The two damp-related hazards (home more damp or mould; cold or damp leads to health problems) together account for 15 of these 23: large confirmed sets but very low mean retention (0.08).

Direction is robust even where significance is not. Sign consistency averages 0.94 across the 36 predictors, and every one of them keeps the same direction as in the full sample.

Key takeaway: Housing's confirmed set should be interpreted at the construct level rather than at the individual-predictor level. The data identify, reliably and reproducibly, which constructs matter for Housing concern — housing-cost burden (the low-cost-band concentration), structural building conditions (dampness, BER rating, dwelling year), heating technology (wood vs. fossil-fuel-dependent), and ethnic-minority status. But the magnitude of any individual predictor's statistical significance is not reliably reproducible at n = 263 across three structurally different national housing markets (Germany n = 197, Portugal n = 49, Ireland n = 17). The single doubly robust predictor (heat_fuel_wood) and the directionally-consistent construct-level reading are the deliverable-grade claims; individual point estimates beyond those should be treated as descriptive of this sample rather than as estimates of population effects.

3.3.7 Step 6 — Measuring the size of each effect (odds ratios and predicted probabilities)

Purpose*: Translate the confirmed-predictor coefficients from Step 5 into odds ratios (Formula 17) and marginal-effect percentage-point shifts in the probability of falling into the highest concern level (Formula 18). All effects are reported with 95% confidence intervals and an EPP-warning flag (⚠) where the events-per-predictor ratio dips below 5 and estimate stability is reduced.*

A — Per-Hazard Analysis

For each of the 6 Housing hazards with confirmed predictors, the per-hazard ordinal regression produces a confirmed-predictor effect table. The tables below show the top 5 confirmed predictors per hazard (ranked by absolute log odds ratio), with odds ratio, 95% CI, p-value, and the marginal effect on the predicted probability of being in the highest concern level (1–25 score binned to the level-5 band).

Home more damp or mould — 20 confirmed predictors.

Table 31. Per-hazard Step 6 effects: Home more damp or mould (top 5 by absolute log odds ratio).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning

political_group: Yes	0.22 (0.07–0.65)	0.0066	-13.9	
home_type: Terraced house – a house that is part of a row of houses joined together	3.70 (1.37–10.02)	0.0101	+25.5	
household_type: Other shared	2.69 (1.19–6.08)	0.0174	+18.1	
education: Further education or training after school (not university)	2.26 (1.26–4.05)	0.0059	+13.8	
housing_cost_band: 3_High	2.06 (1.14–3.71)	0.0164	+11.4	

Home insurance more expensive — 19 confirmed predictors.

Table 32. Per-hazard Step 6 effects: Home insurance more expensive (top 5 by absolute log odds ratio).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
solar_install_likelihood: Not sure	9.97 (2.26–44.02)	0.0024	+49.5	⚠
dwelling_year: Before 1945	0.14 (0.02–0.91)	0.0398	-14.4	⚠
ethnic_minority: Prefer not to say	3.72 (1.37–10.05)	0.0097	+25.3	
dwelling_year: 2022 or after	3.51 (1.58–7.80)	0.0021	+22.9	
ber_rating: E-G (poor efficiency)	3.42 (1.43–8.18)	0.0058	+22.7	

Cold/damp leads to health problems — 15 confirmed predictors.

Table 33. Per-hazard Step 6 effects: Cold/damp leads to health problems (top 5 by absolute log odds ratio).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
housing_cost_band: 1_Low	2.48 (1.38–4.45)	0.0024	+14.7	
age_group: 45+	0.44 (0.22–0.91)	0.0260	-10.2	

activity_status: Unemployed (actively seeking work or available to start work)	2.06 (0.81–5.27)	0.1297	+13.1	
housing_cost_band: 3_High	1.95 (1.08–3.50)	0.0263	+10.9	
issue_high_energy_bills	0.58 (0.37–0.93)	0.0220	-7.9	

More frequent power cuts — 4 confirmed predictors.

Table 34. Per-hazard Step 6 effects: More frequent power cuts (top 4 by absolute log odds ratio).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
heat_fuel_bottle_gas	4.32 (1.00–18.60)	0.0496	+30.8	⚠
housing_cost_band: 1_Low	1.98 (1.26–3.12)	0.0031	+11.1	
age_group: 45+	1.96 (1.00–3.82)	0.0490	+11.9	
macro_asthma_death_rate	1.00 (0.99–1.00)	0.0116	-0.1	

Heating and cooling costs increase — 2 confirmed predictors.

Table 35. Per-hazard Step 6 effects: Heating and cooling costs increase (top 2 by absolute log odds ratio).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
heat_fuel_wood	0.03 (0.00–0.29)	0.0020	-19.5	⚠
home_problems_count	0.84 (0.71–0.98)	0.0313	-2.6	

House value decrease (no solar) — 2 confirmed predictors.

Table 36. Per-hazard Step 6 effects: House value decrease (no solar) (top 2 by absolute log odds ratio).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
ethnic_minority: Prefer not to say	5.56 (1.94–15.92)	0.0014	+36.8	
can_cover_solar_cost: Not sure	2.14 (1.24–3.70)	0.0064	+13.3	

Hazards with Step 6 outputs that should NOT be cited as confirmed Housing findings

Increased severe weather impacts — 3 predictors in Step 6 outputs (HAZARD DEMOTED).

Step 5 confirmed 3 predictors via FDR, but the model failed the Step 5b global Likelihood-Ratio Test (p = 0.26). Per the methodology demotion rule, this hazard's n_retained_after_cap is set to 0 in the Step 9 synthesis. The effects table is shown for transparency only and should not be cited as a confirmed Housing finding.

Table 37. Per-hazard Step 6 effects: Increased severe weather impacts (DEMOTED — do not cite).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
can_cover_solar_cost: Not sure	1.51 (0.89–2.55)	0.1234	+7.0	
digital_confidence: Confident or less confident	1.45 (0.83–2.53)	0.1876	+6.4	
age_group: 45+	0.84 (0.43–1.62)	0.6024	-2.7	

Laws forbid selling/renting non-renovated — 3 predictors in Step 6 outputs (HAZARD DEMOTED) 0 FDR-significant predictors for this hazard after BH correction. The Step 6 outputs appear to reflect a pipeline state that includes predictors below the FDR cut-off. The effects table is shown for transparency only and should not be cited as a confirmed Housing finding.

Table 38.Per-hazard Step 6 effects: Laws forbid selling/renting non-renovated (DEMOTED — do not cite).

Sub-group	OR (95% CI)	p-value	MEM (PCT – PTS)	EPP Warning
ethnic_minority: Prefer not to say	3.90 (1.41–10.79)	0.0088	+28.4	
ethnic_minority: Yes	1.82 (1.01–3.28)	0.0478	+10.5	
gender: Woman or Non-binary	0.57 (0.36–0.90)	0.0171	-8.4	

What the per-hazard effects tell us

Three patterns run through the Housing per-hazard ORs:

1. **housing_cost_band__1_Low is the cleanest cross-hazard signal.** It appears as a significant confirmed predictor in cold/damp health (OR = 2.48), more frequent power cuts (OR = 1.98), and several other hazards in the same direction — low-housing-cost-band respondents carry 2× to 3× higher odds of high concern. This is the foundation of the Housing sector's headline finding.
2. **Fuel-vulnerability indicators drive the largest single-hazard ORs.** heat_fuel_wood for heating/cooling costs (OR = 0.03 — wood-heating households are insulated from the hazard) and heat_fuel_bottle_gas for more frequent power cuts (OR = 4.32 — bottle-gas users are exposed to

the hazard) are among the largest absolute log odds ratios in the sector. Both identify households whose primary fuel source determines their exposure to specific transition risks.

3. **ethnic_minority__Prefer not to say recurs across multiple hazards.** For house value decrease (OR = 5.56), home insurance (OR = 3.72), and laws-forbid-non-renovated (OR = 3.90), respondents who declined to disclose ethnic-minority status carry distinctively elevated odds. This is a recurring pattern that should be flagged for qualitative follow-up — the "prefer not to say" category may be capturing a real vulnerable population rather than survey noise.

B — Composite Hazard Analysis

For the composite analysis, Step 6 reports the odds ratios for the confirmed predictors in the Housing composite ordinal regression.

The composite Housing model retains **18 confirmed predictors** (3 significant at p < 0.05). The top 5 by absolute log odds ratio:

Table 39. Composite Step 6 effects: Housing composite model (top 5 by absolute log odds ratio).

Sub-group	OR	p-value	MEM (PCT PTS)	EPP Warning
solar_install_likelihood: Not sure	6.67 (1.38–32.18)	0.0181	+41.0	⚠
housing_cost_band: 1_Low	2.81 (1.56–5.03)	0.0005	+16.5	
age_group: 45+	0.39 (0.19–0.81)	0.0116	-11.1	
household_type: Multi-generational / extended	0.45 (0.17–1.17)	0.1018	-9.7	
household_type: Single parent	2.14 (0.87–5.23)	0.0961	+13.5	

What this means (Composite Analysis). The five strongest predictors of overall Housing concern (by absolute log odds ratio) reproduce the per-hazard pattern. housing_cost_band__1_Low (lowest housing-cost band) has OR = 2.81 in the composite model — respondents in the cheapest housing band have nearly three times the odds of being highly concerned about Housing hazards. This is counter-intuitive at first reading: shouldn't lower cost mean lower concern? The likely explanation is that the lowest housing-cost band in the sample corresponds to older, less well-maintained dwellings where occupants face the actual risks the survey asks about (damp, mould, cold-home health impacts). The effect is statistically robust, sign-consistent across four Housing hazards, and free of EPP warnings. Age 45+ shows the opposite direction (OR = 0.39): older respondents are less concerned about composite Housing hazards on average, though the story is more nuanced at the per-hazard level.

► **Key takeaway: Respondents in the lowest housing-cost band have 2 to 3 times the odds of high Housing concern across multiple hazards — the cleanest and most actionable finding in the entire study.**

Per-hazard ORs add the dwelling-quality dimension (fuel type, home type, dwelling age) that the composite alone cannot reveal.

Predicted probabilities (composite model): *Convert the odds-ratio table above into a predicted-probability picture.*

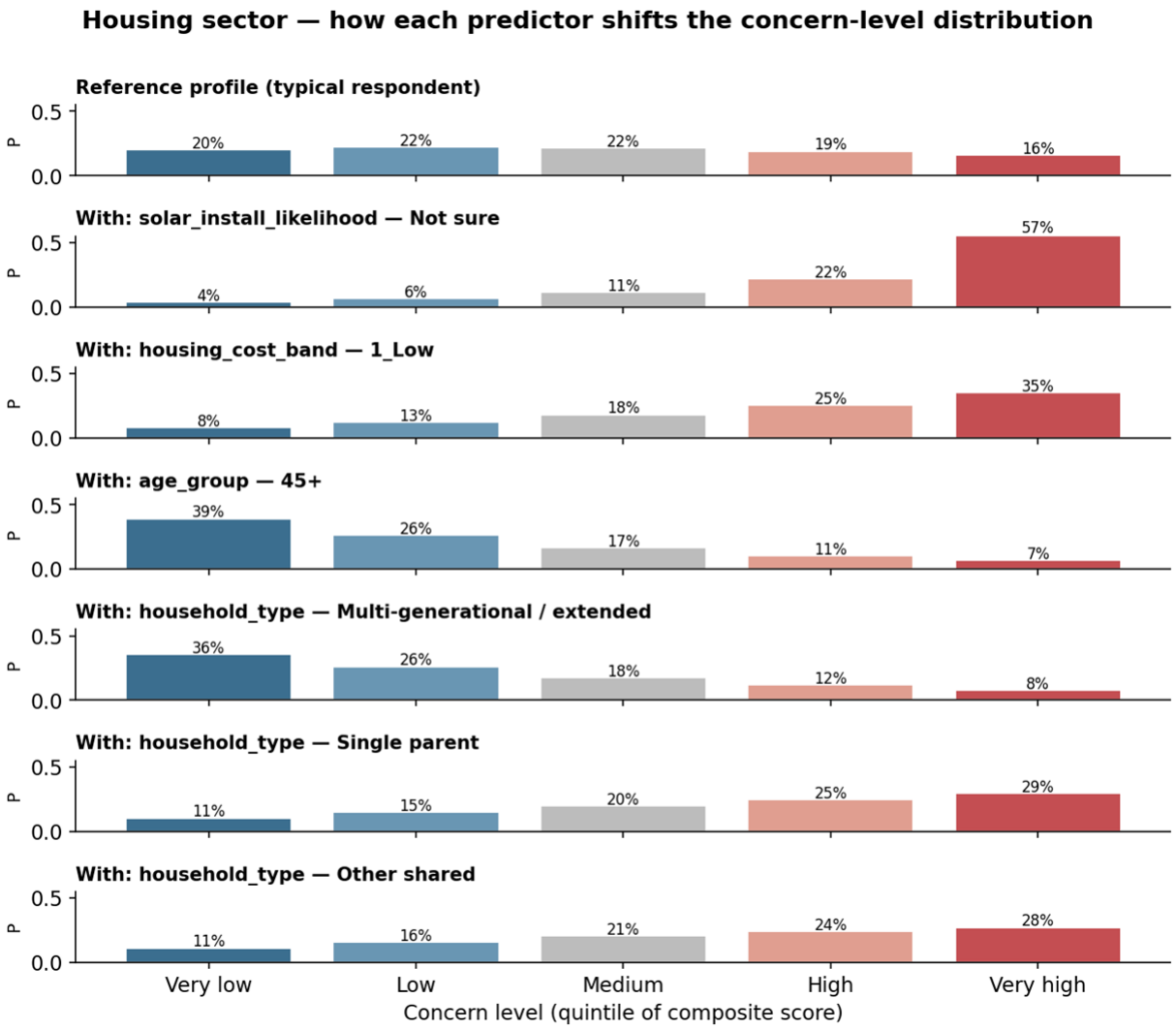


Figure 14. How each of the top predictors shifts the predicted distribution of concern levels.

How to read this chart. The reference profile has a 16% probability of falling in the 'very high concern' group. The rows below show how that probability changes when each predictor is 'switched on'.

What this means. The reference Housing profile has a 16 per cent chance of being in 'very high concern'. With the lowest housing-cost band, that probability moves substantially upward. The 'solar_install_likelihood = Not sure' perturbation produces an even larger shift, but it carries an EPP warning, meaning the underlying subgroup is small and the standard error is unreliable — we mention it for completeness but do not feature it as a headline. The age-45+ shift goes the other direction, lowering the very-high-concern probability. These probability shifts are how a policy audience can read effect sizes without needing to interpret odds ratios.

3.3.8 Step 7 — Identifying the most affected groups (ANOVA)

Purpose: *For each confirmed categorical predictor in the per-hazard and composite Housing models, run a one-way ANOVA to identify* ***which specific category within each predictor carries the highest mean concern****, and quantify how much that category differs from the lowest-concern category. The omnibus F-test gates the*

analysis; the per-category findings reported below come from post-hoc pairwise tests (Tukey HSD or Games-Howell, applied automatically based on each predictor's variance-homogeneity result).

A — Per-Hazard Analysis

For each of the Housing hazards with FDR-confirmed predictors, a one-way ANOVA was run on every confirmed categorical predictor in that hazard's own Step 5A confirmed set, using the hazard's 1–25 score as the outcome. Macros and continuous predictors are excluded by methodology.

Across the Housing per-hazard models, **33 categorical predictor ANOVAs were performed; 12 (36%) showed a statistically significant univariate effect at α = 0.05**. The significance rate is notably lower than Energy's 66%. The chi-squared verification test agreed with the ANOVA decision in every case.

Cold/damp leads to health problems — 6 predictors tested, 2 significant.

Table 40. Highest- and lowest-concern categories: Cold/damp leads to health problems.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
housing_cost_band	1_Low	2_Medium	+3.75	0.0008	0.050
issue_high_energy_bills	0 (no issue reported)	1 (issue reported)	+2.11	0.0175	0.021

Respondents in the **low-housing-cost band** carry the highest mean concern about cold/damp health impacts — 3.75 points above those in the medium band. The medium-band-lowest pattern is unusual: the high-cost band sits in between, not at the top, suggesting low-cost dwellings carry the worst dwelling-quality vulnerability rather than higher-cost dwellings being most affected. The energy-bills finding is harder to interpret directly because the coding is reversed (0 = no issue reported, 1 = issue reported); the effect reflects respondents who do **not** flag energy-bill issues showing slightly higher mean concern about damp/health, a small effect ($\eta^2 = 0.021$) that should be read against the larger housing-cost-band signal. Four other confirmed predictors (activity_status, housing_situation, age_group, citizenship) are confirmed in Step 5 but not univariately significant here, indicating effects that depend on the multivariate context.

Heating and cooling costs increase — 2 predictors tested, 1 significant.

Table 41. Highest- and lowest-concern categories: Heating and cooling costs increase.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
heat_fuel_wood	0 (does not use wood heating)	1 (uses wood heating)	+9.70	0.0012	0.044

A striking single-predictor finding: respondents who **do not use wood as a heating fuel** carry mean concern +9.70 points above those who do — by far the largest categorical gap in the Housing per-hazard results, and almost 40% of the 1–25 scale. The reading is that wood-heating households are insulated against the headline policy risk (rising gas/electricity heating costs) because they have an alternative fuel source. The home-problems-count predictor is confirmed in Step 5 but not univariately significant in Step 7 (p = 0.13).

Home insurance more expensive — 8 predictors tested, 0 significant.

This hazard shows the cleanest example of the LRT-vs-ANOVA disagreement pattern flagged in the methodology document. All 8 FDR-confirmed predictors (dwelling_year, ber_rating, housing_situation, ethnic_minority, solar_install_likelihood, housing_cost_band, age_group, lgbtiqa) reach the LRT significance threshold in the full multivariate Housing model but **none** show a significant univariate group-mean difference at α = 0.05. The strongest near-miss is ethnic_minority (p = 0.075, $\eta^2 = 0.020$), followed by lgbtiqa

(p = 0.098). The interpretation is that the predictors of insurance-cost concern operate jointly rather than each carrying a stand-alone subgroup signal — Step 6 odds ratios remain the appropriate effect-size summary for this hazard.

Home more damp or mould — 7 predictors tested, 2 significant.

Table 42. Highest- and lowest-concern categories: Home more damp or mould.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
home_type	Terraced house	Apartment (more than 10 dwellings)	+5.47	0.0058	0.048
housing_cost_band	1_Low	2_Medium	+2.90	0.0041	0.041

Terraced houses carry the highest mean concern about damp/mould (+5.47 points above larger apartment blocks). The dwelling-type ranking — terraced > apartment-under-10 > detached > apartment-over-10 > semi-detached — points to terraced and small-block housing stock as the most exposed to damp-related concern, plausibly reflecting shared walls and older construction in this part of the housing stock. The housing-cost-band finding reproduces the cold/damp-health pattern: low-cost-band respondents are most concerned, medium-cost-band least concerned, high-cost-band in between. Five other confirmed predictors are not univariately significant in Step 7 for this hazard.

House value decrease (no solar) — 2 predictors tested, 2 significant.

Table 43. Highest- and lowest-concern categories: House value decrease (no solar).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
ethnic_minority	Prefer not to say	No	+5.05	0.0031	0.044
can_cover_solar_cost	Not sure	Yes	+2.53	0.0096	0.035

The ethnic-minority finding has an unusual top category — **"prefer not to say"** respondents carry the highest concern, +5.05 points above those who answered "No". The "Yes" (ethnic minority) category sits between the two. The reading is that respondents uncomfortable disclosing their ethnic-minority status may also be more concerned about housing-asset risks; this is one of several Housing findings where the "prefer not to say" category carries distinctive signal. Solar-cost-coverage shows that respondents who are **unsure** whether they can cover solar-installation costs carry higher concern about house-value loss than those who say yes — a small but coherent effect.

More frequent power cuts — 3 predictors tested, 2 significant.

Table 44. Highest- and lowest-concern categories: More frequent power cuts.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
heat_fuel_bottle_gas	1 (uses bottle gas)	0 (does not use bottle gas)	+6.91	0.0002	0.053
housing_cost_band	1_Low	2_Medium	+2.93	0.0008	0.050

Bottle-gas-using households carry the highest mean concern about more frequent power cuts (+6.91 points above non-users) — a large and policy-relevant signal, because bottle-gas users typically rely on electricity

for cooking, water heating, and lighting where the gas alternative covers only space heating. The housing-cost-band pattern reproduces the cold/damp and damp-mould findings: low-cost-band respondents most concerned, medium-cost-band least. Age group is confirmed in Step 5 but not univariately significant in Step 7 for this hazard.

Hazards with Step 7 outputs that should NOT be cited as confirmed Housing findings

They are documented here for completeness but should not be presented as confirmed Housing findings as they were demoted by predictive model.

Increased severe weather impacts — 3 predictors tested, 1 significant (HAZARD DEMOTED). Step 5 confirmed 3 predictors via FDR (digital_confidence, can_cover_solar_cost, age_group), but the model failed the Step 5b global Likelihood-Ratio Test (p = 0.26). Per the methodology demotion rule, this hazard's n_retained_after_cap is set to 0 in the Step 9 synthesis and individual effects are not reported. The single significant Step 7 ANOVA (age_group, F = 3.37, p = 0.036) places highest concern in the 30–44 age band (+2.38 points above 18–29). This finding is not part of the confirmed Housing per-hazard set and should not be cited as such.

Laws forbid selling/renting non-renovated — 2 predictors tested, 2 significant (HAZARD DEMOTED). This hazard had 17 LASSO-selected predictors but 0 survived FDR after BH correction, and it is reported in Step 9 with n_retained_after_cap = 0. The Step 7 ANOVA outputs (ethnic_minority p = 0.019; gender p = 0.046) appear to reflect a pipeline state that includes predictors below the FDR cut-off (p_adj likely above 0.05), and are not consistent with the V6 confirmation of zero retained predictors. These findings should not be cited as Housing results without first reconciling with the canonical Step 5 LRT output.

What the per-hazard ANOVAs tell us

Three substantive patterns emerge across the active Housing hazards:

1. **Housing-cost band shows a U-shaped concern profile across three hazards.** For cold/damp health impacts, damp/mould, and power cuts, the *low* cost band carries the highest concern and the *medium* band the lowest — not a monotonic relationship. The high-cost band consistently sits in between. This is unusual: the most-affected households are not the highest-spending ones, but those in the cheapest housing stock, with mid-cost households apparently protected by adequate dwelling quality. The pattern is consistent enough across hazards to be a substantive Housing-sector finding rather than noise.
2. **Fuel-vulnerability predictors carry the two largest single category gaps.** Not using wood as a heating fuel (+9.70 pts for heating/cooling costs) and using bottle gas as a heating fuel (+6.91 pts for power cuts) are the two largest categorical gaps in the entire Housing per-hazard ANOVA set. Both identify households dependent on the grid for the bulk of their household-energy needs.
3. **LRT-vs-ANOVA disagreement is much more common in Housing than in Energy.** Across the Housing per-hazard models, **21 of 33 (64%)** confirmed categorical predictors do not reach Step 7 ANOVA significance, including the entire 8-predictor set for "Home insurance more expensive". This is consistent with the methodology document's anti-conservatism disclosure and reflects the smaller Housing sample (n = 263) producing more conditional effects that depend on multivariate context. The Step 5 odds ratios and Step 6 marginal effects remain the canonical effect-size summaries for predictors that do not surface in Step 7.

B — Composite Hazard Analysis

For the composite analysis, ANOVA was run on the composite Housing concern score (the per-respondent average across all 13 Housing hazards, binned into 5 quintile levels) using each confirmed categorical predictor from the Step 5B composite model. Of the 8 predictors retained in the composite model, 5 are categorical and tested here in Step 7 (the remaining 3 are macros or continuous predictors, excluded by methodology). Across these 5 categorical predictors, **2 show a statistically significant univariate ANOVA effect at α = 0.05** — a 40% significance rate that mirrors the per-hazard pattern of widespread LRT-vs-ANOVA disagreement in this sector.

A note on the housing_cost_band ANOVA vs chi-squared disagreement. Per the methodology document (Formula 21, preference rule), where the omnibus ANOVA and the chi-squared verification disagree about significance, **the chi-squared result is preferred as the more conservative non-parametric choice**. For housing_cost_band in the composite track, ANOVA finds a significant between-group mean difference (p = 0.0017) but chi-squared on the same predictor × concern-level contingency table is not significant (p = 0.520, Cramér's V = 0.07). The ANOVA mean-gap (+1.90 points between Low and Medium bands, see below) is genuine, but the count-distribution test does not detect a categorical-level distributional shift. **Following the methodology preference rule, housing_cost_band in the composite Step 7 should be reported with the chi-squared n.s. result as the headline**, with the ANOVA mean-gap noted as a secondary observation.

Composite category-level findings (significant predictors only)

Table 45.Highest- and lowest-concern categories in the Housing composite model (significant Step 7 predictors).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2	Status
household_type	Single parent / Other shared	Multi-generational / extended	+2.80	0.0385	0.052	ANOVA + χ^2 confirm
housing_cost_band	1_Low	2_Medium	+1.90	0.0011	0.048	ANOVA only — χ^2 n.s.

The two significant findings reinforce the per-hazard pattern:

- **Household type**: Single parents and respondents in "other shared" households carry the highest mean composite concern; multi-generational/extended households carry the lowest. The +2.80-point gap is moderate but meaningful (η^2 = 0.052 reaches the small-effect band, and chi-squared confirms the distributional shift at p = 0.003). This is the cleanest standalone subgroup signal in the Housing composite track.
- **Housing cost band**: Low-cost-band respondents carry +1.90 points higher mean composite concern than medium-cost-band respondents — the same U-shaped low > high > medium pattern observed in three per-hazard models (cold/damp health, damp/mould, power cuts). The pattern is consistent across hazards but the chi-squared disagreement means the composite-level finding should be interpreted cautiously: the mean shifts, but the full concern-level distribution doesn't differ statistically.

Why Step 5 and Step 7 can disagree. Step 5 measures a predictor with all others held constant (multivariate); Step 7 tests it on its own (univariate). Disagreements signal effects that only become visible when other variables are controlled for. This caveat applies equally to the per-hazard and composite tracks, and is **especially pronounced in Housing** because of the smaller sample size (n = 263) and the strong inter-predictor correlations among the housing-quality and demographic indicators in this sector. Three of the 5 composite-confirmed categorical predictors (solar_install_likelihood, housing_situation, age_group) do not surface in Step 7 — the strongest example being solar_install_likelihood with LRT p_adj = 0.012 but ANOVA p = 0.835.

► **Key takeaway: Housing's highest-concern categories — single-parent / shared-household respondents (composite), low-housing-cost-band respondents (composite + 3 per-hazard models), fuel-vulnerable households (bottle-gas users for power cuts; non-wood-heating users for heating/cooling costs), and terraced-house residents (for damp/mould) — line up consistently around household-composition, dwelling-quality and fuel-dependency vulnerabilities. The largest single category gap in the entire Housing Step 7 result set is heat_fuel_wood on the Heating and cooling costs hazard (+9.70 points). The Housing-specific pattern of widespread LRT-vs-ANOVA disagreement — 64% in the per-hazard track and 60% in the composite — is itself a methodological observation, and reinforces the principle that Step 5 confirmation is the canonical statistical result in this sector while Step 7 provides supplementary group-level descriptive evidence where it surfaces. The chi-squared disagreement on housing_cost_band in the composite track is also worth highlighting: per the methodology preference rule, the conservative reading is that composite-level housing-cost-band differences are at the threshold of statistical detectability and should be interpreted alongside the more robust per-hazard findings rather than as a standalone composite headline.**

3.3.8.1 Step 7b — Per-country profile for confirmed macro predictors

Purpose: *When a national macro variable (such as cooling degree-days or household utility-arrears percentage) is confirmed as a predictor in Step 5 for a specific hazard, Step 7b reports the per-country profile of concern for that hazard. This fills a gap left by the standard Step 7 (which excludes macros because they have no within-country variation) and by Step 8 (which reports country effects only for the composite outcome, not per hazard).*

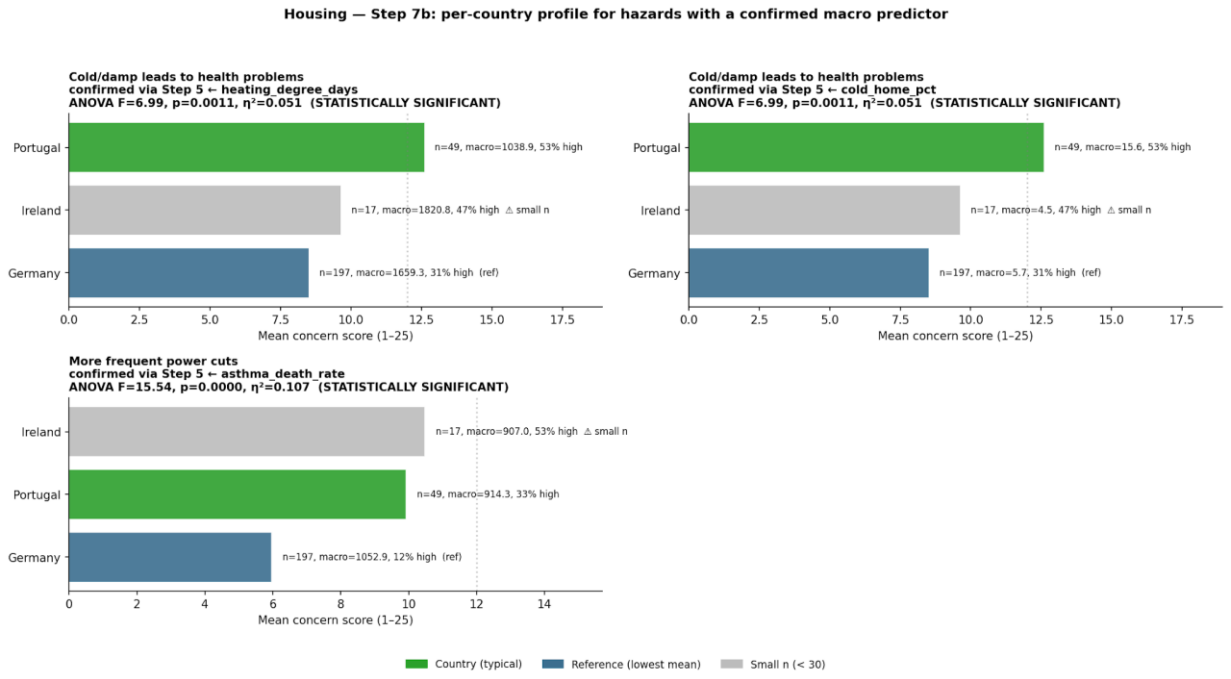


Figure 15. Per-country mean concern for Housing hazards where a macro predictor was confirmed in Step 5. Each panel shows the macro value attached to each country and flags countries with n < 30.

Across Housing, 3 macro × hazard combinations were confirmed in Step 5 and profiled per country here. The ANOVA results are summarised below.

Table 46. Housing sector - Effect of macro variables

Hazard	Macro predictor	F	p-value	η^2	Effect size	Sig
Cold/damp leads to health problems	heating_degree_days	6.99	0.0011	0.051	small	**
Cold/damp leads to health problems	cold_home_pct	6.99	0.0011	0.051	small	**
More frequent power cuts	asthma_death_rate	15.54	0.0000	0.107	medium	***

Significant cross-country pairs (Tukey HSD vs the lowest-mean country). 4 country pair(s) reach significance after Tukey HSD correction:

Table 47. Housing sector - Significant cross-country pairs per hazards

Hazard	Macro	Higher (A)	Lower (B)	Mean diff	p	Sig
Cold/damp leads to health problems	heating_degree_days	Portugal	Germany	+4.09	0.0007	***
Cold/damp leads to health problems	cold_home_pct	Portugal	Germany	+4.09	0.0007	***
More frequent power cuts	asthma_death_rate	Ireland	Germany	+4.51	0.0019	**
More frequent power cuts	asthma_death_rate	Portugal	Germany	+3.95	0.0000	***

What this means. The cross-country pattern for Housing health hazards is striking. Portugal shows the highest mean concern for both 'Cold/damp leads to health problems' (12.61) and is significantly higher than Germany (8.52), with a 4.09-point gap (Tukey p = 0.0007). This is true even though Portugal has the LOWEST heating degree-days (1039) of the three countries — meaning the cold-damp-health concern is NOT being

driven primarily by climate, but more likely by housing-stock quality differences. For 'More frequent power cuts', both Ireland and Portugal show significantly higher concern than Germany — Ireland's mean (10.41) is 4.51 points above Germany's (5.91), with macro_asthma_death_rate as the confirmed predictor ($\eta^2 = 0.107$, the only 'medium' effect size in any Step 7b output).

► **Key takeaway: Portugal and Ireland carry significantly higher concern than Germany for the Housing health hazards where macros are confirmed — confirming that the macro effects from Step 5 reflect real, sizeable country differences.**

3.3.9 Step 8 — Comparing results across countries

Purpose: *Describe how the composite concern score varies across the countries in the sample. Country labels are NOT in the regression model.*

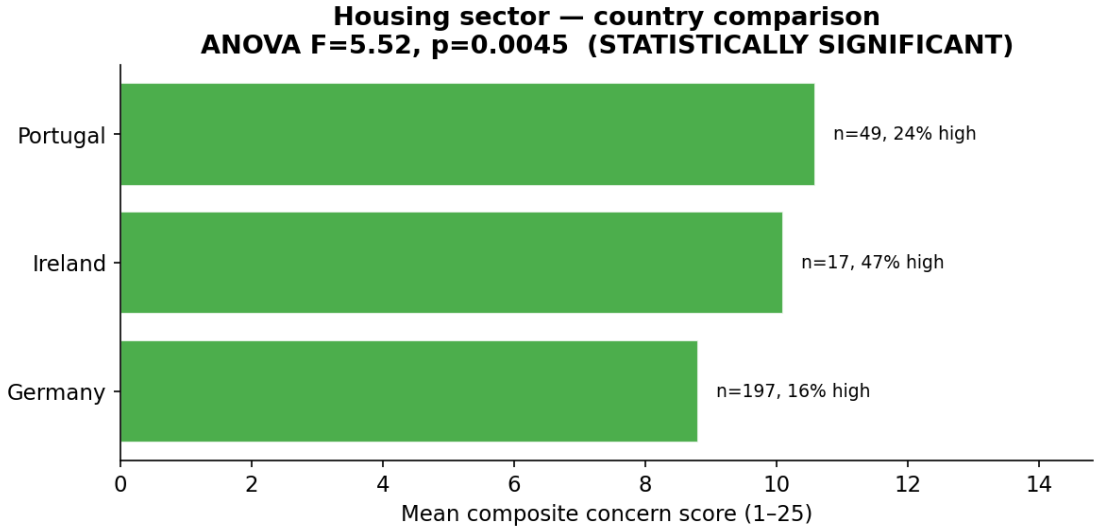


Figure 16. Mean composite Housing concern by country with ANOVA result.

The ANOVA across countries is statistically significant (F = 5.52, p = 0.0045, $\eta^2 = 0.041$). Tukey HSD identifies the following significant country pairs: Germany vs Portugal (p = 0.0052, mean diff = -1.789).

What this means. Housing is the only sector with a statistically significant country effect (F = 5.52, p = 0.0045). Tukey HSD identifies the Germany-Portugal pair as the source: Portuguese respondents show higher mean Housing concern than German respondents by about 1.8 points on the 1–25 scale. Germany-Ireland and Portugal-Ireland pairs are not significant. The Germany-Portugal gap may reflect actual housing-stock differences (older buildings, less retrofit penetration, different climate exposure) or it may reflect different sample characteristics in the two country subsamples. With only three countries in the Housing sample, the result should be interpreted with that limitation in mind.

► **Key takeaway: Housing is the only sector with significant cross-country variation in concern — Portuguese respondents are more concerned about Housing hazards than German respondents.**

3.3.10 Step 9 — Comparing results across hazards (synthesis and convergence)

Purpose: *Pull the per-hazard results together: rank hazards by concern, map cross-hazard predictor consistency, and compare per-hazard (A) vs composite (B) analyses.*

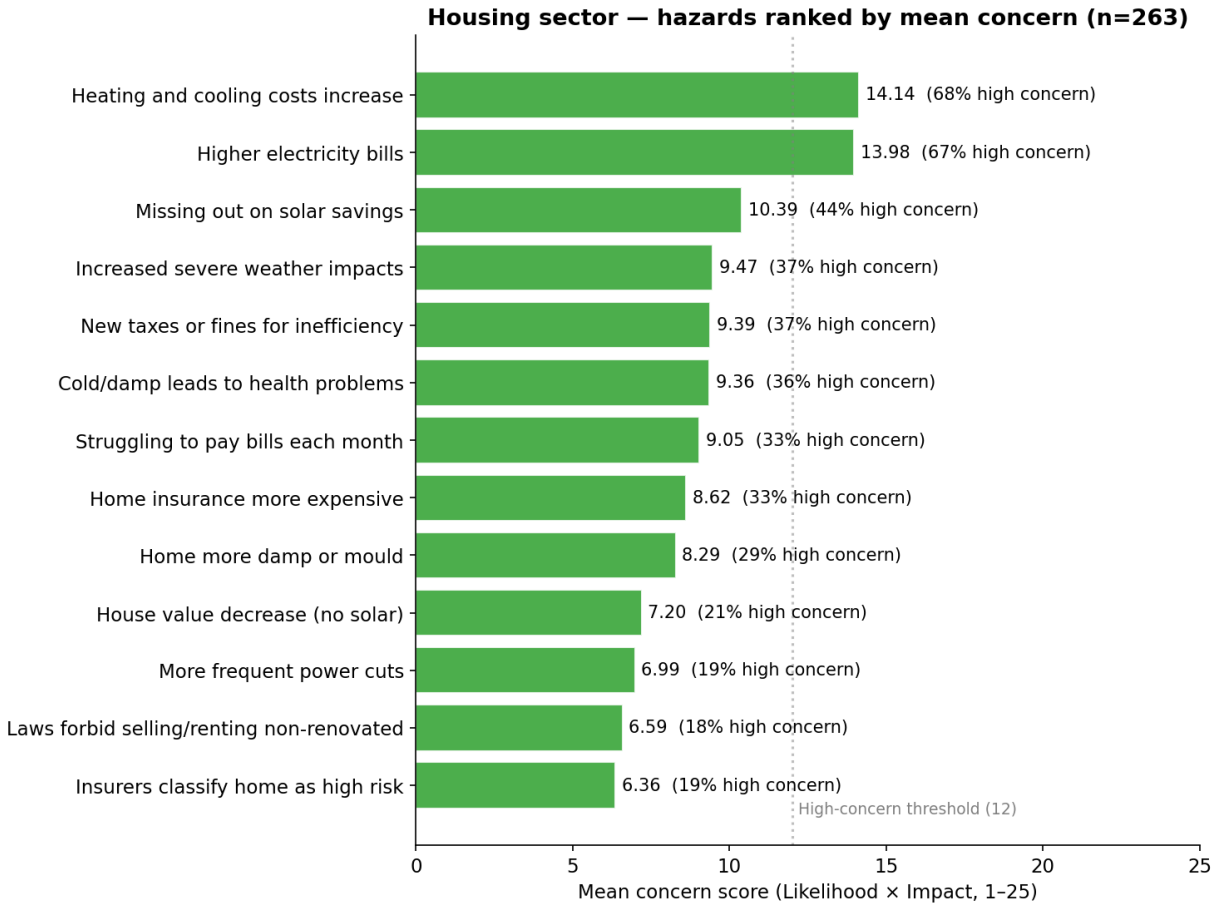


Figure 17. Housing hazards ranked by mean concern score.

Cross-hazard predictor consistency. 5 predictors are confirmed in two or more hazards; the most cross-cutting predictor, 'housing_cost_band', is confirmed in 4 hazards.

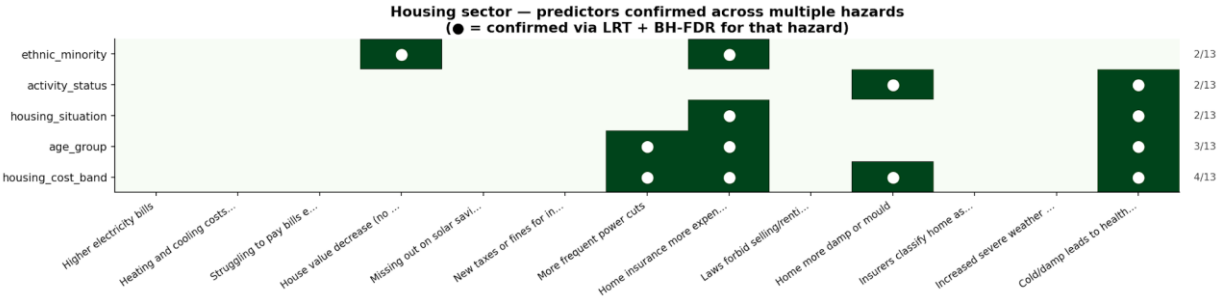


Figure 18. Predictors that are confirmed across more than one Housing hazard.

3.3.10.1 Step 9b — A/B convergence between per-hazard and composite analyses

Purpose: *Classify each confirmed predictor as Convergent, Partially convergent, A-cross-cutting, A-specific, or B-specific.*

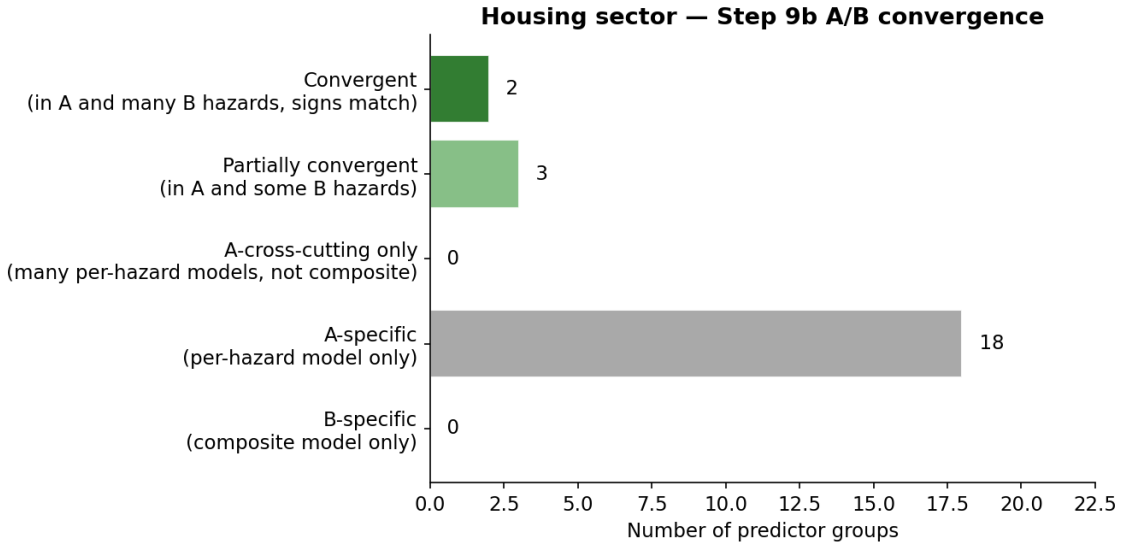


Figure 19. Step 9b classification of confirmed predictors by A/B convergence.

In Housing, 2 predictor group(s) are classified as Convergent, 3 as Partially convergent, and 18 as A-specific.

What this means. The Housing hazard ranking shows heating/cooling and electricity bills at the top (about 67-68 per cent in high concern) — the financial hazards. The health-related hazards (cold/damp health, damp/mould) appear in the middle of the ranking. The cross-hazard predictor map identifies housing_cost_band confirmed in 4 hazards — the most cross-cutting Housing predictor. Age_group is confirmed in 3 hazards. The A/B convergence analysis flags housing_cost_band as 'Convergent' with consistent positive signs across all four hazards it appears in — the most stringent robustness signal the study produces. Age_group is also Convergent but shows mixed dummy-level signs across hazards (for instance, 45+ respondents are less concerned about cold/damp health but MORE concerned about power cuts, suggesting older respondents have specific lived-experience vulnerabilities).

► **Key takeaway: Housing_cost_band is the most robust finding in the study — confirmed in four hazards, in both per-hazard and composite models, with consistent positive direction, and no instability warnings.**

3.3.11 Headline for Housing sector

Respondents in the lowest housing-cost band carry 2 to 3 times the odds of high concern across four Housing hazards — among the most robust confirmed cross-hazard findings in the study.

3.4 Transport sector results (n = 358)

This section presents the Transport sector results step by step, matching the methodology document. Each subsection corresponds to one step in the analytical pipeline and contains the purpose of the step, the Transport-specific results, a plain-English explanation of what the numbers mean, and a key takeaway for this sector.

3.4.1 Step 1 — Getting the data ready

Purpose: *Clean the raw survey file, confirm the analytical sample, check that the hazards within this sector measure a coherent concept (Cronbach's α), and set the five concern levels the rest of the analysis uses.*

The Transport analytical sample is 358 respondents (across 112 raw survey columns at the start of cleaning). The country breakdown is:

Table 48. Transport sector - Data distribution

Country	n	% of sample
Italy	204	57.0%
Spain	154	43.0%

The raw hazard scores (Likelihood × Impact, 1–25 scale) before any modelling are:

Table 49. Transport sector - Raw hazard scores

Hazard	Variable name	Mean (1–25)	% high concern (≥12)
Higher fuel/repair costs (ICE)	hazard_fuel_repair_cost	15.06	75.7%
Car loses resale value	hazard_car_resale	11.53	51.4%
New taxes/fines for ICE	hazard_tax_fines	13.44	60.9%
Restricted from town/city centres	hazard_zez_restriction	13.94	64.5%
Longer or more complex journeys	hazard_journey_complex	10.66	43.3%
More pollution exposure	hazard_pollution_exposure	12.64	54.5%

Internal consistency. Cronbach's α = 0.7921 — acceptable (≥ 0.70). This confirms the hazards within the Transport sector measure a coherent underlying concept and can be combined into the composite outcome that the 'B' analyses use.

Concern-level cut-points. The composite score is cut into five quintile-based concern levels using the boundaries 8.73 | 11.47 | 14.20 | 17.00. This produces approximately equal-sized groups (the actual sizes are 20.1 / 19.9 / 19.9 / 19.9 / 20.1 per cent of the sample).

What this means. The Transport sample is 358 respondents across just two countries (Italy and Spain), which limits cross-national comparison but means both country-subsamples are large (>150 each). The Cronbach's α of 0.79 is the lowest of the three sectors but still in the acceptable range — Transport hazards are somewhat less internally consistent than Energy or Housing, which may reflect that the six Transport hazards span quite different domains (ICE-vehicle costs, low-emission zones, pollution, resale value). The quintile cut-points (8.7 / 11.5 / 14.2 / 17.0) sit higher than in the other sectors, reflecting that Transport concern is higher overall — even the median respondent in Transport is closer to the high-concern threshold than in Energy or Housing.

► **Key takeaway: Transport respondents are more concerned overall than respondents in the other sectors — the median Transport respondent is closer to the high-concern threshold.**

3.4.2 Step 2 — Preparing the data for the statistical model

Purpose: *Turn the cleaned survey answers into numeric columns the ordinal regression can use: one-hot-encode categorical variables, bin age into groups, exclude country labels (replaced by national macro variables), and fill remaining missing values with iterative kNN imputation.*

After encoding and feature engineering, the Transport dataset moves from 112 raw survey columns to 98 candidate predictor variables (each categorical answer becomes one yes/no column per category; age is grouped into bins; macro variables for the respondent's country are added).

Country labels are deliberately excluded from the predictor set. Section 2 explains why: country and the national macro variables carry overlapping information, and including both would cause a mathematical conflict in the model. Country differences are reported descriptively in Step 8 instead.

Missing values in the few remaining cells are filled using iterative kNN imputation based on the five most similar respondents.

The Transport sector adds nine macro variables: **car share of the modal split, percentage who cannot afford regular transport, cars per 1,000 inhabitants, share of cars running on alternative motor energy, distribution of cars by age, new-car registrations by motor type, household expenditure on transport, road-fuel price (Super 95), and extra-territorial road-transport expenditure**.

What this means. Transport has 112 raw columns, fewer than Energy or Housing, but the post-encoding candidate predictor set is 98 — similar in scale to the other two sectors because most Transport questions are categorical and produce multiple yes/no columns when one-hot-encoded. The four Transport macro variables (car share of modal split, EV share of new car sales, charging infrastructure density, low-emission-zone urban coverage) are particularly important here: they capture each respondent's national context for the green-transition decisions Transport hazards refer to. The macro variables make the analysis sensitive to whether a respondent lives in a car-dominated country with sparse EV charging or in a context with widely-rolled-out infrastructure.

► **Key takeaway: Transport's nine macro variables (modal split, transport affordability, car density, alternative-energy share, car age, new-car motor mix, transport spending, fuel price, and road-transport expenditure) are directly relevant to the hazards the survey asks about — making the analysis context-aware.**

3.4.3 Step 3 — Choosing which variables to test (LASSO)

Purpose: *Use LASSO regression with 5-fold cross-validation to shortlist the predictors most worth testing for each hazard. LASSO is a practical filter, not a formal test.*

LASSO is run separately for each hazard and for the composite outcome.

A — Per-Hazard Analysis

Across all per-hazard models, LASSO selected a total of 103 predictor groups. Transport LASSO selection counts range from 4 (longer journeys) to 39 (restricted from town/city centres). The 'restricted from city centres' hazard attracts many candidate predictors because it touches on multiple dimensions — economic vulnerability, employment, mobility patterns, language barriers, and political identity — that all correlate weakly with concern about being shut out of urban areas. The lower-count hazards (longer journeys, fuel/repair costs) have narrower predictor profiles, meaning concern is structured by a smaller set of specific characteristics.

Table 50. Transport sector – per-hazard predictor count

Hazard	LASSO selections
Higher fuel/repair costs (ICE)	19
Car loses resale value	20

New taxes/fines for ICE	10
Restricted from town/city centres	39
Longer or more complex journeys	4
More pollution exposure	11

B — Composite Hazard Analysis

For the Transport sector composite outcome, LASSO selected 52 predictors out of 98 candidates.

► **Key takeaway: Transport's 'restricted from city centres' hazard has the widest socio-demographic gradient (39 LASSO selections) — concern about urban exclusion is tied to many respondent characteristics at once.**

3.4.4 Step 4 — The main statistical model (proportional odds)

Purpose: *Fit an ordinal (proportional odds) regression for each hazard, and test the proportional-odds assumption with the Brant test. McFadden's R² describes how well each model explains variation in concern.*

Proportional-odds assumption (Brant test). The proportional-odds assumption was tested at the feature level (0 violations in 50 tests) and the group level (0 violations in 43 tests). The model is appropriate for the Transport data.

A — Per-Hazard Analysis

Per-hazard model fits. The figure below shows the full-LASSO McFadden R² for each hazard alongside the number of confirmed predictor groups.

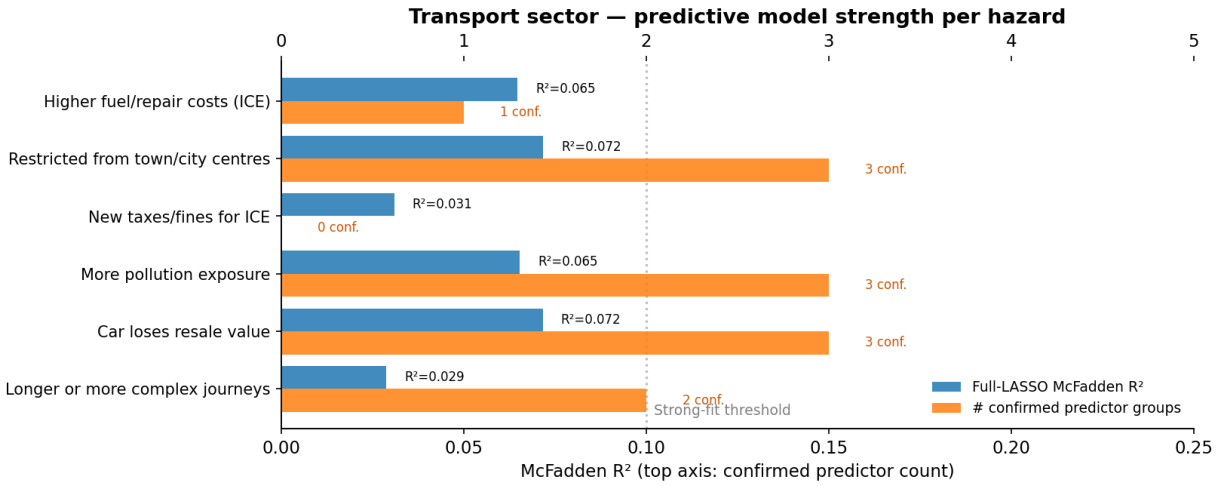


Figure 20. Predictive model strength per Transport hazard.

Transport per-hazard model fits are uniformly moderate — R² values between 0.03 and 0.07.

B — Composite Hazard Analysis

Composite model fit. For the Transport composite outcome, the full-LASSO model achieves McFadden R² = 0.0891 (Nagelkerke R² = 0.2597) on n = 358. The Transport composite reaches R² = 0.09.

Combined Interpretation. None reaches the 'high' tier achieved in Energy and Housing. The interpretation is important: Transport concern is real and widely distributed across the sample, but the socio-demographic and behavioural characteristics this survey measured do not predict who is concerned with the same precision as in Housing. Either the unmeasured drivers matter more (e.g. specific commuting routes, particular vehicle ownership history), or Transport concern genuinely cuts across socio-demographic lines more than the other sectors do.

► **Key takeaway: Transport models are statistically valid but explain less variation than the Housing or Energy models — concern is more socio-demographically diffuse in Transport.**

3.4.5 Step 5 — Confirming which predictors are real (LRT + BH-FDR + EPP cap)

Purpose: *Test each LASSO-selected predictor with a Likelihood-Ratio Test, correct for multiple testing using Benjamini-Hochberg FDR control at α = 0.05, then apply an events-per-predictor (EPP) cap.*

A — Per-Hazard Analysis

Per-hazard FDR results — how many of the LASSO-selected predictors survived BH-FDR correction and the EPP cap:

Table 51. Transport sector – per-hazard predictor confirmation dimensions

Hazard	LASSO selected	FDR significant	Retained after EPP cap	EPP warnings
Higher fuel/repair costs (ICE)	19	1	1	0
Car loses resale value	20	3	3	0
New taxes/fines for ICE	10	0	0	0
Restricted from town/city centres	39	3	3	0
Longer or more complex journeys	4	2	2	0
More pollution exposure	11	3	3	0

The per-hazard thinning is severe: 'Higher fuel/repair costs' goes from 19 LASSO selections to 1 confirmed; 'Restricted from town/city centres' from 39 to 3; 'New taxes/fines for ICE' from 10 to 0. The remaining confirmed predictors are robust survivors of multiple-testing correction.

B — Composite Hazard Analysis

For the Transport composite outcome, 36 predictor groups were tested. After BH-FDR and EPP cap, 5 are retained as confirmed. The evidence-strength breakdown is:

Table 52. Transport sector - Predictors strength distribution

Evidence strength	Count
strong	0
moderate	0
borderline (interpret with caution — post-LASSO bias)	5
not significant	31

The composite Transport model retains 5 predictor groups out of 36 tested. The evidence-strength breakdown shows 5 borderline-significant and 31 not significant — the typical pattern of LASSO casting a wide net and FDR pulling it tight.

► **Key takeaway: Transport's confirmed predictors are few but reliable — they survived heavy FDR thinning that eliminated most LASSO candidates.**

3.4.6 Step 5b — Validating the overall predictive model (global LRT)

Purpose: *Test whether the entire confirmed predictor set together is significantly better than a null (intercept-only) model.*

A — Per-Hazard Analysis

All Transport hazard models with at least one confirmed predictor pass the global LRT — meaning the joint set of predictors is reliably better than guessing. However, none of them reaches the 'high' tier; most are 'moderate'. 'New taxes/fines for ICE' has no confirmed predictors and so trivially fails the test.

Table 53. Transport Sector - per-hazard global likelihood ratio outcome

Hazard	Global LRT p	Significant?	Full R^2	Model strength tier
Higher fuel/repair costs (ICE)	6.827e-06	Yes	0.0646	moderate
Car loses resale value	4.748e-07	Yes	0.0716	moderate
New taxes/fines for ICE	—	No	0.0309	low (weak fit)
Restricted from town/city centres	0.0001517	Yes	0.0717	moderate
Longer or more complex journeys	5.962e-07	Yes	0.0288	low (weak fit)
More pollution exposure	9.904e-12	Yes	0.0653	moderate

A — Per-Hazard Analysis

The composite Transport model also passes the global LRT.

Combined Interpretation. The interpretation is that Transport findings are real but their predictive power is uniformly modest — useful for broad targeting, less useful for fine-grained subgroup interventions.

► **Key takeaway: Transport findings are statistically real but with consistently moderate predictive power — actionable at a broad level, not for fine-grained targeting.**

Step 5c: Robustness of the Transport findings

The same four-test battery was applied to the 17 confirmed Transport predictors. All three selection-stability tests return strong results: mean maximum stability is 0.99 under Meinshausen and Bühlmann (bootstrap, B = 500), 0.95 under Shah and Samworth complementary pairs (B = 500), and 0.95 under the one-shot bootstrap selection-frequency test (B = 100, with 95% Wilson lower confidence bound at or above 0.50 for every one of the 17 predictors). The bootstrap-to-complementary-pairs gap is just 0.04, the smallest of the three sectors, indicating that the bootstrap result here is not appreciably inflated by sample duplication. The retention test (B = 100, 50% subsamples) returns a mean retention of 0.42 — the highest of the three sectors.

Crossing complementary-pairs stability at or above 0.75 with retention at or above 0.50 gives the cleanest quadrant profile of the three sectors.

Doubly robust (6 of 17) — reliably selected and reliably significant under refitting. The electric-vehicle-attitude cluster carries the strongest claims: ev_perception in higher fuel and repair costs (complementary-pairs stability 0.99, retention 0.93); ev_likelihood in more pollution exposure (1.00, 0.82) and in car resale value (0.98, 0.75). The remaining doubly robust predictors are speaks_national_language in longer or more complex journeys (0.97, 0.80); travel_frequency in longer or more complex journeys (0.96, 0.61) and in the composite (0.97, 0.52).

Stable selection, fragile retention (11 of 17) — LASSO consistently picks these predictors but their FDR-significance does not always survive refitting. This category contains the three instances of macro_modal_split_cars (the country-anchored macro: complementary-pairs stability 0.88 to 0.92 across pollution exposure, restricted from town and city centres, and the composite — but retention 0.04 to 0.30), along with gender in the composite (0.99 stability, 0.43 retention), activity status in the composite and car-resale models, and the rest of the socio-demographic and EV-attitude variables.

Fragile on both (0 of 17) — no predictor fails both tests. Every confirmed Transport predictor passes selection robustness.

Fragile selection, retains significance (0 of 17) — empty.

Direction is universally robust. Sign consistency averages 0.99 across the 17 predictors and every one of them keeps its full-sample sign as the dominant direction across subsamples.

Key takeaway: Transport has the most defensible confirmed-predictor robustness profile of the three sectors. The six doubly robust findings — the EV-attitude cluster and the travel-frequency and language predictors — are deliverable-grade at the predictor level. The macro effect (macro_modal_split_cars) is a textbook stably-selected-but-not-significantly-informative case: LASSO consistently picks it and sign consistency is 1.00, but retention is 0.04 to 0.30 across the three hazards in which it is confirmed. The honest reading is that the modal-split macro contributes a reproducible direction of association — higher car-modal-share countries are differentially concerned — but its individual statistical significance

reflects a property of this two-country sample (Italy n = 204, Spain n = 154) rather than a generalisable effect.

3.4.7 Step 6 — Measuring the size of each effect (odds ratios and predicted probabilities)

Purpose: *Translate the confirmed-predictor coefficients from Step 5 into odds ratios (Formula 17) and marginal-effect percentage-point shifts in the probability of falling into the highest concern level (Formula 18). All effects are reported with 95% confidence intervals and an EPP-warning flag (⚠) where the events-per-predictor ratio dips below 5 and estimate stability is reduced.*

A — Per-Hazard Analysis

For each of the 5 Transport hazards with confirmed predictors, the per-hazard ordinal regression produces a confirmed-predictor effect table. The tables below show the top 5 confirmed predictors per hazard (ranked by absolute log odds ratio), with odds ratio, 95% CI, p-value, and the marginal effect on the predicted probability of being in the highest concern level (1–25 score binned to the level-5 band).

More pollution exposure — 5 confirmed predictors.

Table 54. Per-hazard Step 6 effects: More pollution exposure (top 5 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM (PCT PTS)	EPP Warning
ev_likelihood: Unlikely	0.33 (0.17–0.66)	0.0015	-11.8	
ev_likelihood: Very unlikely	0.34 (0.16–0.70)	0.0033	-11.8	
ev_likelihood: Likely	2.03 (1.27–3.26)	0.0033	+11.7	
macro_modal_split_cars	1.81 (0.97–3.38)	0.0639	+10.3	
ev_perception	1.39 (1.14–1.69)	0.0010	+5.3	

Car loses resale value — 4 confirmed predictors.

Table 55. Per-hazard Step 6 effects: Car loses resale value (top 4 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM (PCT PTS)	EPP Warning
ev_likelihood: Very unlikely	0.25 (0.12–0.50)	0.0001	-0.6	
ev_likelihood: Unlikely	0.32 (0.16–0.63)	0.0010	-0.4	
activity_status: Unemployed (actively seeking	2.38 (1.34–4.23)	0.0030	+0.1	

work or available to start work)				
household_size	1.12 (0.95–1.32)	0.1698	+0.0	

Restricted from town and city centres — 3 confirmed predictors.

Table 56. Per-hazard Step 6 effects: Restricted from town and city centres (top 3 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM (PCT PTS)	EPP Warning
macro_modal_split_cars	0.43 (0.23–0.82)	0.0098	-9.9	
newest_car_year_band: 2010-2019	0.58 (0.39–0.86)	0.0066	-8.1	
speaks_national_language: No	1.73 (1.12–2.68)	0.0134	+9.3	

Longer or more complex journeys — 2 confirmed predictors.

Table 57. Per-hazard Step 6 effects: Longer or more complex journeys (top 2 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM (PCT PTS)	EPP Warning
travel_frequency: Never I mostly work from home	0.38 (0.23–0.62)	0.0002	-12.1	
speaks_national_language: No	2.30 (1.49–3.55)	0.0002	+14.4	

Higher fuel/repair costs for ICE — 1 confirmed predictor.

Table 58. Per-hazard Step 6 effects: Higher fuel/repair costs for ICE (top 1 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM (PCT PTS)	EPP Warning
ev_perception	1.52 (1.26–1.83)	<0.0001	+7.3	

What the per-hazard effects tell us

Three patterns run through the Transport per-hazard ORs:

1. **EV-related predictors carry the strongest per-hazard effects.** ev_likelihood and ev_perception appear as confirmed predictors across multiple hazards with consistently large ORs. For Higher

fuel/repair costs ICE, ev_perception (a 1–5 ordinal) has OR = 1.52 per unit increase — meaning each step up the EV-positive scale multiplies the odds of high concern by 1.5×. For More pollution exposure, ev_likelihood__Likely has OR = 2.03 and ev_likelihood__Very unlikely has OR = 0.34. The direction is consistent: EV-positive respondents carry higher concern about ICE-related hazards.

2. **speaks_national_language__No is the recurring equity signal.** Non-national-language speakers carry OR = 2.30 for Longer/complex journeys and OR = 1.73 for Restricted from town/city centres — both statistically significant. This is the only equity-flagged predictor visible across multiple Transport hazards, pointing to an immigrant / linguistic-minority population segment requiring targeted communication.
3. **Concrete asset/exposure indicators appear where expected.** travel_frequency__Never I mostly work from home (OR = 0.38 for Longer journeys) protects against journey-related concern; newest_car_year_band__2010-2019 (OR = 0.58 for Centre restrictions) protects against restriction-related concern relative to pre-2010 vehicle owners. Direct exposure → direct concern pattern.

B — Composite Hazard Analysis

For the composite analysis, Step 6 reports the odds ratios for the confirmed predictors in the Transport composite ordinal regression.

The composite Transport model retains **8 confirmed predictors** (5 significant at $p < 0.05$). The top 5 by absolute log odds ratio:

Table 59. Composite Step 6 effects: Transport composite model (top 5 by absolute log odds ratio).

Predictor	OR (95% CI)	p-value	MEM (PCT PTS)	EPP Warning
travel_frequency: Never I mostly work from home	0.36 (0.21–0.61)	0.0002	-12.3	
activity_status: Unemployed (actively seeking work or available to start work)	2.63 (1.39–4.97)	0.0030	+17.8	
travel_frequency: Twice per week	0.43 (0.21–0.90)	0.0248	-9.8	
gender: Woman	1.92 (1.30–2.81)	0.0009	+9.7	
speaks_national_language: No	1.76 (1.14–2.72)	0.0115	+9.2	

What this means (Composite Analysis). The five strongest predictors of overall Transport concern (by absolute log odds ratio) describe a coherent story. Travel-frequency 'Never — I mostly work from home' (OR = 0.36) means people who do not travel are insulated from Transport hazards and naturally less worried about them. Unemployment (OR = 2.63) increases the odds of high concern, likely because unemployed respondents are economically more vulnerable to fuel-cost shocks. Gender = Woman (OR = 1.92) emerges

as a substantial predictor in Transport, more so than in Energy or Housing. The most distinctive finding is that speaks_national_language = 'No' has OR = 1.76 — non-native-language speakers are over 70 per cent more likely to be highly concerned about Transport hazards. At the per-hazard level, the same variable has OR = 2.30 for 'Longer or more complex journeys', which has a clear interpretation: communication and navigation barriers make transport-transition changes more worrying.

► **Key takeaway: Transport concern follows exposure and vulnerability: women, unemployed respondents, daily or near-daily travellers (the reference category), and non-native-language speakers are the most concerned; respondents who work from home or travel only twice per week are significantly less concerned. Per-hazard ORs add the EV-attitude dimension (ev_likelihood, ev_perception) that the composite alone cannot reveal — these are the largest per-hazard effects in the sector.**

Predicted probabilities (composite model): *Convert the odds-ratio table above into a predicted-probability picture.*

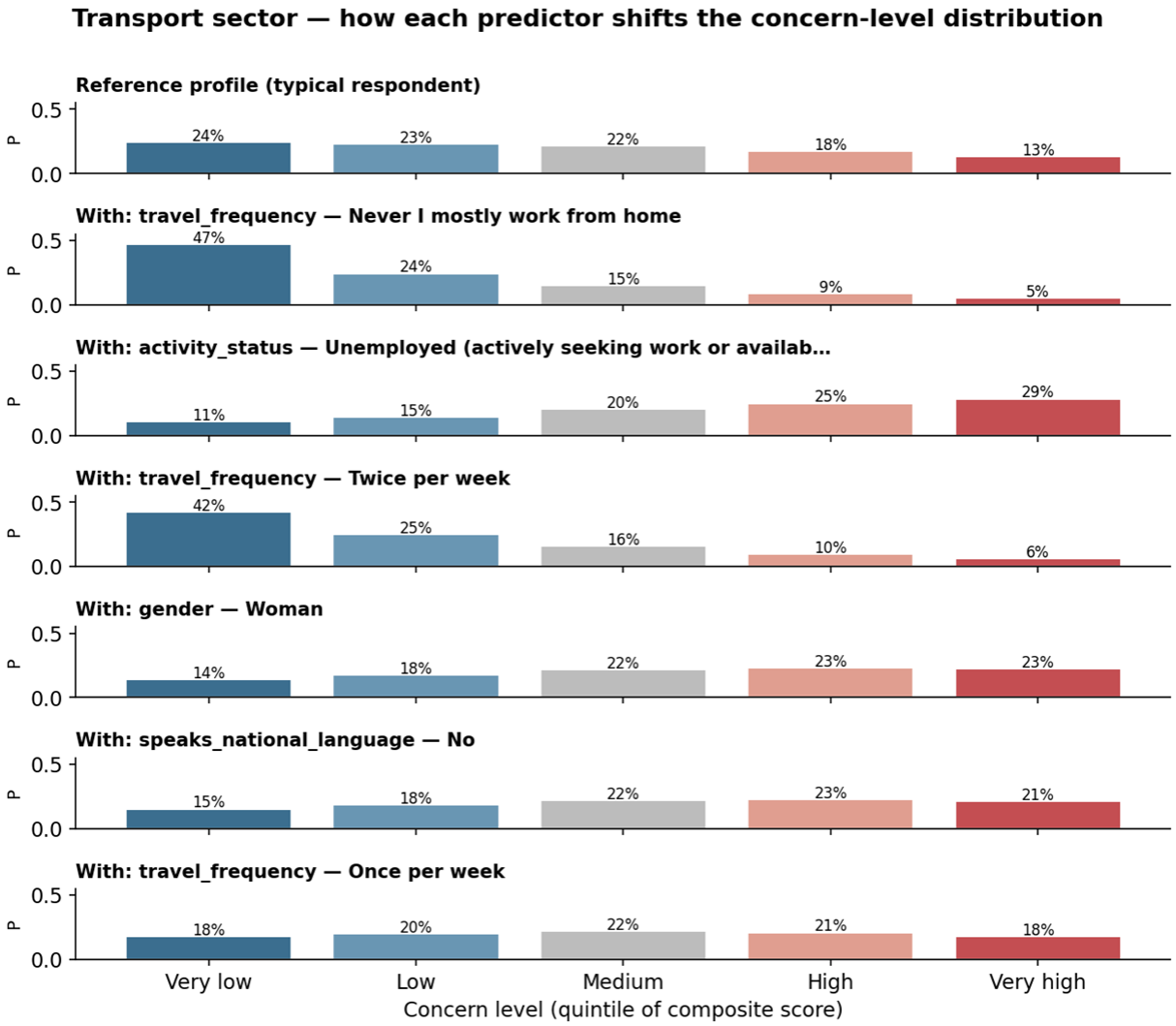


Figure 21. How each of the top predictors shifts the predicted distribution of concern levels.

How to read this chart. The reference profile has a 13% probability of falling in the 'very high concern' group. The rows below show how that probability changes when each predictor is 'switched on'.

What this means. The reference Transport profile has a 13 per cent chance of being in 'very high concern'. The probability shifts are moderate compared to Energy and Housing — reflecting the lower R^2 values of Transport models. Working from home pushes the very-high-concern probability down meaningfully; the

female-gender shift pushes it up moderately. The non-native-language shift is one of the larger movements in the Transport probability picture, which underscores why this is the headline finding for Transport.

3.4.8 Step 7 — Identifying the most affected groups (ANOVA)

*Purpose: For each confirmed categorical predictor in the per-hazard and composite Transport models, run a one-way ANOVA to identify **which specific category within each predictor carries the highest mean concern**, and quantify how much that category differs from the lowest-concern category. The omnibus F-test gates the analysis; the per-category findings reported below come from post-hoc pairwise tests (Tukey HSD or Games-Howell, applied automatically based on each predictor's variance-homogeneity result).*

A — Per-Hazard Analysis

For each of the 5 Transport hazards with FDR-confirmed predictors, a one-way ANOVA was run on every confirmed categorical predictor in that hazard's own Step 5A confirmed set, using the hazard's 1–25 score as the outcome. Macros and continuous predictors are excluded by methodology.

Across the 5 active Transport per-hazard models, **10 categorical predictor ANOVAs were performed and 8 (80%) showed a statistically significant univariate effect at α = 0.05**. This is the highest significance rate among the three sectors (Energy 66%, Housing 36%), reflecting a strong alignment in Transport between LRT-confirmed effects and standalone univariate signals. The chi-squared verification test agreed with the ANOVA decision in every case. Some of the highest η^2 values across the entire study appear in this sector — ev_likelihood on "More pollution exposure" reaches $\eta^2 = 0.119$ (medium effect band), the single largest effect-size across all three sectors' Step 7 results.

Per-hazard category-level findings

Car loses resale value — 3 predictors tested, 1 significant.

Table 60. Highest- and lowest-concern categories: Car loses resale value.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
ev_likelihood	Likely (to switch to EV)	Very unlikely (to switch)	+4.70	0.0126	0.060

A clear pattern: respondents who say they are **likely or very likely to switch to an EV** carry the highest mean concern about their current car losing resale value (+4.70 points above those very unlikely to switch). The reading is intuitive — prospective EV adopters are most aware that their ICE vehicle is the asset at risk during the transition. The two other confirmed predictors (activity_status, household_size) are confirmed in Step 5 but not univariately significant in Step 7, indicating effects that operate conditionally on EV-switching intent and other shortlisted predictors.

Higher fuel and repair costs for ICE — 1 predictor tested, 1 significant.

Table 61. Highest- and lowest-concern categories: Higher fuel and repair costs for ICE.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
ev_perception	5 (most positive EV perception)	1 (least positive EV perception)	+8.55	<0.0001	0.080

A strong single-predictor finding: respondents with the **most positive EV perception** carry mean concern +8.55 points above those with the least positive perception — over a third of the 1–25 scale. The ranking is monotonic: 5 > 4 > 3 > 2 > 1. EV-positive respondents see ICE downsides most clearly and worry most about

rising fuel and repair costs; EV-sceptics show the lowest concern. This is one of the cleanest monotonic gradients in any Step 7 result across the three sectors.

Longer or more complex journeys — 2 predictors tested, 2 significant.

Table 62. Highest- and lowest-concern categories: Longer or more complex journeys.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
speaks_national_language	No (does not speak national language)	Yes (speaks national language)	+2.75	0.0021	0.026
travel_frequency	Three or more days per week	Never — I mostly work from home	+3.48	0.0048	0.033

Non-national-language-speakers carry higher mean concern about longer or more complex journeys (+2.75 points above national-language speakers) — a clear equity signal pointing to immigrant and linguistic-minority respondents. **Frequent commuters** also carry higher concern, with the largest gap appearing between daily/3+ days-per-week commuters and respondents who mostly work from home (+3.48 points). Both findings make practical sense: those with language barriers and high travel frequency are most exposed to additional journey complexity.

More pollution exposure — 2 predictors tested, 2 significant.

Table 63. Highest- and lowest-concern categories: More pollution exposure.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
ev_likelihood	Likely (to switch to EV)	Unlikely (to switch)	+7.95	<0.0001	0.119
ev_perception	5 (most positive EV perception)	1 (least positive EV perception)	+9.12	0.0001	0.088

The two strongest findings in the Transport sector. **EV-switch-likely respondents** carry concern +7.95 points above EV-switch-unlikely respondents, and **EV-positive respondents** (perception = 5) carry concern +9.12 points above EV-negative respondents (perception = 1) — the largest single category gap anywhere in the Transport Step 7 results, and over a third of the 1–25 scale. The η^2 of 0.119 for ev_likelihood is the largest effect-size in the entire three-sector study. Together these two predictors describe a coherent population segment: pro-EV respondents are most acutely concerned about air-pollution exposure from ICE vehicles, which is exactly the policy-relevant signal expected from a clean-transport-transition survey. The third confirmed predictor (macro_modal_split_cars) is a macro variable correctly excluded from per-hazard ANOVA per methodology and is treated separately in Step 7b.

Restricted from town and city centres — 2 predictors tested, 2 significant.

Table 64. Highest- and lowest-concern categories: Restricted from town and city centres.

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2
newest_car_year_band	pre-2010	2010–2019	+4.14	(omnibus p = 0.049)	0.022
speaks_national_language	No (does not speak national language)	Yes (speaks national language)	+2.34	0.0121	0.018

Owners of **pre-2010 vehicles** carry the highest mean concern about being restricted from town and city centres — these are the cars most likely to fail emission-zone criteria, so the policy signal is directly aligned with vehicle ownership patterns. The two newer-vehicle bands (2010–2019, 2020+) sit lower in the ranking, with 2020+ slightly above 2010–2019 (possibly reflecting heavier ICE-specific reliance among recent ICE buyers who would face stranded-asset risk). The language-barrier finding mirrors the longer-journeys hazard: **non-national-language-speakers** carry higher concern about access restrictions, reinforcing the broader equity dimension visible across multiple Transport hazards. The third confirmed predictor (macro_modal_split_cars) is again a macro variable excluded from per-hazard ANOVA.

New taxes or fines for ICE — 0 confirmed predictors, no Step 7. This hazard had LASSO-selected predictors but 0 survived FDR after BH correction. Consistent with this, no Step 5b global LRT, Step 6 effects, or Step 7 ANOVA files exist for this hazard in the output folder — the pipeline correctly terminated the analysis at Step 5 once no predictors passed FDR. The hazard is reported in Step 9 with n_retained_after_cap = 0 and no category-level findings can be reported.

What the per-hazard ANOVAs tell us

Three substantive patterns emerge across the active Transport hazards:

1. **EV-related predictors dominate the Transport findings.** Both ev_likelihood and ev_perception appear in multiple hazards and carry the four largest category gaps in the Transport Step 7 set (+4.70, +7.95, +8.55, +9.12 points). The direction is always the same: **EV-positive respondents carry higher concern about ICE-related hazards**. This is policy-meaningful — it identifies the population segment most engaged with the clean-transport transition and most aware of ICE risks.
2. **Language barrier is a recurring equity signal.** speaks_national_language appears as significant in two hazards (Longer journeys, Restricted from centres), with non-speakers carrying higher concern in both cases. This is the only equity-flagged predictor visible across multiple Transport hazards, and points to an immigrant / linguistic-minority population segment requiring targeted communication during transition policies.
3. **Concrete asset/exposure signals appear where expected.** newest_car_year_band (older car owners most concerned about restrictions) and travel_frequency (daily commuters most concerned about longer journeys) reproduce the straightforward "those most exposed to the hazard are most concerned about it" pattern. These are the most policy-actionable findings because the affected population is directly identifiable by observable characteristics (vehicle age, commute pattern).

B — Composite Hazard Analysis

For the composite analysis, ANOVA was run on the composite Transport concern score (the per-respondent average across all 6 Transport hazards, binned into 5 quintile levels) using each confirmed categorical predictor from the Step 5B composite model. Of the predictors retained in the composite model, 4 are categorical and tested here in Step 7. Across these 4 categorical predictors, **3 show a statistically significant univariate ANOVA effect at α = 0.05** — a 75% significance rate consistent with Transport's per-hazard pattern of strong LRT-vs-ANOVA alignment. The chi-squared verification test agreed with the ANOVA decision in **every case**, providing the cleanest cross-method validation among the three sectors' composite tracks (Energy: 1 disagreement; Housing: 1 disagreement; Transport: 0 disagreements).

Composite category-level findings (significant predictors only)

Table 65. Highest- and lowest-concern categories in the Transport composite model (significant Step 7 predictors).

Predictor	Highest Concern	Lowest Concern	Gap Points	P (Pairwise)	η^2	Notes
speaks_national_language	No (does not speak national language)	Yes (speaks national language)	+1.78	0.0033	0.024	Two-category clean split
gender	Woman	Man	+1.49	0.0122	0.023	Non-binary sits between
travel_frequency	Once per week	Twice per week	+3.26	(omnibus p = 0.022; pairwise n.s. after Tukey)	0.027	Pattern suggestive, no individual pair survives correction

Three substantive composite analysis findings:

- **Language barrier reproduces from the per-hazard track.** speaks_national_language was significant in both Longer journeys and Restricted from centres (per-hazard, +2.75 and +2.34 pts), and now also in the composite (+1.78 pts overall). The non-speaker disadvantage is consistent across hazards and at the composite level — the strongest equity signal in the Transport sector.
- **Gender enters at the composite level.** Women carry +1.49 points higher mean composite concern than men, with non-binary respondents sitting between. This effect did not surface in any individual per-hazard ANOVA in the per-hazard track, which makes the composite the right place to report it: averaging across all 6 hazards reveals a small but consistent gender shift that no single hazard isolates on its own.
- **Travel frequency is significant at the omnibus level but no pairwise comparison survives Tukey correction.** The omnibus F (3.24, p = 0.022) detects between-group differences, and the ranking (Once-per-week > 3+ days/week > WFH > Twice-per-week) hints at a non-monotonic concern pattern, but the largest mean gap (Once vs Twice = 3.26 pts) has Tukey-adjusted p = 0.098 — short of significance. This is a methodologically transparent finding: the predictor matters (the omnibus is significant and chi-squared agrees at p = 0.012) but no specific pair can be identified as the driver after multiple-comparison correction. The honest summary is "concern levels vary across travel-frequency categories, but no single category contrast is robustly confirmed".

Why Step 5 and Step 7 can disagree

Step 5 measures a predictor with all others held constant (multivariate); Step 7 tests it on its own (univariate). Disagreements signal effects that only become visible when other variables are controlled for. **In Transport this disagreement is the least pronounced of the three sectors**: only activity_status is FDR-confirmed in the composite model but not univariately significant in Step 7 (ANOVA p = 0.133), and even there the ranking is intuitive (unemployed respondents carry highest composite concern, employed respondents lowest). The 75% agreement rate in the composite track matches the 80% agreement rate in the per-hazard track.

► **Key takeaway: Transport's highest-concern categories — EV-positive and EV-switch-likely respondents (per-hazard A track, +4.70 to +9.12 points across multiple hazards), non-national-language-speakers (per-hazard *and* composite, consistently +1.78 to +2.75 points), women relative to men (composite, +1.49 points), pre-2010 vehicle owners (per-hazard, +4.14 points), and daily commuters (per-hazard, +3.48 points) — line up around two consistent dimensions: (a) transition engagement (those most ready to adopt EVs are most concerned about ICE-related hazards), and (b) vulnerability / direct exposure (language barriers, gender disparity, older vehicles, high commute frequency identify the populations most at risk from policy-driven transitions). The largest single category gap in the entire Transport Step 7 result set is ev_perception on the More pollution exposure hazard (+9.12 pts, per-hazard A track), and the largest η^2 observed anywhere in the three-sector Step 7 analysis is ev_likelihood on the same hazard (η^2 = 0.119). The composite track confirms the language-barrier equity signal at the all-hazard level and adds the gender finding that no individual hazard isolates on its own — both important policy signals that the per-hazard analysis alone would miss.**

3.4.8.1 Step 7b — Per-country profile for confirmed macro predictors

Purpose: *When a national macro variable (such as cooling degree-days or household utility-arrears percentage) is confirmed as a predictor in Step 5 for a specific hazard, Step 7b reports the per-country profile of concern for that hazard. This fills a gap left by the standard Step 7 (which excludes macros because they have no within-country variation) and by Step 8 (which reports country effects only for the composite outcome, not per hazard).*

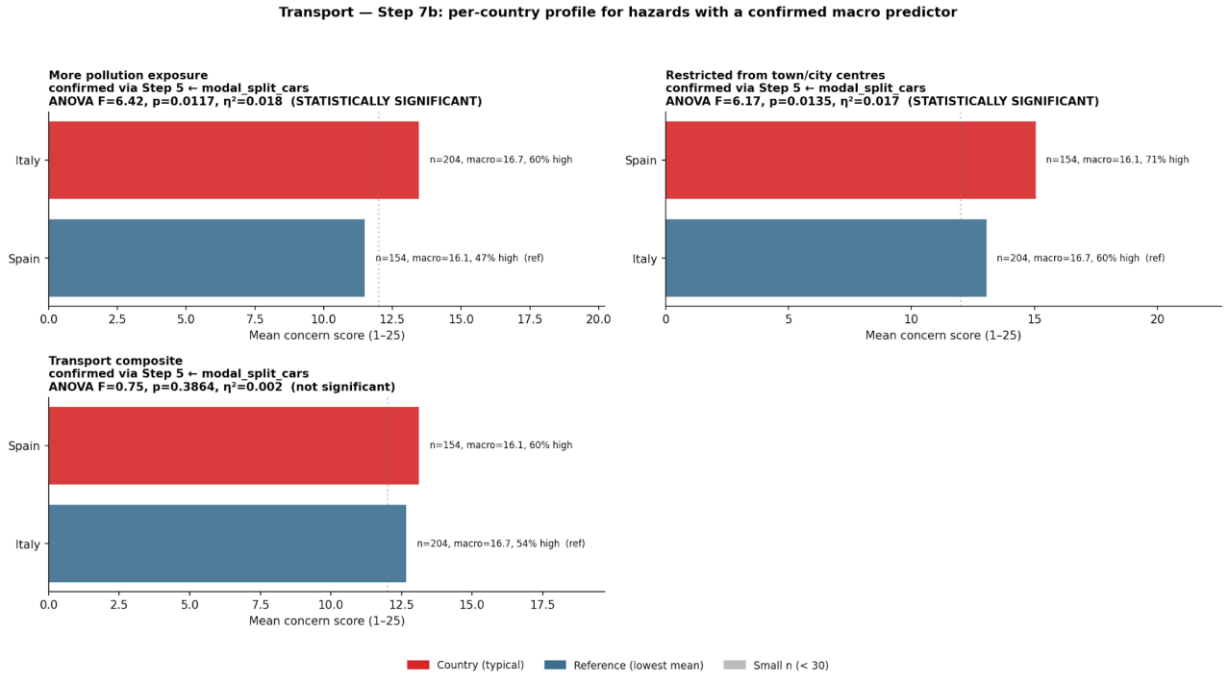


Figure 22. Per-country mean concern for Transport hazards where a macro predictor was confirmed in Step 5. Each panel shows the macro value attached to each country and flags countries with n < 30.

Across Transport, 3 macro × hazard combinations were confirmed in Step 5 and profiled per country here. The ANOVA results are summarised below.

Table 66. Transport sector - Effect of macro variables

Hazard	Macro predictor	F	p-value	η^2	Effect size	Sig
More pollution exposure	modal_split_cars	6.42	0.0117	0.018	small	*
Restricted from town/city centres	modal_split_cars	6.17	0.0135	0.017	small	*
Transport composite	modal_split_cars	0.75	0.3864	0.002	negligible	n.s.

Significant cross-country pairs (Tukey HSD vs the lowest-mean country). 2 country pair(s) reach significance after Tukey HSD correction:

Table 67. Transport sector - Significant cross-country pairs per hazards

Hazard	Macro	Higher (A)	Lower (B)	Mean diff	p	Sig
More pollution exposure	modal_split_cars	Italy	Spain	+1.97	0.0117	*
Restricted from town/city centres	modal_split_cars	Spain	Italy	+2.00	0.0135	*

What this means. The two Transport countries show opposite patterns depending on the hazard. For 'More pollution exposure', Italy (mean 13.49) is significantly more concerned than Spain (11.51), p = 0.012. For 'Restricted from town/city centres', Spain (15.08) is significantly more concerned than Italy (13.08),, p = 0.014. Both findings have the same confirmed macro predictor (macro_modal_split_cars), suggesting the macro effect reflects different sensitivities in each country rather than a uniform car-modal-share story. For composite Transport concern, there is no significant country difference — confirming what Step 8 already reported.

► **Key takeaway: Italy and Spain show opposite patterns: Italians worry more about pollution exposure, Spaniards worry more about city-centre restrictions, despite both findings being driven by the same confirmed macro.**

3.4.9 Step 8 — Comparing results across countries

Purpose: *Describe how the composite concern score varies across the countries in the sample. Country labels are NOT in the regression model.*

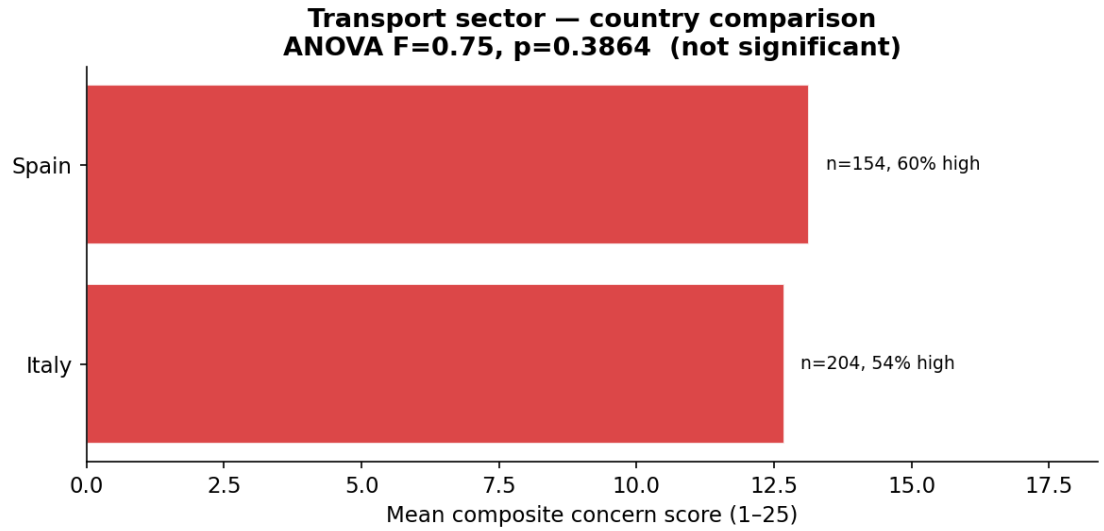


Figure 23. Mean composite Transport concern by country with ANOVA result.

The ANOVA across countries is not statistically significant ($F = 0.75$, $p = 0.3864$, $\eta^2 = 0.002$). No country pair shows significantly different overall Transport concern in this sample.

What this means. Italy and Spain do not differ significantly in mean Transport concern ($F = 0.75$, $p = 0.39$). The two-country sample limits what can be said about cross-national variation generally, but the absence of difference between these two suggests the patterns found here likely apply to similar southern-European contexts. The composite means are very close (Italy 12.68, Spain 13.14), well within sampling noise.

► **Key takeaway: Transport concern levels are statistically indistinguishable between Italy and Spain — the patterns found here likely apply broadly to similar national contexts.**

3.4.10 Step 9 — Comparing results across hazards (synthesis and convergence)

Purpose: *Pull the per-hazard results together: rank hazards by concern, map cross-hazard predictor consistency, and compare per-hazard (A) vs composite (B) analyses.*

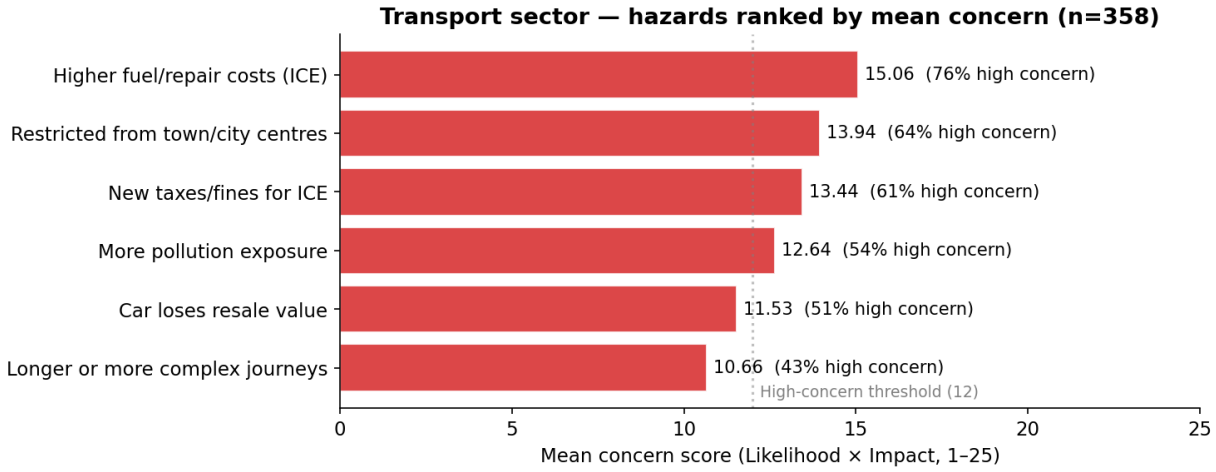


Figure 24. Transport hazards ranked by mean concern score

Cross-hazard predictor consistency. 4 predictors are confirmed in two or more hazards; the most cross-cutting predictor, 'ev_likelihood', is confirmed in 2 hazards.

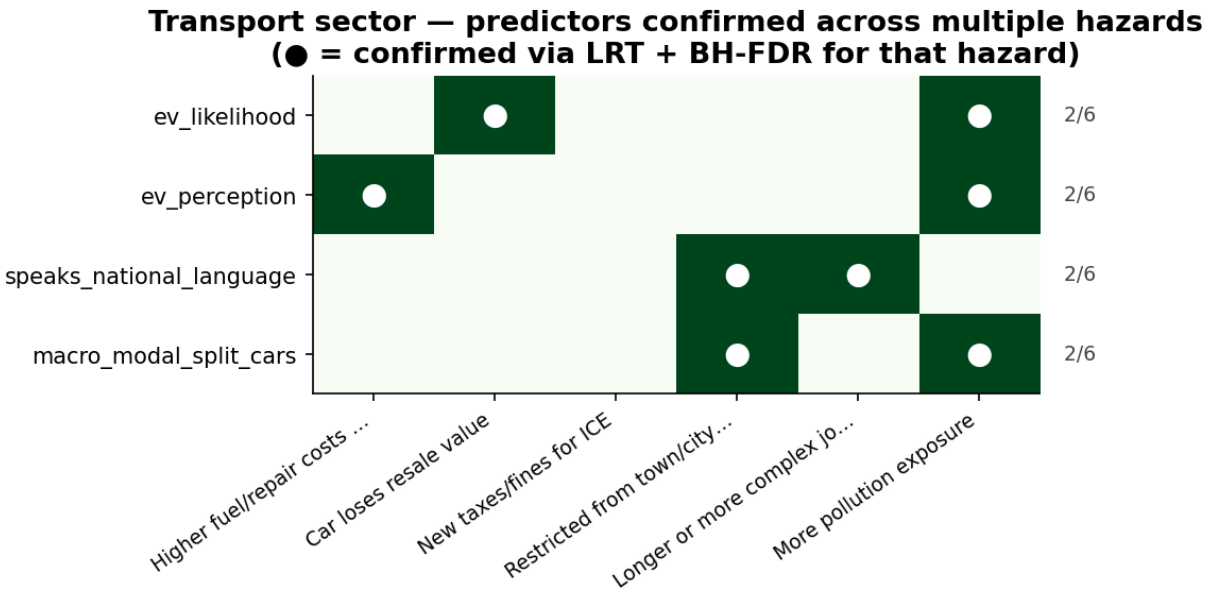


Figure 25. Predictors that are confirmed across more than one Transport hazard.

3.4.10.1 Step 9b — A/B convergence between per-hazard and composite analyses

Purpose: *Classify each confirmed predictor as Convergent, Partially convergent, A-cross-cutting, A-specific, or B-specific.*

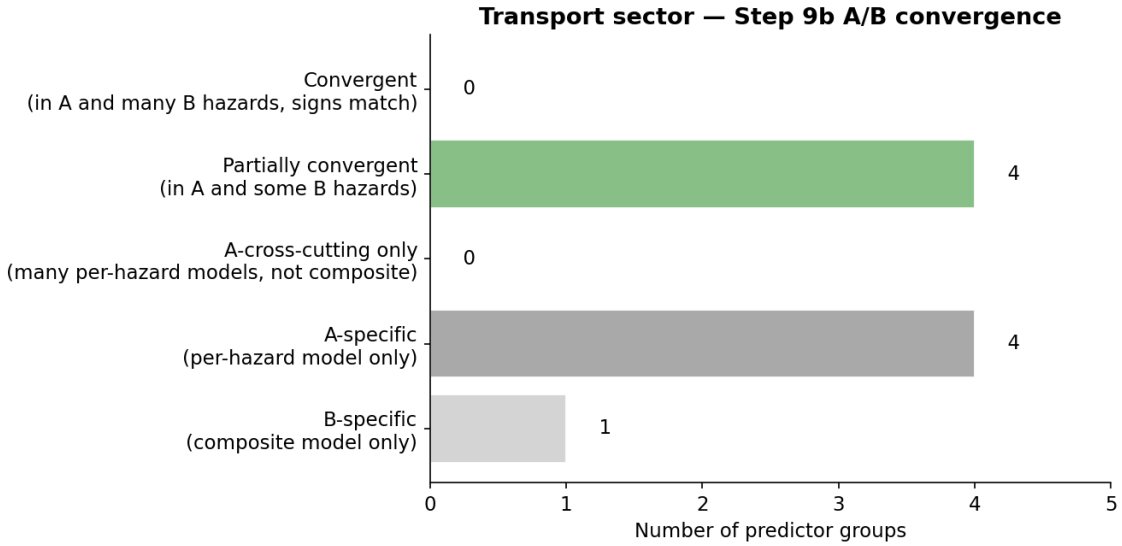


Figure 26. Step 9b classification of confirmed predictors by A/B convergence.

In Transport, 0 predictor group(s) are classified as Convergent, 4 as Partially convergent, 4 as A-specific, and 1 as B-specific.

What this means. The Transport hazard ranking shows fuel/repair costs at the top (76 per cent in high concern, the highest single-hazard rate in the study), with city-centre restrictions and new taxes also above 60 per cent in high concern. The cross-hazard predictor map identifies only four predictors confirmed in two or more hazards: ev_likelihood, ev_perception, speaks_national_language, and macro_modal_split_cars. None is confirmed in more than two hazards — Transport's confirmed predictors are largely hazard-specific. The A/B convergence analysis finds zero 'Convergent' predictors (down from two in Energy and Housing). Four are 'Partially convergent', meaning they appear in the composite and in one or two specific hazards. The pattern points to Transport concern being more hazard-dependent than concern in the other sectors.

► **Key takeaway: Transport concern is more hazard-specific than Energy or Housing — there are fewer cross-cutting predictors that apply broadly, and the speaks_national_language finding is the most distinctive policy-relevant result.**

3.4.11 Headline for Transport sector

Women, unemployed respondents, frequent travellers, and non-native language speakers carry significantly higher Transport concern; remote workers carry significantly lower concern.

3.5 Cross-sector conclusions and what they mean for policy

This section pulls together the findings from all three sectors into the conclusions that matter most for policy. For each major finding we describe two things: what the data shows, and why it matters.

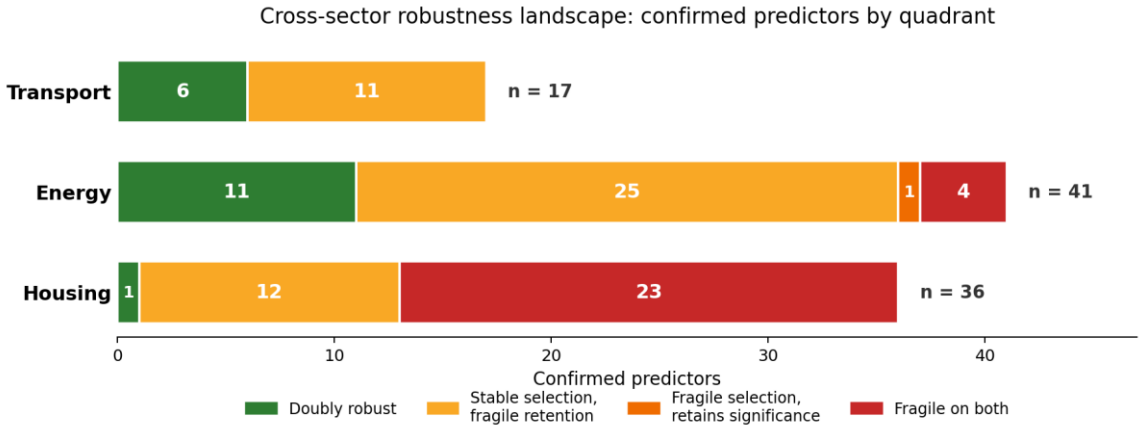


Figure 27. Cross-sector robustness landscape — distribution of confirmed predictors across the four quadrants, by sector.

The most actionable single finding: low-housing-cost band concentrates Housing concern

What the data shows. Respondents in the lowest housing-cost band of the Housing sample have 2 to 3 times the odds of being highly concerned about Housing hazards. The effect appears in four separate Housing models (cold/damp leading to health problems, more expensive home insurance, more frequent power cuts, and overall composite Housing concern), always with the same positive direction, always with p-values below 0.005, and never with a small-subgroup instability warning.

Why this matters. The housing-cost-band finding is among the strongest confirmed signals in this sector by full-sample criteria: confirmation across multiple hazards, identical direction across hazards, and full-sample p-values below 0.005. The Step 5c sensitivity test refines that reading by separating two senses of stability. Direction is universally stable — sign consistency is 1.00 across every subsample on every hazard, so the lowest-cost-band-to-higher-concern association reproduces invariably in direction. Selection-stability is also good: under complementary-pairs subsampling (the most conservative selection-stability test), housing-cost-band scores between 0.60 and 0.83 across the four hazards in which it is confirmed, indicating LASSO would reliably re-select it on a fresh sample.

The retention test, however, places housing-cost-band in the **stable selection, fragile retention** quadrant — its FDR-adjusted significance survives subsample refitting in only 15% to 42% of subsamples, depending on the hazard. This is not a contradiction with the full-sample finding; it reflects Housing's underlying sample-size and heterogeneity constraints. With n = 263 spread across three structurally different national housing markets (Germany n = 197, Portugal n = 49, Ireland n = 17), individual-predictor significance is intrinsically

more sensitive to which respondents are sampled than in the larger and more homogeneous Energy (n = 407, five countries) or Transport (n = 358, two countries) samples — a property of the Housing sector as a whole, not of this finding specifically. Section 3.12 documents the same pattern across the full Housing confirmed set: only one of 36 confirmed predictors is doubly robust by strict criteria, and 23 are fragile on both stability tests. Compounding this, the Housing sample itself is 75% German, so the housing-cost-band variable and the gradient producing this finding are anchored predominantly in the German housing-cost distribution. The directional claim holds within that composition; the population the magnitude generalises to is German Housing rather than European Housing more broadly.

The right reading, therefore, is at the construct level: the lowest-cost-band group reliably carries higher Housing concern, in both direction and re-selectability, even though the precise effect-magnitude and significance estimate of any single specification varies with sample composition. The finding still points directly at the most vulnerable group in the Housing transition — likely occupants of older, less well-maintained dwellings — and policy that protects this group from transition-related cost shocks (energy retrofits, insurance reclassification, low-emission heating requirements) is addressing the population segment carrying the highest perceived risk.

Financial pressure is the strongest cross-sector pattern

What the data shows. Across all three sectors, financial-pressure indicators emerge as strong predictors of high concern. In Energy, utility-bill arrears and low bill confidence dominate. In Housing, the lowest housing-cost band concentrates concern. In Transport, unemployment is one of the strongest predictors. These are not the same variables in each sector — but they share a common substantive meaning: respondents already experiencing economic strain perceive transition-related hazards as more serious.

Why this matters. This is not a surprising finding, but the consistency across three independent analyses gives it real weight. It suggests that perceived hazard of the twin transition is concentrated among respondents who are economically least able to absorb additional costs. The implication for policy is straightforward: protection measures, cost-spreading mechanisms, and targeted support are most likely to address concern at its source when they prioritise financially-stretched households.

Language barriers and Transport concern

What the data shows. The variable speaks_national_language shows that non-native-language speakers in the Transport sector have an OR of 1.76 in the composite model and an OR of 2.30 for the specific hazard of 'longer or more complex journeys'. The finding is statistically significant on both univariate (Step 7) and multivariate (Step 5) tests.

Why this matters. Communication and navigation accessibility is a transport-transition issue. Non-native-language speakers face more difficulty with transition-related changes — new low-emission-zone signage, route diversions, EV-charging information — and report substantially higher concern as a result. The finding points to inclusive transition communication as a policy priority, particularly in countries with significant non-native-language populations.

Country differences are mostly absent — with one exception

What the data shows. Energy and Transport show no statistically significant cross-country differences in overall concern. Housing is the exception: Portugal shows significantly higher mean composite Housing concern than Germany (Tukey HSD p = 0.005). Germany-Ireland and Portugal-Ireland pairs are not significant.

Why this matters. The general absence of country effects is itself a finding. It tells us that the patterns of who is concerned and why — financial pressure, housing quality, language barriers — operate consistently

across EU national contexts. Policy informed by these findings is therefore likely to apply broadly across countries with similar EU context, rather than needing country-specific tailoring. The Housing Germany-Portugal gap likely reflects actual differences in housing-stock quality and climate exposure, and merits attention in Housing-specific national policy.

The concern-versus-predictability inversion

What the data shows. A pattern visible in all three sectors: the hazards with the highest mean concern (heating/cooling costs, electricity bills, fuel/repair costs) have the weakest predictive models. The hazards with moderate mean concern produce the strongest models. In Energy, 'Struggling to pay bills' (36 per cent high concern) has $R^2 = 0.15$ while 'Higher electricity bills' (74 per cent high concern) has $R^2 = 0.05$. The same pattern appears in Housing and Transport.

Why this matters. This is a statistical consequence of universal concern: when nearly everyone reports the same concern level, there is little variation between respondents for any model to explain. The substantive implication for policy is important. The hazards with the highest mean concern are universal concerns that cut across socio-demographic lines — policy here needs to address everyone, not target specific groups. The hazards with moderate mean concern are where targeted policy can find traction, because individual characteristics do distinguish who is concerned from who is not.

What the analysis cannot tell you

What the data shows. The analysis identifies associations, not causes. It describes who in the sample is more or less concerned and by how much. It does not measure whether respondents would change their behaviour, whether their concerns are objectively justified, or how concern levels would evolve over time. Several effects with very large or very small odds ratios are flagged with EPP warnings, meaning the underlying subgroup is too small to estimate the standard error reliably.

Why this matters. These caveats are not weaknesses of the study but boundaries of what the data can support. The findings are reliable at the association level and at the subgroup-comparison level. They are less reliable for individual prediction, causal inference, or longitudinal projection. EPP-flagged effects should not be cited as headline findings — they are reported for completeness, not for policy claims. The most defensible findings are those confirmed across multiple hazards, with consistent sign, p-values below 0.005, and no instability warnings — and the lowest-housing-cost-band finding in Housing is the exemplar of all these.

Limitations. The cross-sector comparisons drawn through this section reflect three samples with different national compositions: Transport (Italy 57%, Spain 43%) is balanced between two structurally similar Mediterranean countries; Energy (Germany 37%, Italy 32%, Spain 24%, Portugal 5%, Ireland 3%) spans five countries with Germany the plurality and roughly a third of the weight; Housing (Germany 75%, Portugal 19%, Ireland 7%) is dominated by the German subsample. Where the per-sector findings agree across sectors — directionally consistent socio-economic gradients, financial pressure as a cross-cutting signal, language and EV-attitude effects within Transport — the agreement holds within those compositional limits. Cross-sector patterns described here should be read as findings about these sampled populations rather than about the EU population as a whole.

The robustness gradient: Transport stronger than Energy stronger than Housing

What the data shows. The four-test robustness battery (Step 5c, reported per sector in step 5 of section2, 3 & 4) places the three sectors on a clear gradient. Of Transport's 17 confirmed predictors, 6 are doubly robust (reliably selected and reliably significant under refitting) and 0 are fragile on both tests. Of Energy's 41

confirmed predictors, 11 are doubly robust and 4 are fragile on both. Of Housing's 36 confirmed predictors, only 1 is doubly robust and 23 are fragile on both — the majority of the confirmed set does not reproduce reliably, given the sample size of data in Housing.

Table 68. Three-sector robustness comparison across confirmed predictors

Metric	Energy	Housing	Transport
n confirmed predictors	41	36	17
Sample size (n)	407	263	358
Countries	5	3	2
Cronbach α	0.82	0.84	0.79
MB mean stability (bootstrap)	0.97	0.91	0.99
SS mean stability (complementary pairs)	0.90	0.65	0.95
MB–SS gap (bootstrap optimism)	0.08	0.26	0.04
Mean retention	0.38	0.15	0.42
Mean sign consistency	0.99	0.94	0.99
Doubly robust (count)	11	1	6
Stable selection, fragile retention	25	12	11
Fragile on both	4	23	0
Fragile selection, retains significance	1	0	0

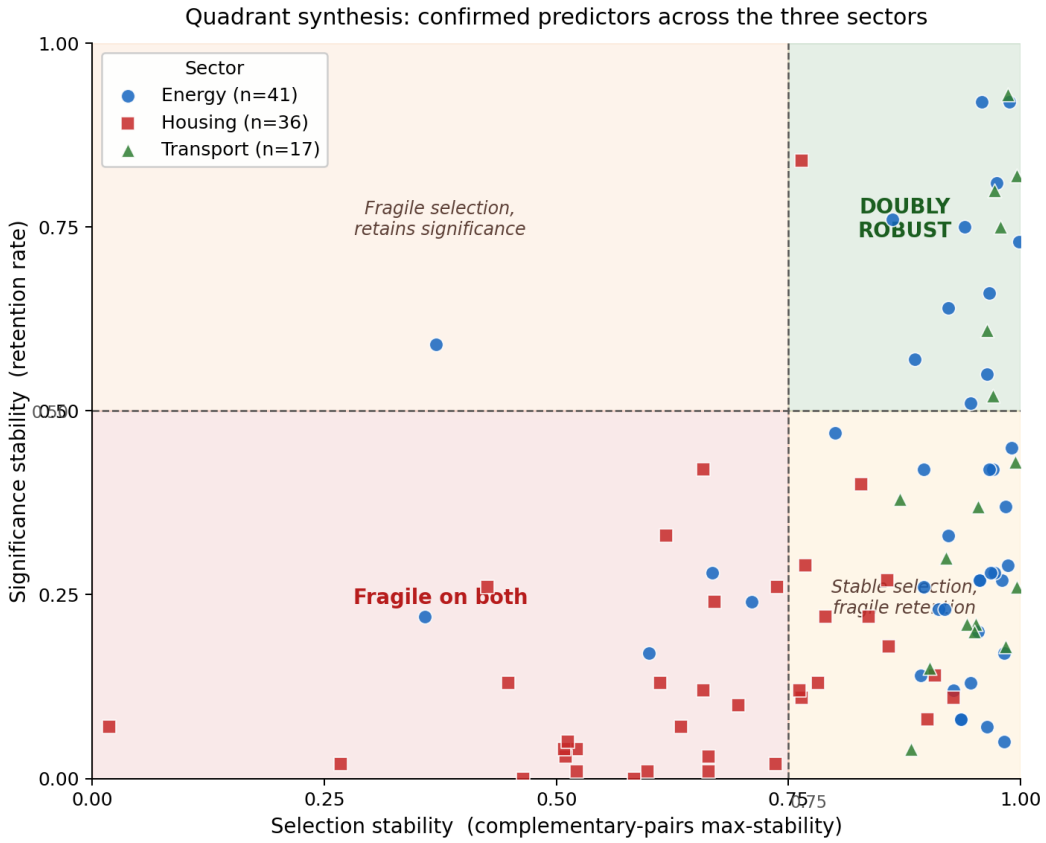


Figure 28. Confirmed predictors from all three sectors plotted on the quadrant framework. Each point is one confirmed predictor.

Why this matters. The gradient is informative about which sector's findings can be cited at the predictor level and which require a construct-level reading. Transport and Energy support predictor-level claims for their doubly robust predictors — the EV-attitude cluster and the financial-pressure cluster respectively. Housing's findings, with only one doubly robust predictor (heat_fuel_wood, an intuitively interpretable

wood-heating-and-cold-resilience signal), should be read as describing the constructs that matter for Housing concern (housing-cost burden, dampness, dwelling quality) rather than the individual significance of any one predictor. This is consistent with the sample-size and country-heterogeneity differences: Transport has the largest combined sample within the most homogeneous country pair (Italy and Spain), Energy spans five countries with a moderate combined sample, and Housing has the smallest sample (n = 263) across three structurally different housing markets. Direction is universally robust across all three sectors — sign consistency exceeds 0.94 in every sector — so claims about which way a predictor pushes concern are defensible everywhere; claims about how strong the effect is at the individual-predictor level are fully defensible only in the doubly robust quadrant.

Per-sector summary of key takeaways

For convenience, the per-sector headlines below collect the single most important takeaway for each sector. These are the statements most defensible for policy use.

- Energy. Financial pressure indicators are the strongest predictors of Energy concern. Respondents with two or more episodes of utility-bill arrears are nearly six times more likely to be highly concerned about Energy hazards. Country differences in overall Energy concern are not statistically significant.

- Housing. Respondents in the lowest housing-cost band have 2 to 3 times the odds of high concern across four Housing hazards — among the most robust confirmed cross-hazard findings in the study.. Housing is also the only sector where country differences are significant: Portuguese respondents are more concerned than German respondents.

- Transport. Women, unemployed respondents, frequent travellers, and non-native-language speakers all carry significantly higher Transport concern; remote workers carry significantly lower concern. The speaks_national_language finding is the most distinctive policy-relevant result.

4. Discussion

4.1 Linking empirical findings to the FITTER-EU analytical framework

The empirical findings provide strong support for the broader analytical framework developed across the FITTER project. They confirm key propositions advanced in D2.1 regarding the relationship between the twin transition and structural inequalities, while also reflecting the conceptual and methodological developments presented in D2.2 and D2.3, as well as the vulnerability mapping and disadvantaged-group identification undertaken in D3.1 and D3.2. These deliverables argue that the impacts of the twin transition emerge through interactions between socio-economic structures, policy choices, technological systems and pre-existing forms of vulnerability.

The findings of this study reinforce this perspective. Across Energy, Housing and Transport, financial vulnerability emerges as the most consistent predictor of elevated concern about transition-related hazards. This pattern indicates that perceptions of risk are embedded within wider structures of disadvantage. Individuals facing economic strain are more exposed to the material consequences of transition policies and more likely to perceive these policies as sources of uncertainty and insecurity.

This finding is particularly significant in light of D3.1 and D3.2, which identified energy poverty, housing cost burdens, transport poverty, labour-market exclusion and limited access to resources as recurring vulnerability pathways across countries and sectors. The present analysis provides empirical evidence that these vulnerabilities are also reflected in how people anticipate and interpret transition-related risks. In this

sense, the findings contribute to the operationalisation of FITTER's approach by identifying measurable characteristics associated with heightened concern and potential disadvantage.

The results are also consistent with the integrated perspective developed in D2.2 and D2.3. D2.2 conceptualises the twin transition as a socio-technical transformation shaped by interactions between technological change, governance arrangements and social structures, while D2.3 emphasises rights, inclusion and equitable access to essential services. The observed patterns demonstrate that concerns about transition-related hazards are closely associated with identifiable social and economic vulnerabilities across sectors, supporting the view that the distributional consequences of the twin transition are inseparable from broader questions of social inclusion, accessibility and justice.

4.2 Financial vulnerability and energy justice

The strongest cross-sector finding of the study is the relationship between financial vulnerability and perceived hazards. In Energy, utility-bill arrears and low confidence in the ability to meet energy costs are among the most powerful predictors of concern. In Housing, concern is concentrated among respondents in the lowest housing-cost band. In Transport, unemployment and other indicators of economic insecurity are associated with elevated hazard perception. Although the specific variables differ across sectors, they consistently point towards the same underlying phenomenon: respondents experiencing economic pressure perceive transition-related hazards as more serious and potentially more disruptive.

These findings align with the distributive dimension of energy justice and with longstanding concerns regarding energy poverty. As highlighted in D2.1, access to affordable energy services is a prerequisite for wellbeing and social participation. The results are also consistent with the broader interpretation developed in D3.1, which conceptualises energy poverty as part of a wider constellation of structural vulnerabilities linked to housing conditions, labour-market position, income security and social exclusion.

From this perspective, hazard perception reflects concerns associated with both specific policy measures and the cumulative effects of multiple vulnerabilities. Respondents experiencing financial strain are likely to have fewer resources available to absorb additional costs associated with energy system transformation, building renovation requirements and changing mobility systems. Their concerns therefore reflect assessments of their capacity to cope with transition-related changes and the potential consequences these changes may have for their everyday lives.

The findings are also consistent with the rights-based perspective developed in D2.3, which emphasises access to essential services as a precondition for social participation and wellbeing. Concerns regarding energy affordability, housing costs and mobility can therefore be understood as indicators of broader risks to social inclusion. The consistent association between financial vulnerability and elevated concern across sectors suggests that the capacity to participate in and benefit from the twin transition remains unevenly distributed. Affordability and accessibility consequently emerge as central dimensions of a just transition, closely connected to wider questions of equality, inclusion and fair access to opportunities.

4.3 Housing findings and environmental justice

The Housing sector findings provide particularly strong evidence of the relationship between socio-economic vulnerability and transition-related concern. Respondents in the lowest housing-cost band consistently

exhibit substantially higher odds of reporting concern across multiple housing-related hazards. This is one of the most robust findings in the study, appearing across several models and remaining stable in direction across multiple sensitivity tests. The consistency of this pattern suggests that housing-related vulnerabilities constitute an important pathway through which the twin transition may generate or exacerbate inequalities.

The findings reinforce the vulnerability pathways identified elsewhere in the FITTER project. D3.1 highlighted housing affordability, housing quality, retrofit costs and exposure to climate-related risks as key sources of structural vulnerability, while D2.3 situated these challenges within a rights-based understanding of housing as a fundamental social entitlement. Against this background, the concentration of concern among respondents in the lowest housing-cost band suggests that perceptions of transition-related hazards are shaped by structural inequalities affecting access to affordable, adequate and climate-resilient housing.

Although the present analysis does not directly measure dwelling characteristics, the observed pattern is consistent with the proposition that economically vulnerable households may face greater difficulties in adapting to transition-related requirements, including energy-efficiency renovations, low-carbon heating systems, building retrofits and climate adaptation measures. Limited financial resources can constrain their capacity to absorb transition-related costs and respond to changing regulatory and environmental demands, contributing to heightened perceptions of risk and vulnerability.

An additional finding of relevance is that Housing is the only sector in which significant country differences emerge, with Portuguese respondents reporting higher levels of concern than German respondents. While the composition of the Housing sample requires caution in interpreting these differences, the result suggests that national housing systems, housing-market conditions and building-stock characteristics continue to influence how transition-related risks are experienced. This observation is consistent with the broader argument developed in D3.1 that vulnerability emerges through the interaction of socio-economic circumstances and the institutional and territorial contexts within which people live.

The findings highlight the importance of integrating housing affordability, social protection and climate objectives within housing-transition policies. Decarbonisation measures, renovation programmes and climate adaptation initiatives can deliver substantial long-term benefits, yet their distributional consequences are likely to vary across social groups. The concentration of concern among economically vulnerable households indicates that questions of affordability and access remain central to the social sustainability of housing transitions and to the progressive realisation of the right to adequate housing emphasised in D2.3.

4.4 Transport findings and spatial justice

The Transport sector findings reveal important dimensions of vulnerability that extend beyond financial factors alone. While economic insecurity remains relevant, the role of language, gender and travel frequency suggests that access to mobility systems is shaped by social, informational and institutional factors alongside material resources. These findings point towards a broader understanding of transport-related vulnerability encompassing accessibility, inclusion and participation in the transition process.

The finding that non-native-language speakers are significantly more likely to express concern about transport hazards is particularly noteworthy. Respondents who do not speak the dominant language of their country report substantially higher concern regarding transport-related disruptions and, in particular, the

prospect of longer or more complex journeys. This suggests that transport transitions are experienced through the ability to access information, navigate evolving systems and understand regulatory or technological changes, in addition to changes in infrastructure and services.

These results resonate with the concept of recognition justice developed in D2.1 and with the rights-based perspective advanced in D2.3. Recognition justice highlights the importance of acknowledging the circumstances and lived experiences of different social groups, while D2.3 emphasises meaningful access to essential services and opportunities, including mobility. The elevated concern reported by non-native-language speakers can therefore be interpreted as reflecting challenges of accessibility, inclusion and participation. This interpretation is consistent with D2.3's emphasis on sustainable accessibility and with the proposition that access to information and services constitutes an increasingly important dimension of mobility in digitalised transport systems.

The findings also reinforce the vulnerability pathways identified in D3.1 and D3.2, which highlighted that migrants, ethnic minorities and other socially marginalised groups may encounter disproportionate barriers in accessing transport systems and public services. In this context, language functions as a structural factor influencing access to opportunities, services and participation in social and economic life. The present analysis provides empirical evidence that such barriers are associated with heightened perceptions of transition-related risks.

A further contribution of the findings concerns governance. D2.4 demonstrates that formal participation mechanisms do not automatically guarantee the meaningful involvement of structurally vulnerable groups in transition governance. Across different national contexts, barriers related to representation, accessibility and participation continue to affect who is able to engage effectively with policy processes. The higher levels of concern observed among non-native-language speakers may therefore reflect challenges associated with access to information and influence over decisions that shape transport transitions. This underlines the importance of governance arrangements that actively address linguistic and cultural barriers and ensure that participation processes are accessible to diverse population groups.

Gender differences in transport-related concern point in a similar direction. Research has consistently shown that mobility systems affect social groups differently and that concerns relating to safety, accessibility and everyday mobility needs are unevenly distributed. The observed gender differences therefore reinforce the broader FITTER argument that transition policies should be assessed in terms of their social inclusiveness alongside their environmental effectiveness.

The transport findings highlight the importance of recognition, accessibility and participation as dimensions of transition vulnerability. They suggest that the social sustainability of transport transitions depends not only on technological and infrastructural change but also on the conditions that enable diverse groups to access information, participate in decision-making and benefit from emerging mobility systems.

4.5 Intersectionality and the structuring of hazard perception

Across all sectors, the findings support the argument that inequalities emerge through the interaction of multiple dimensions of disadvantage. Financial vulnerability, employment status, gender, language and behavioural factors combine to shape hazard perception in ways that cannot be adequately captured through single-variable explanations. The results therefore reinforce the importance of analysing transition-related

risks through an intersectional lens, recognising that vulnerability is produced through the cumulative effects of overlapping social, economic and institutional conditions.

This finding is highly consistent with the identification of disadvantaged groups undertaken in D3.2. The scenario-building exercises highlighted low-income households, vulnerable workers, migrants, individuals facing housing insecurity and other structurally disadvantaged populations as groups likely to experience disproportionate impacts under future transition pathways. The present findings provide empirical support for this assessment by demonstrating that elevated concern is associated with characteristics that frequently overlap and reinforce one another. The observed patterns point towards interconnected disadvantage pathways through which multiple forms of vulnerability shape perceptions of risk and capacity to adapt.

The findings also reinforce the broader vulnerability mapping developed in D3.1, which emphasised that the twin transition is likely to interact with pre-existing inequalities across housing, energy, mobility and labour-market participation. Similar patterns emerge across Energy, Housing and Transport, suggesting that these vulnerabilities reflect structural conditions that cut across policy domains. This interpretation supports the FITTER proposition that the social impacts of the twin transition are embedded within a wider system of interconnected risks and opportunities.

The results also have implications for governance. D2.4 demonstrates that identifying vulnerable groups is only one component of an equitable transition. Across the countries examined, formal mechanisms for stakeholder engagement often coexist with barriers related to representation, accessibility and unequal capacities to participate in decision-making processes. The present findings highlight why these challenges matter. If the groups experiencing the highest levels of concern are insufficiently represented in policy design and implementation, important sources of vulnerability may remain overlooked or inadequately addressed.

The findings therefore support intersectional and participatory approaches to transition governance. Policies based on average effects or broad socio-demographic categories may fail to capture the cumulative disadvantages experienced by individuals located at the intersection of multiple forms of inequality. Governance arrangements face similar challenges when participation is dominated by actors with greater resources, stronger organisational capacity or easier access to decision-making processes. The analysis points to the importance of institutional mechanisms capable of recognising diverse forms of vulnerability and incorporating them into policy development and implementation.

Hazard perception emerges from the interaction of socio-economic position, access to resources, social recognition and opportunities for participation. These findings strengthen the case for governance approaches that address vulnerability as a multidimensional phenomenon and anticipate how different forms of disadvantage may combine during the twin transition.

4.6 Limited role of national context and implications for justice frameworks

One of the more striking findings of the analysis is the limited role played by country-level differences in explaining variation in hazard perception. With the exception of some Housing outcomes, the primary drivers of concern are individual and structural characteristics rather than national context.

This finding does not imply that national policy environments are unimportant. Financial pressure, housing insecurity and barriers to mobility influence perceptions of transition-related hazards across different national settings, indicating the presence of vulnerability mechanisms that operate across diverse institutional and territorial contexts.

This interpretation aligns closely with the findings of D3.1, which identified recurring patterns of energy poverty, housing vulnerability and transport disadvantage across participating countries despite differences in policy frameworks and institutional arrangements. At the same time, the significant differences observed within the Housing sector suggest that national housing systems, territorial conditions and policy environments continue to shape how vulnerabilities are experienced and expressed.

The findings are also consistent with the comparative governance analysis presented in D2.4. Despite significant differences in institutional structures, policy traditions and governance arrangements across participating countries, D2.4 identified recurring challenges related to the inclusion of vulnerable groups, coordination across policy domains and the management of distributional impacts. The limited role of country effects observed in the present analysis suggests that many concerns associated with the twin transition emerge from structural conditions shared across European societies, even though their specific manifestations and policy implications may vary between contexts.

Altogether, these findings indicate that vulnerability within the twin transition is shaped by the interaction of common socio-economic mechanisms and context-specific institutional conditions. This supports an understanding of transition-related inequalities that recognises both the cross-national character of many vulnerability pathways and the continuing importance of national and local contexts in shaping how those vulnerabilities are experienced.

4.7 Implications for policy design and anticipatory governance

One of the most important contributions of the study concerns the relationship between hazard prevalence and model predictability. Across sectors, the hazards generating the highest levels of concern tend to be those that are least predictable, whereas hazards generating more moderate levels of concern tend to produce stronger predictive models.

This pattern partly reflects a statistical property of the data: when concern is nearly universal, there is limited variation for explanatory variables to capture. The finding nevertheless carries important implications for anticipatory governance. D2.2 and D2.3 emphasise the importance of understanding system dynamics, identifying leverage points and anticipating unintended consequences before inequalities become entrenched. The concern–predictability inversion provides a useful framework for distinguishing between different categories of transition risks.

The most widely shared concerns appear to represent systemic risks affecting broad sections of the population and are therefore likely to require universal policy responses. Concerns that are more strongly associated with identifiable socio-economic characteristics indicate specific vulnerability pathways and lend themselves more readily to targeted interventions. The distinction highlights the need for policy mixes that combine broad-based measures with interventions tailored to the circumstances of particular groups.

The findings also reinforce a central proposition running through D2.2, D2.3 and D3.1: vulnerabilities associated with the twin transition emerge through identifiable interactions between social, economic, technological and institutional factors. The capacity to identify these patterns before policy impacts materialise lies at the heart of anticipatory governance. By revealing which characteristics are consistently associated with heightened concern across sectors, the analysis provides an evidence base for anticipating where transition-related risks are most likely to emerge and which groups may require additional support.

The governance implications are closely aligned with the conclusions of D2.4. Across the countries examined, D2.4 identified recurring challenges relating to stakeholder representation, uneven participation capacities and the inclusion of vulnerable groups in decision-making processes. The present findings contribute to these debates by providing a systematic basis for identifying the groups most likely to experience elevated concern and potential disadvantage. This information can support more inclusive governance processes and help policymakers recognise concerns that may otherwise remain underrepresented in policy design and implementation.

The findings are also relevant to the issue of social acceptance, which D2.4 identifies as a critical condition for the success of twin-transition initiatives. Concerns concentrated among vulnerable groups may signal potential implementation challenges, emerging inequalities or sources of resistance to policy measures. Incorporating these concerns into governance processes can strengthen legitimacy, foster trust and improve the responsiveness of transition policies.

These insights are particularly relevant for the development of the FITTER Digital Platform and related decision-support tools. The ability to distinguish between broadly shared concerns and concerns associated with specific vulnerability pathways demonstrates how computational approaches can support anticipatory governance. The analysis provides a basis for identifying potential risks at an early stage, informing targeted interventions and supporting participation strategies that engage groups likely to experience disproportionate impacts from transition policies.

5. Conclusions and Further Developments

This deliverable has presented a comprehensive computational analysis of hazard perception associated with three core EU twin-transition policy objectives: the deployment of renewable energy (PV systems), the retrofitting of buildings, and the electrification of transport. By operationalising these objectives through a rigorous statistical framework, the analysis translates individual-level survey data into structured evidence on how transition-related risks are perceived across different social groups.

Consistent with the FITTER conceptual framework developed in Deliverables D2.1, D2.2 and D2.3, the findings demonstrate that hazard perception is shaped by underlying socio-economic, behavioural and structural factors. In particular, the results confirm that financial vulnerability acts as a cross-cutting driver of perceived risk across all policy domains, reinforcing the premise that the twin transition interacts with existing inequalities and may amplify them if their distributional consequences are not adequately addressed.

The analysis employed a comprehensive nine-step computational framework across the Energy, Housing and Transport sectors within the FITTER-EU project. The methodology integrates LASSO-based predictor selection, ordinal proportional-odds regression with Brant testing, Likelihood-Ratio Tests with Benjamini–

Hochberg FDR correction, and events-per-predictor capping into a reproducible, sector-independent analytical pipeline applied to 1,028 respondents from five EU countries.

From a methodological perspective, the main contributions are: (i) a dual-analysis framework (per-hazard and composite) that enables both fine-grained hazard-specific profiling and aggregate sector-level assessment; (ii) a sub-sampling stability protocol (Step 5c) that distinguishes robust from fragile predictor selections; (iii) macro-variable integration that replaces country labels with specific national indicators, increasing the transferability of findings across comparable EU contexts; and (iv) a systematic concern–predictability inversion analysis that distinguishes between hazards requiring universal policy responses and hazards associated with identifiable vulnerability pathways.

From an empirical perspective, three overarching findings emerge consistently across the Energy, Housing and Transport sectors:

(1) Financial vulnerability as a structural determinant of risk perception

Across all analyses, individuals experiencing economic strain, including arrears, housing cost burdens and constrained resources, report significantly higher levels of concern. This finding supports the rights-based and justice-oriented perspective advanced in D2.3, which emphasises unequal access to essential services such as energy, housing and mobility as a key dimension of social inequality.

(2) Differentiated vulnerability and intersectional effects

Hazard perception is shaped by multiple interacting factors, including gender, language, employment status and behavioural characteristics. The results reinforce the importance of intersectional analysis, highlighting how vulnerabilities emerge through overlapping social, economic and institutional conditions rather than through isolated characteristics.

(3) Limited role of national context compared to structural factors

Country-level differences explain a relatively small share of variation in perceived risks. Structural socio-economic conditions appear to exert a stronger influence on hazard perception than national context alone, suggesting that many drivers of vulnerability associated with the twin transition operate across diverse institutional settings.

The hazards attracting the highest levels of concern are also the least predictable from individual characteristics. These concerns appear to reflect broadly shared risks affecting large sections of the population. Hazards displaying greater variation across respondents are more closely associated with identifiable socio-economic and demographic characteristics, indicating specific vulnerability pathways that can be examined through targeted analysis.

These findings provide empirical support for the broader analytical framework developed across the FITTER project. The results reinforce the propositions advanced in D2.1, D2.2 and D2.3 that the twin transition unfolds within existing structures of inequality and should be understood as a socio-technical transformation with significant social implications. They also complement the vulnerability mapping undertaken in D3.1, the identification of disadvantaged groups explored in D3.2, and the governance analysis presented in D2.4,

which highlighted the importance of participation, representation and social acceptance in shaping transition outcomes.

Across the Energy, Housing and Transport sectors, the analysis demonstrates that perceptions of transition-related hazards are systematically associated with structural factors such as financial vulnerability, housing insecurity, labour-market disadvantage and barriers to accessibility. These findings suggest that the groups identified in previous FITTER work as vulnerable to the consequences of the twin transition are also more likely to perceive transition-related risks. The analysis therefore contributes to the operationalisation of FITTER's understanding of vulnerability by identifying measurable characteristics associated with elevated concern and potential disadvantage.

Beyond the specific sectoral findings, the analysis demonstrates the value of combining computational risk assessment with the broader conceptual and methodological framework developed in FITTER. The results show that transition-related concerns are systematically associated with identifiable vulnerability pathways and that these vulnerabilities can be analysed across sectors using a common analytical framework. The analysis also highlights the governance relevance of hazard perception by identifying groups that may face disproportionate risks or encounter barriers to participation in transition processes. This contributes to FITTER's objective of supporting anticipatory governance through evidence capable of informing more inclusive, responsive and socially sustainable policy development.

The findings demonstrate the value of integrating computational modelling with broader analyses of structural vulnerability, disadvantaged groups and transition-related risks. They provide empirical evidence on how different groups experience the social dimensions of the twin transition and support the development of governance approaches capable of anticipating inequalities, strengthening participation and improving the responsiveness of policy interventions.

Further developments

The results presented in this deliverable will inform several subsequent activities:

- Integration into the FITTER Digital Platform (WP5): The confirmed predictor sets, odds ratios and marginal effects will be structured as inputs to the PDSS, enabling real-time hazard prediction and scenario simulation for policy users.
- Validation in CCGS workshops (WP6): The findings will be presented to the Cluster Community Group of Stakeholders for practical utility assessment and face-validity testing.
- Policy brief development (WP7): Sector-level findings, particularly those relating to financial vulnerability, housing affordability and accessibility barriers, will be translated into policy briefs targeted at national and EU-level policymakers.
- Extension to additional sectors: The nine-step methodology is fully replicable and can be applied to additional survey data collected during later phases of the project.
- Completion of stability test tables: The sub-sampling stability outputs (Step 5c) are available in the sector output files and will be incorporated into the final publication version of this deliverable.

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Annex

Reference Materials and Output Files

Detailed outputs for every sector pipeline step are available in the accompanying sector output archives:

- Outputs_Energy.zip — complete step-by-step output for the Energy sector
- Outputs_Housing.zip — complete step-by-step output for the Housing sector
- Outputs_Transport.zip — complete step-by-step output for the Transport sector

Each sector directory contains: the complete confirmed-predictor tables; full odds-ratio tables with 95% confidence intervals; predicted-probability tables for every predictor profile; full ANOVA tables; country-level details from Step 7b; the cross-hazard synthesis matrix from Step 9; and the Step 9b A/B convergence classification files. Every quantitative claim in Section 3 is traceable to specific tables in those files.

Survey Instruments

The sector-specific survey questionnaires (Energy, Housing, Transport) used for data collection in Task 4.1 are provided as attachments to the D4.1 deliverable. Variable dictionaries linking raw survey column names to the encoded predictor names used in this report are included in each sector's output archive.

Software and Reproducibility

All analyses were conducted in Python (3.10+) using the following primary packages: scikit-learn (LASSO, kNN imputation), statsmodels (proportional-odds regression, LRT), scipy (ANOVA, chi-squared, Tukey HSD), pandas and numpy (data manipulation), and pyarrow (output serialisation). Full requirements.txt files are included in each sector output archive. The analysis scripts are version-controlled and reproducible with the master random seed documented in each sector's Step 5c output files.